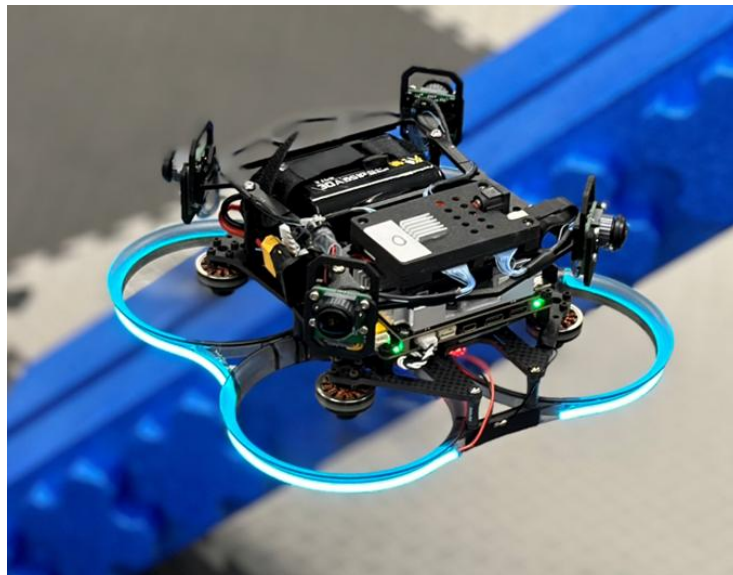


Introduction to Aerial Robotics

Lab Tutorial 1



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TAs: Yang Xu, Pusen Gao

Dept. of ECE, HKUST

18 Mar. 2026

Introduction

Logistics of Lab Tutorials

- Lab tutorial 1 (18 March): Drone assembling and manual flight
- Lab tutorial 2 (25 March): Development environment and fly in API mode with OptiTrack
- Lab tutorial 3 (29 April): State estimation and trajectory planning by your own

Today's Objective

- Assemble your quadrotor
- Let TA check your quadrotor
- Fly the quadrotor under manual control (Position Mode)

Prerequisite Knowledge

- Common sense of mechanical assembly
 - Types of screws
 - Properly tightening screws
- Soldering

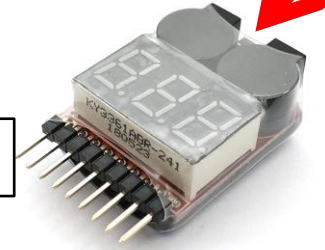
Safety Reminder

Safety Reminder

Low-voltage BB Warning Buzzer

button

set to 3.7V per cell



- Battery
 - Use BB Buzzer to check battery every time before flight
 - !!! At least 23.0V before flight, duration should be less than 5 minutes
 - !!! Stop use the battery immediately when the buzzer starts “b~b~b~b~”
 - Do not charge it by yourself
 - !!! Improper battery charging will burn Robotics Institute
 - !!! Do not use swollen batteries for flight
 - Put used batteries into the “Used Battery” box
 - Try not to make it less than 22.2V



Safety Reminder



- Soldering
 - !!! Do not touch the working soldering iron (400 °C)
 - !!! Do not touch the soldering joint immediately after soldering
 - Put idle soldering iron in stand
 - Turn soldering iron off immediately when you finish your work
- Hot-melt adhesive
 - !!! Do not touch the iron head
 - !!! Do not touch the adhesive that is melt

Safety Reminder

- Electric Tools
 - Do not use electric tools if you have no experience
 - !!! Wrong way to use electric tools may cause a wrist sprain
- Propellers
 - Do not install any propellers until TA checks your UAV
 - !!! Spining propellers can cause severe damage
 - Also beware of rotors spining under high speed even without propellers

Assembled Quadrotor (Reference)



Quadrotor BOM

- Provided in equipment box, also uploaded to Canvas
- Be careful during your experiments, as the total cost of your quadrotor is more than **HKD 5,000!**
- Please check your equipment with the BOM list first.
Contact TAs if you find any part missing.

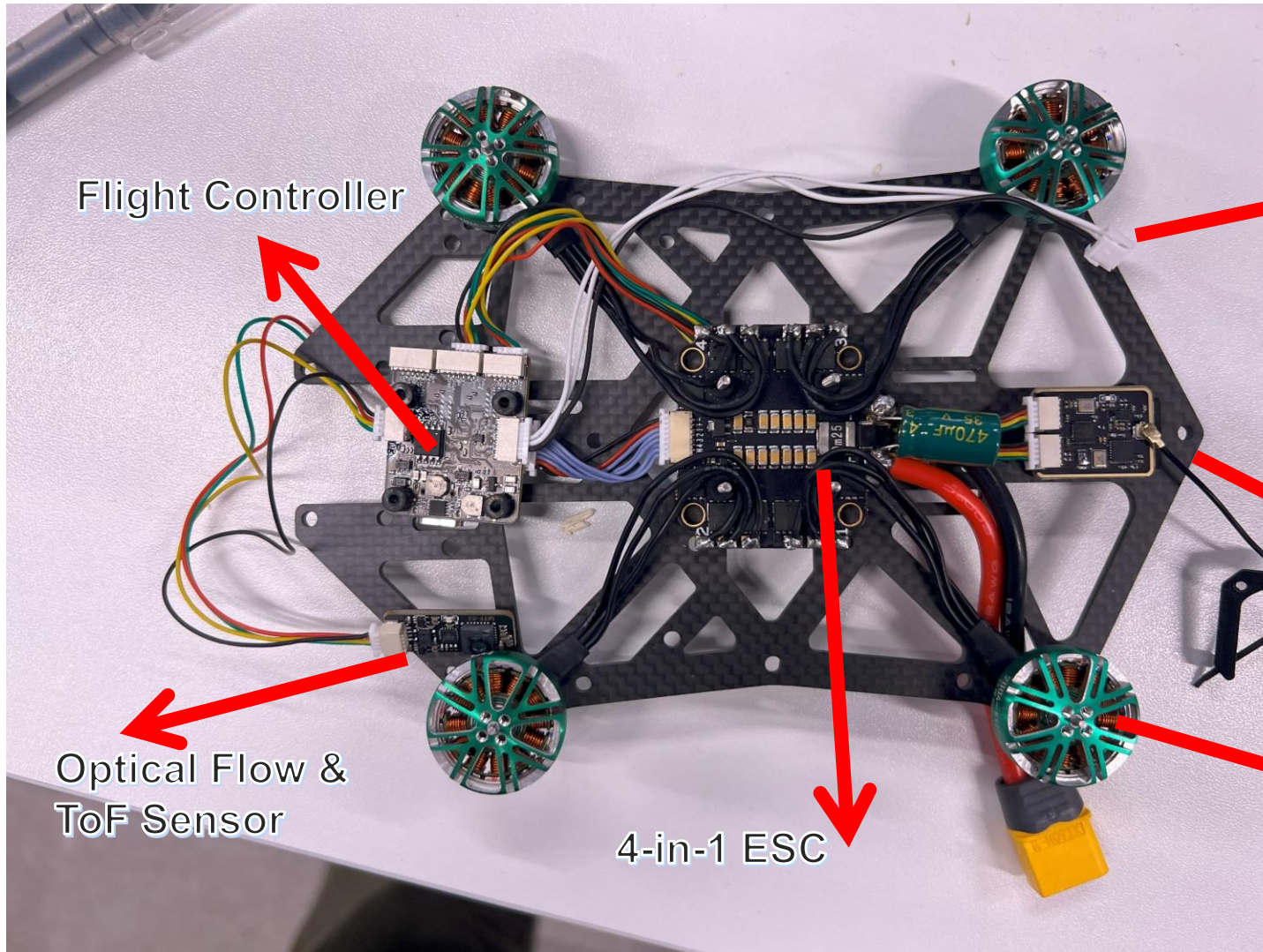
Part 1

Quadrotor Assembly Part 1

- Equipment List:

Item	Quant.
Base Frame	1
4 in 1 ESC	1
NXT Flight Controller	1
TRS Receiver	1
Optical Flow & ToF Sensor	1
Motors	4
Serial Wire	1

Wiring



Flight Controller

Optical Flow &
ToF Sensor

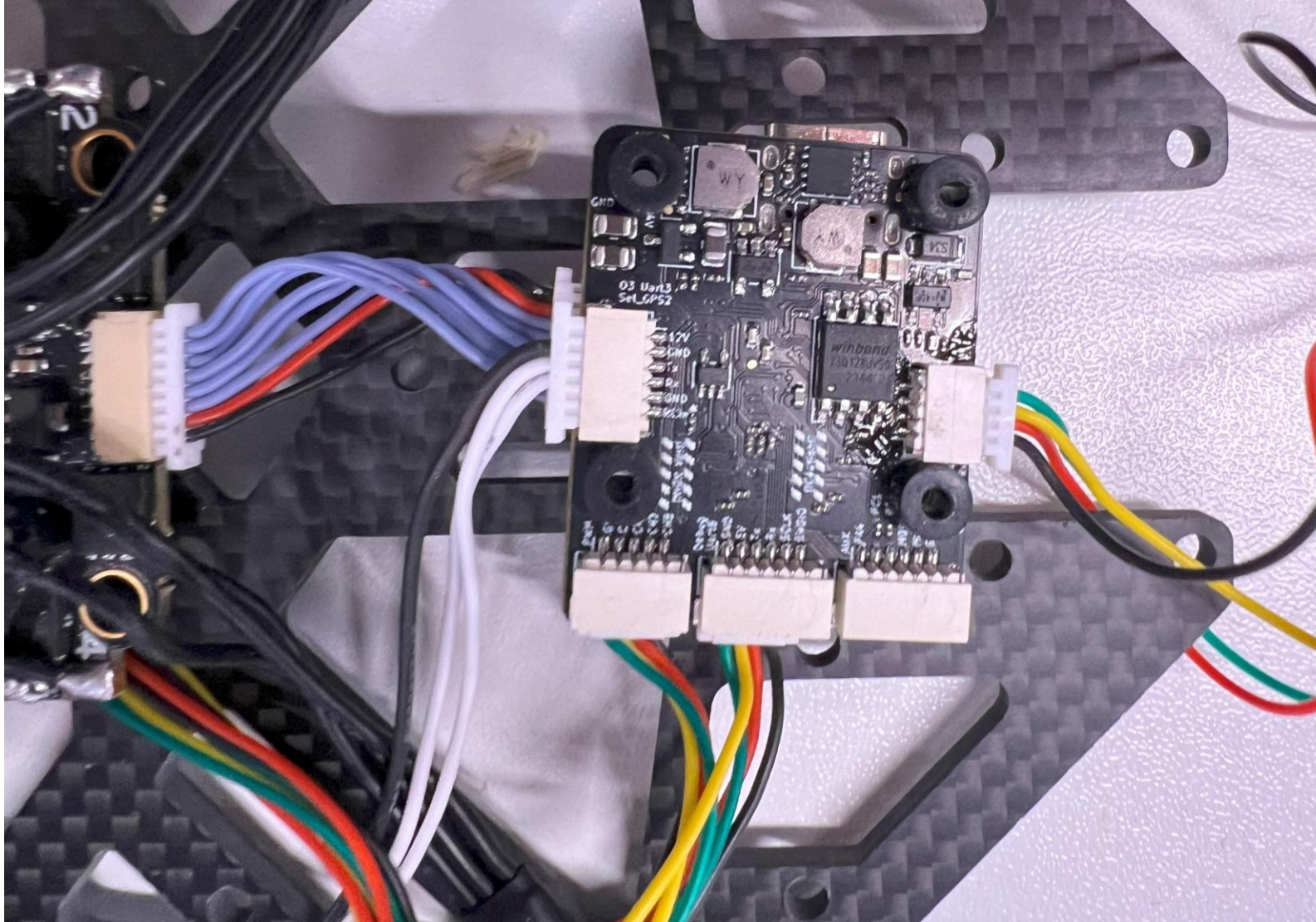
4-in-1 ESC

Serial: Connect to
onboard computer

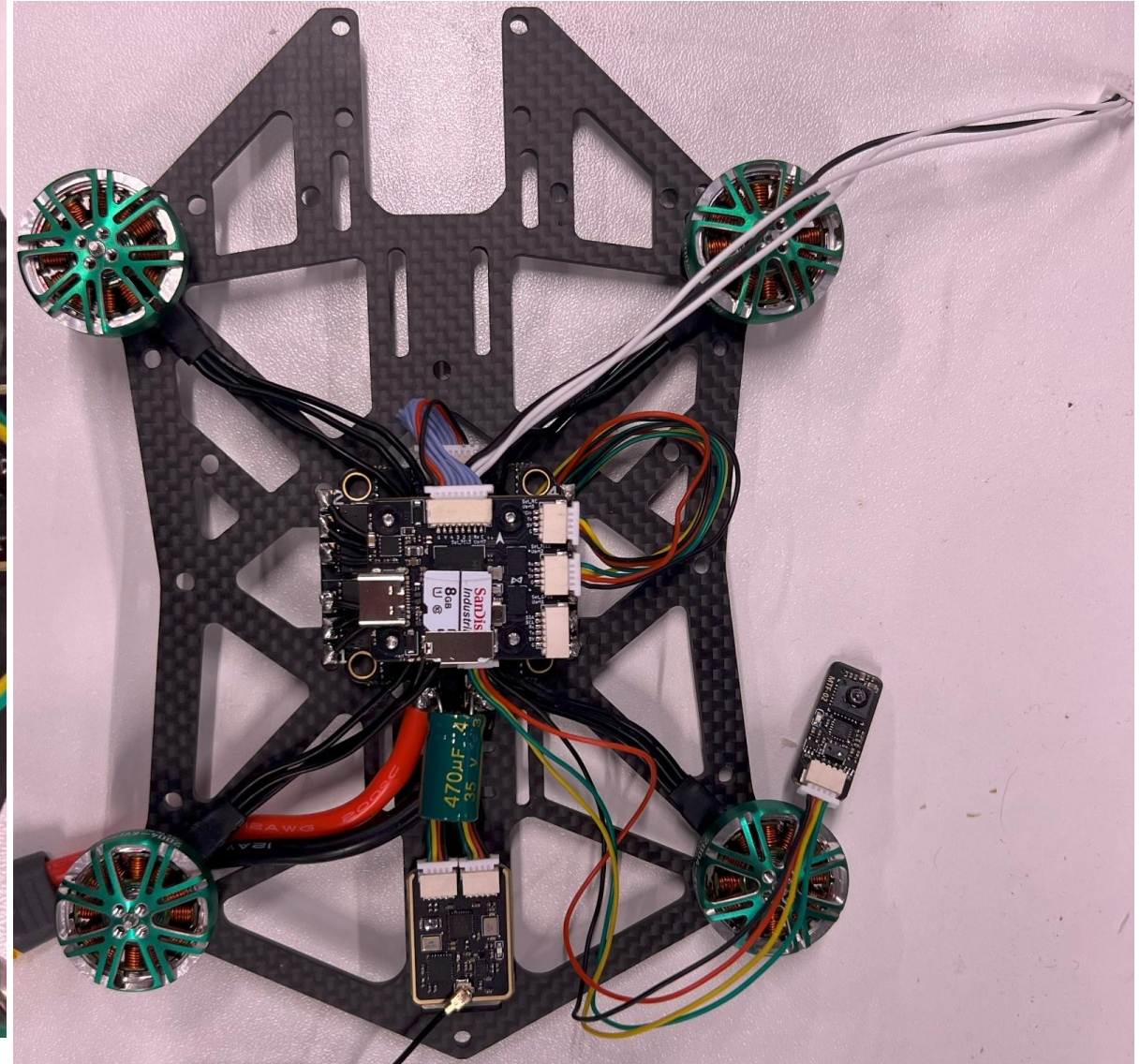
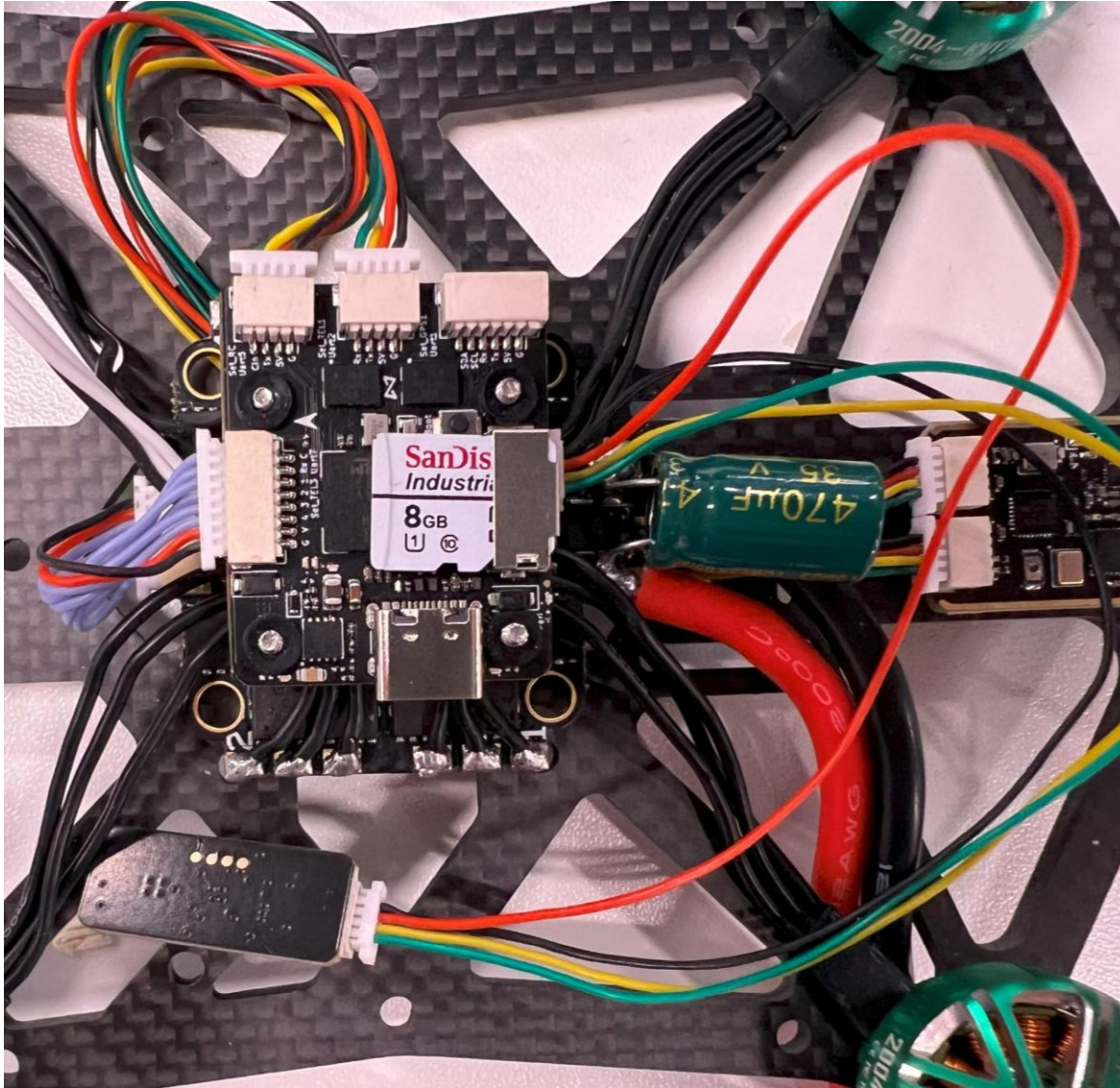
Receiver

Motors

Wiring



Wiring

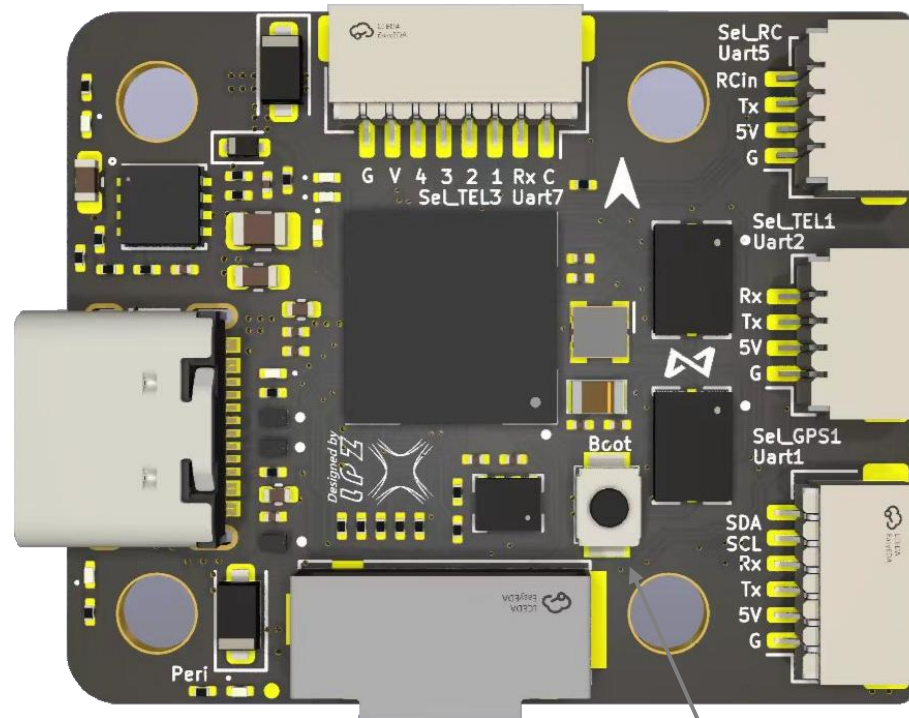


ESC

VBAT is power input (2-6S)

GND	VBAT	PWM 4	PWM 3	PWM 2	PWM 1	Rx 7	Current
-----	------	-------	-------	-------	-------	------	---------

TELEM3(PX4)
SERIAL5(AP)



Rx5	RC(PX4) SERIAL6(A P)
Tx5	
5V	
GND	

Receiver
(SBUS, CRSF, ELRS)

Rx2	TELEM1(PX 4) SERIAL1(A P)
Tx2	
5V	
GND	

SDA	GPS1(PX4) SERIAL3(A P)
SCL	
Rx1	
Tx1	
5V	
GND	

**GPS &
Compass**

USB-Type-C

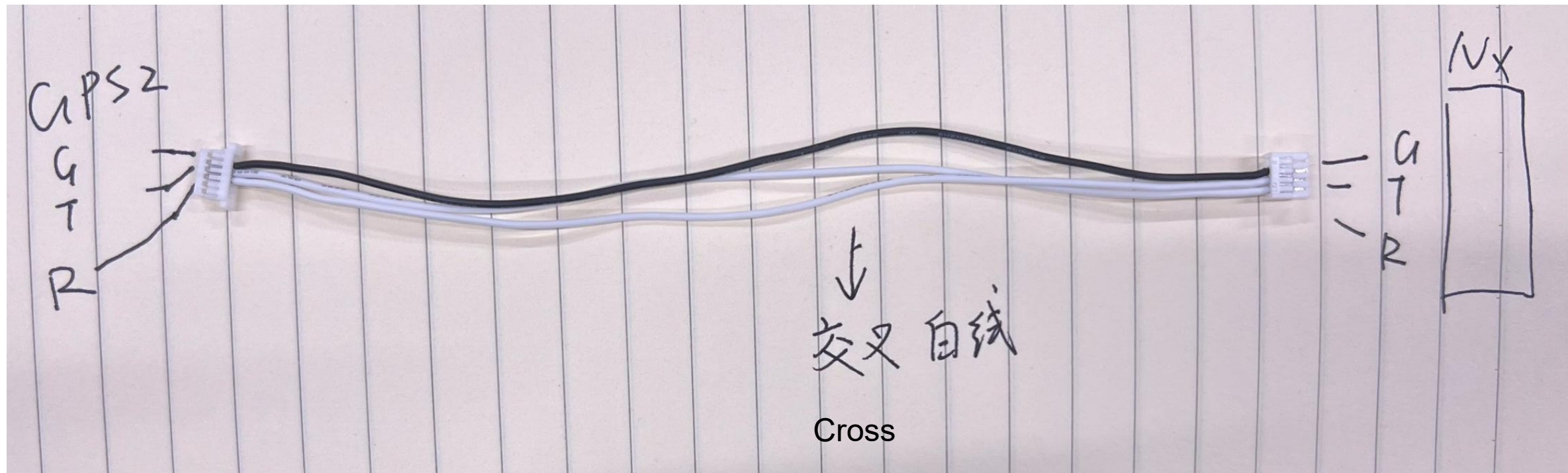
For setting parameters
and update firmware.
**USB power can not
drive other sensors**

TF Card

BOOT

GN D	5V	Tx	Rx
TELEM2(PX4) SERIAL2(AP)			

Serial line (TA Prepared)

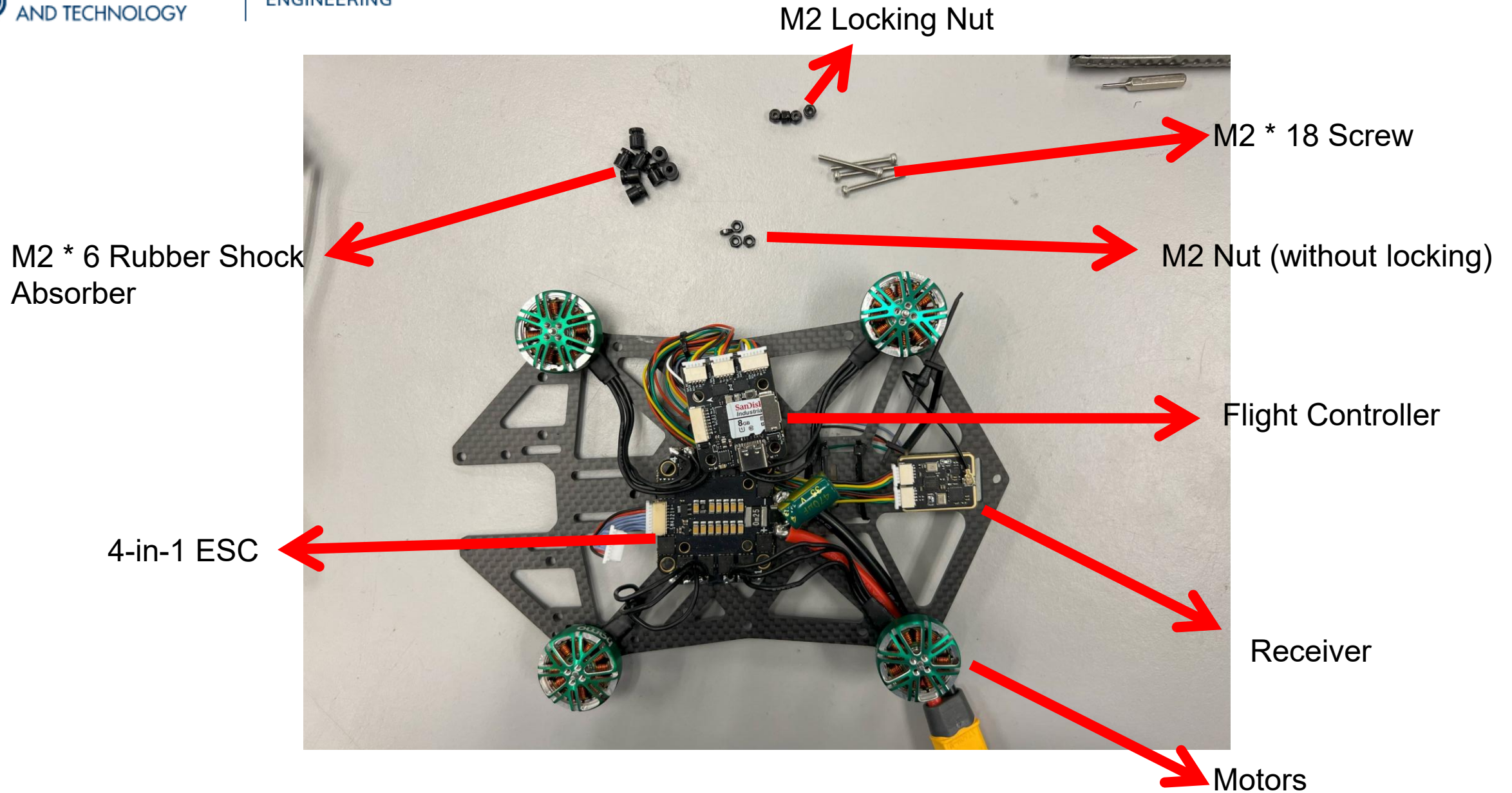


Part 2

Quadrotor Assembly Part 2

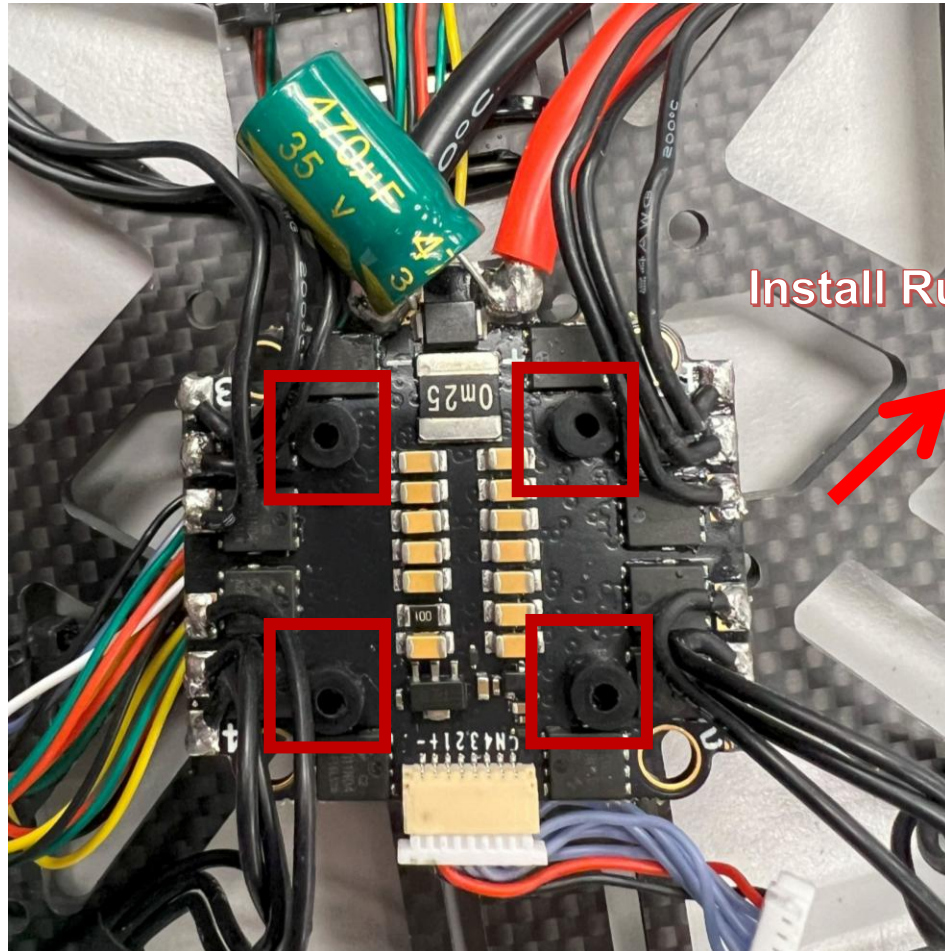
- Equipment List:

Item	Quant.
M2 * 6 Rubber Shock Absorber	8
M2 Locking Nut	4
M2 * 18 Screw	4
M2 Nut (without locking)	4

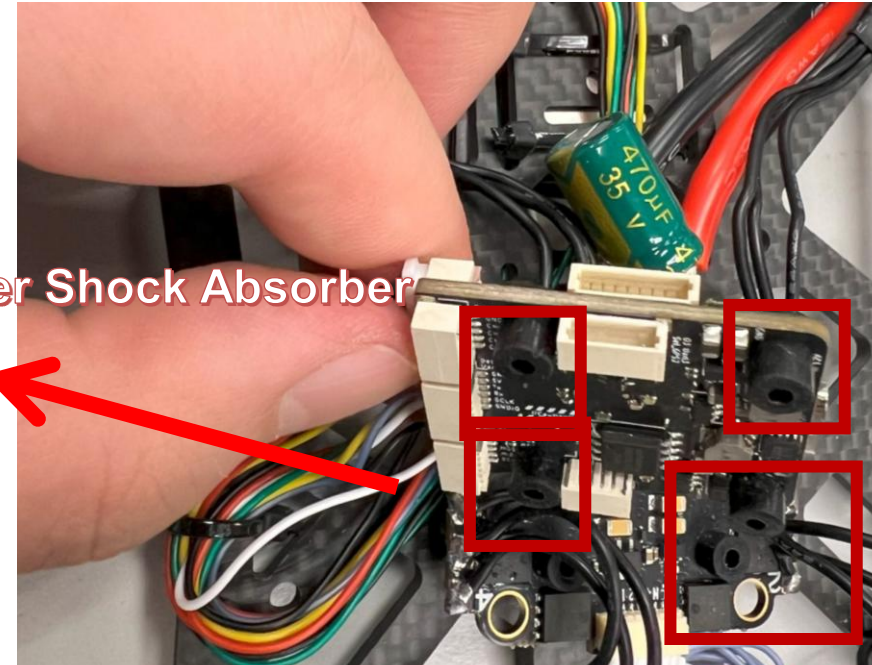


Baseboard

Baseboard (ESC & Flight Controller)



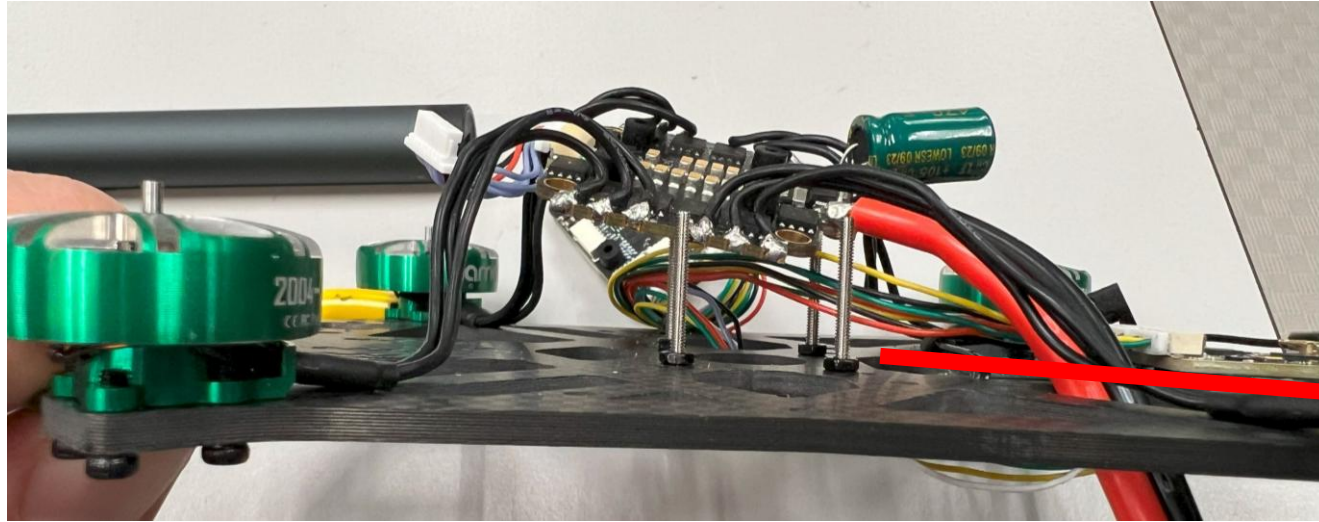
Install Rubber Shock Absorber



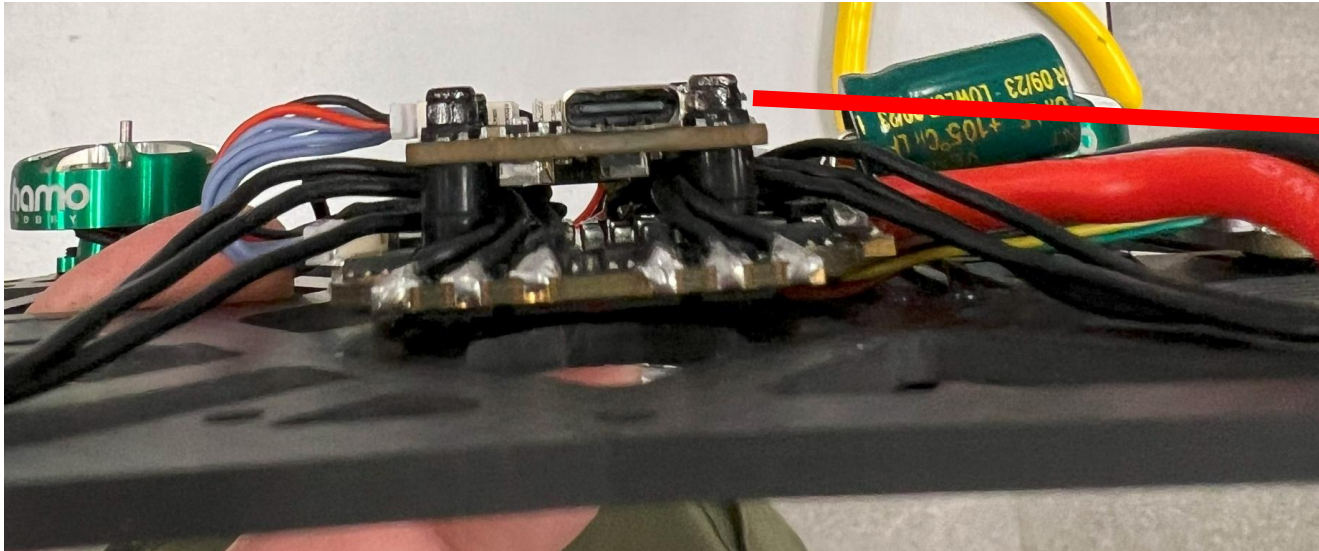
Be careful with the rubber direction,



Baseboard

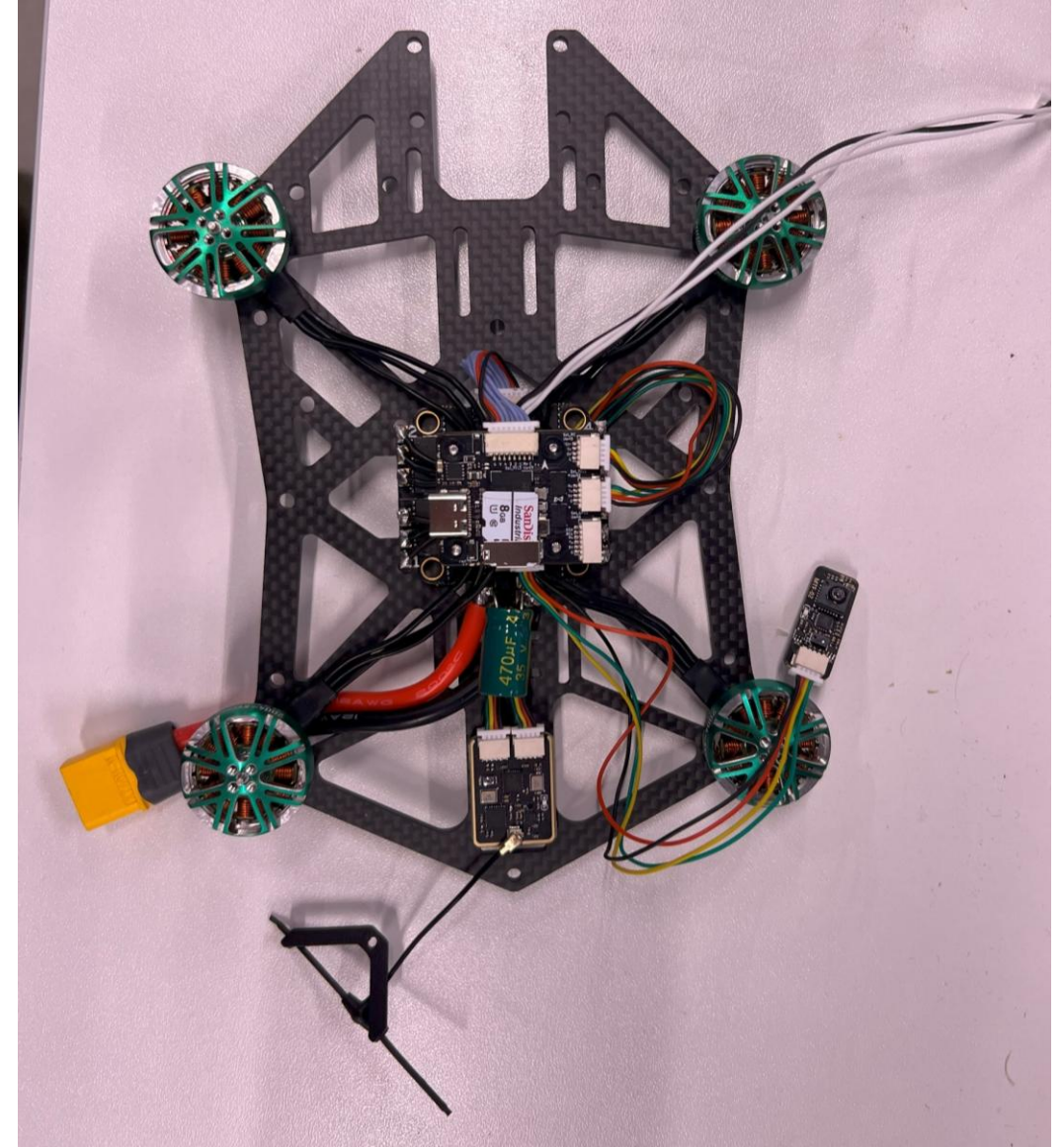
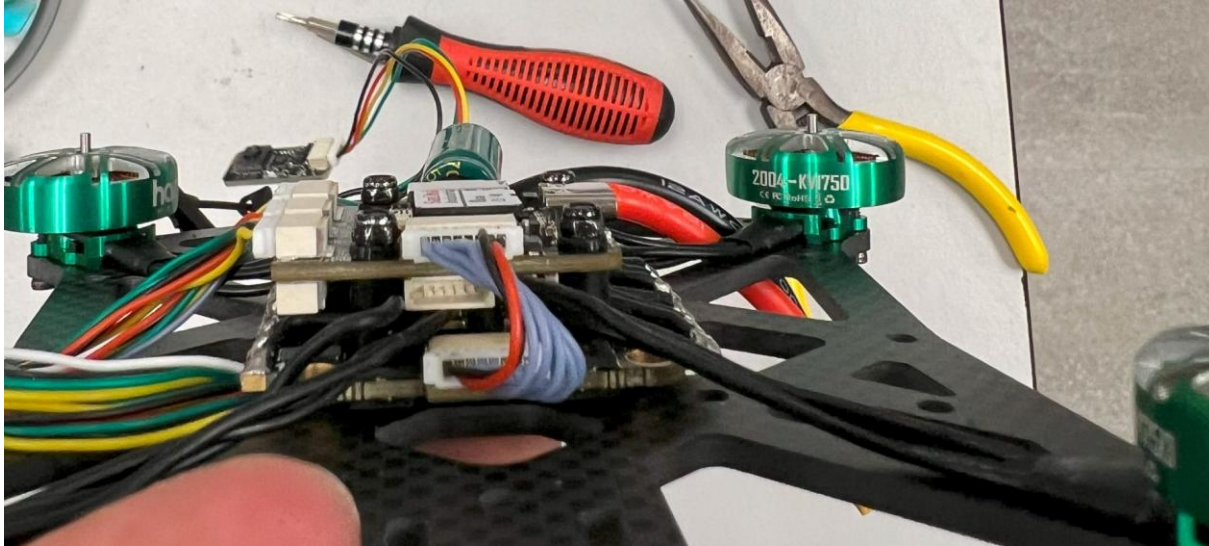


M2 Nut (without locking)



M2 Locking Nut

Flight Controller & ESC



Part 3

Quadrotor Assembly Part 3

- Equipment List:

Item	Quant.
Radio Controller (Radio Master Pocket)	1
External RF (For TRS Receiver)	1

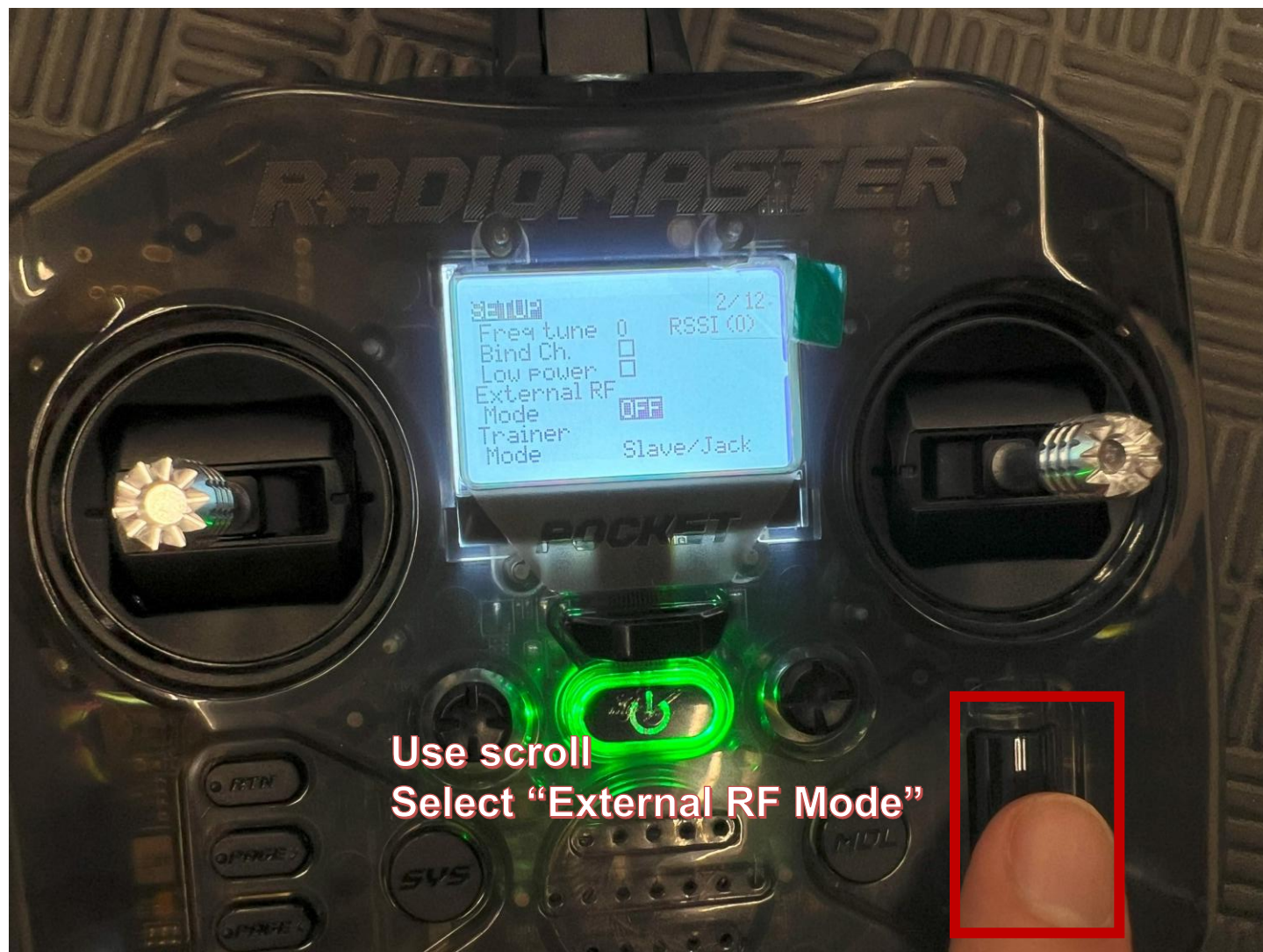
Set Up Radio



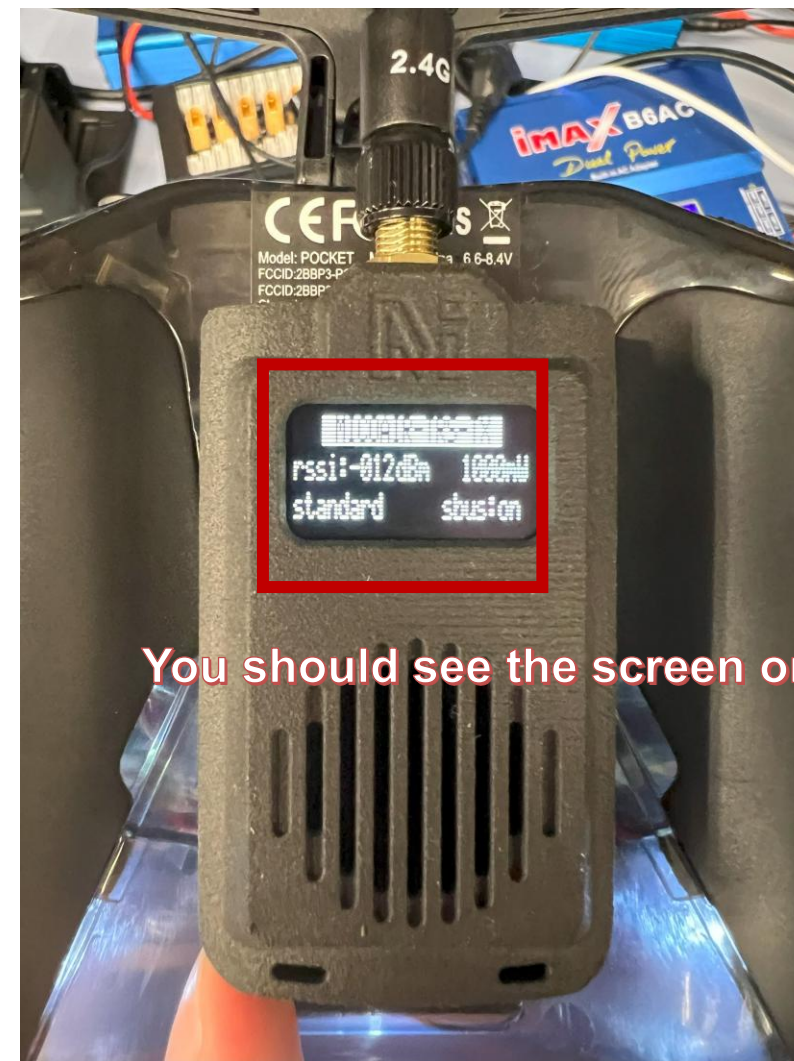
Set Up Radio



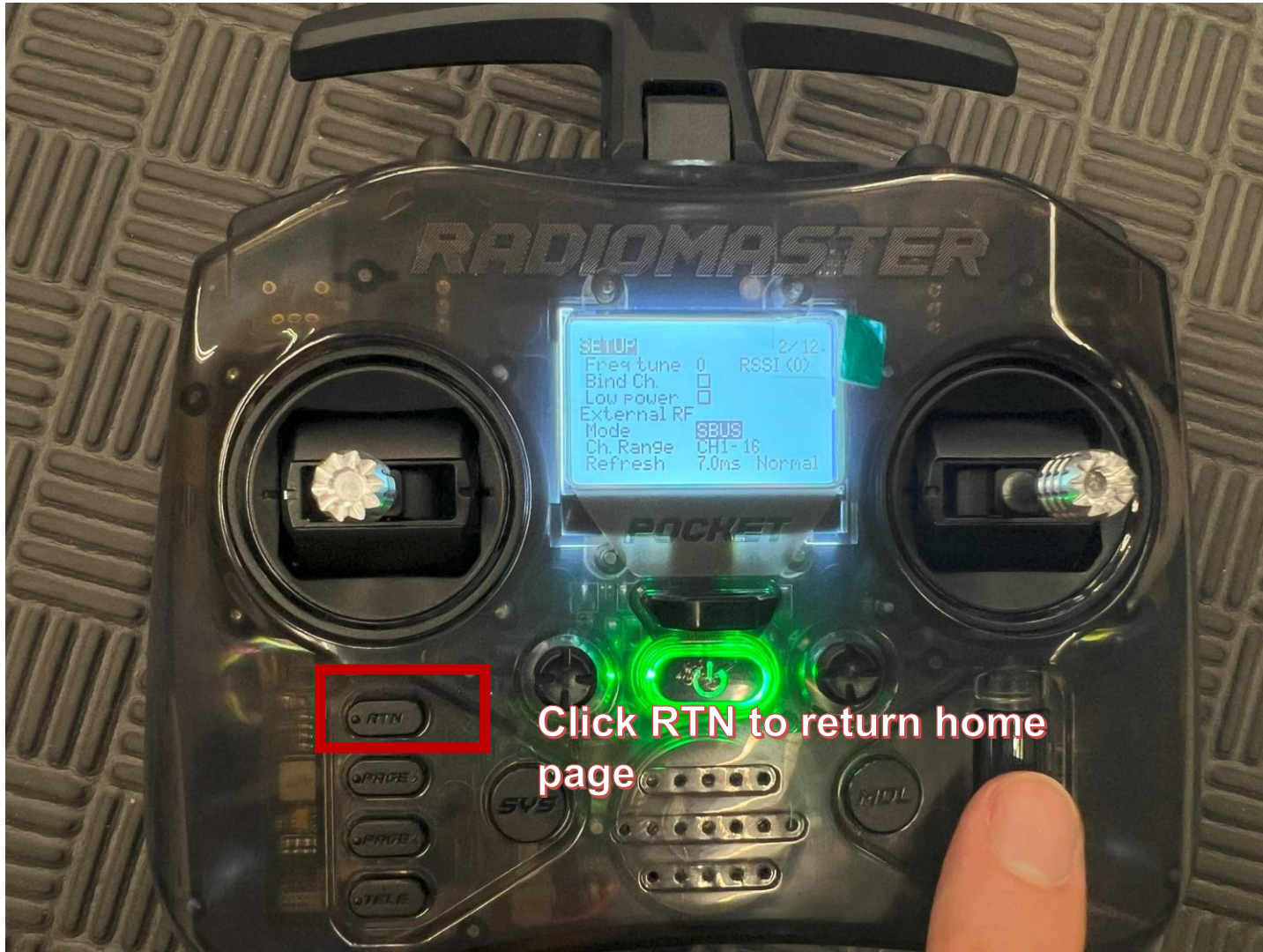
Set Up Radio



Set Up Radio



Set Up Radio



Bind Radio

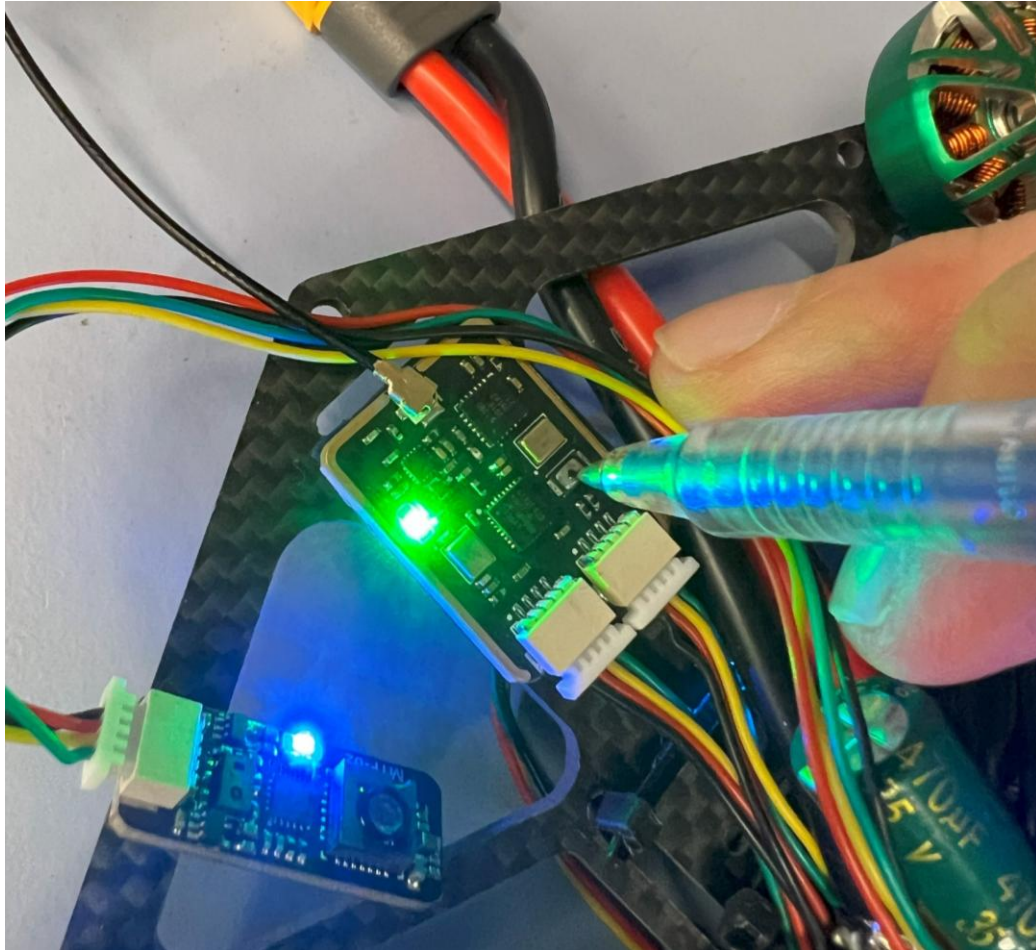
1. Remove the power
2. Click the bind button
3. Plug in the Power

You should see the green LED blinks fast

Then Switch on your Radio Controller

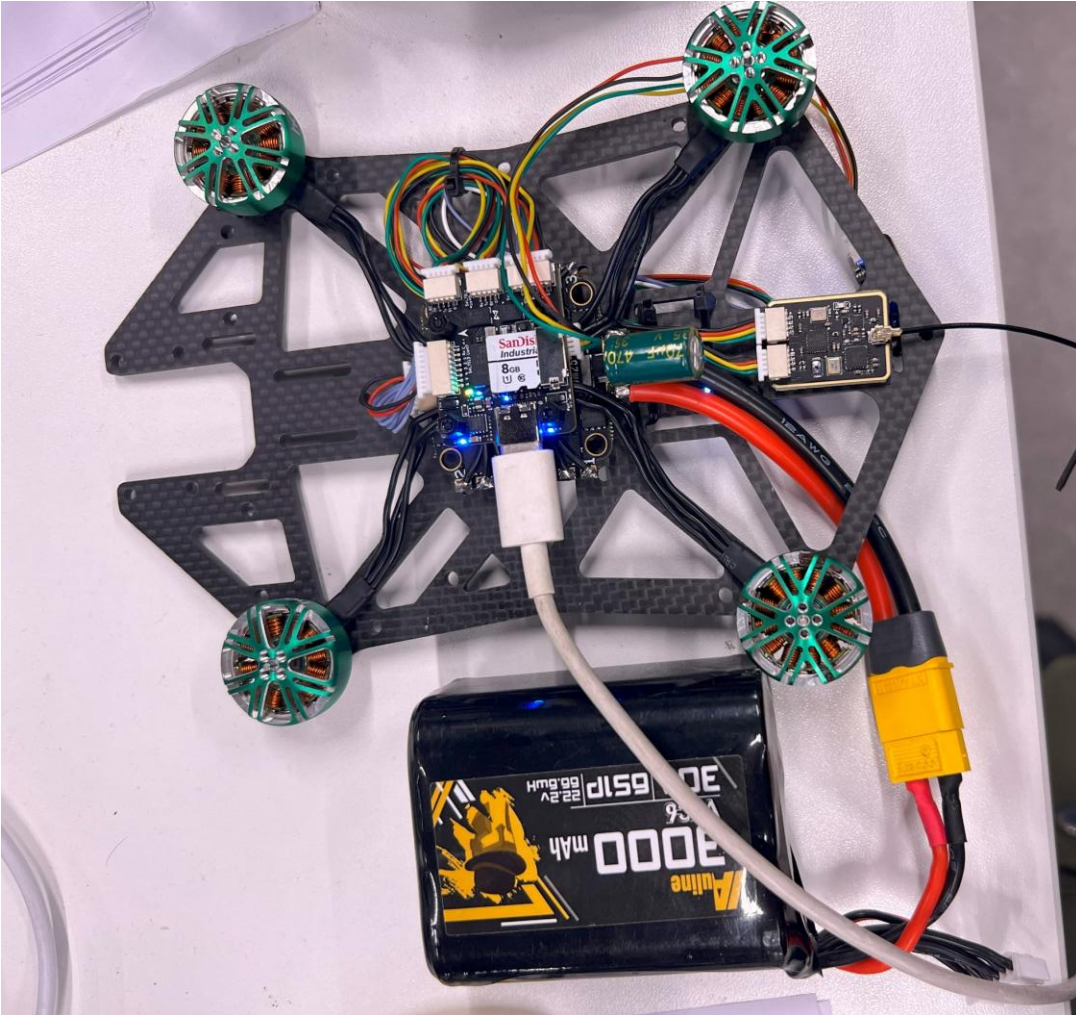
You should see the green LED be always on

Notably, this operation is only allowed for one group at the same time!



You should see the RSSI has a value

Connect to your Computer



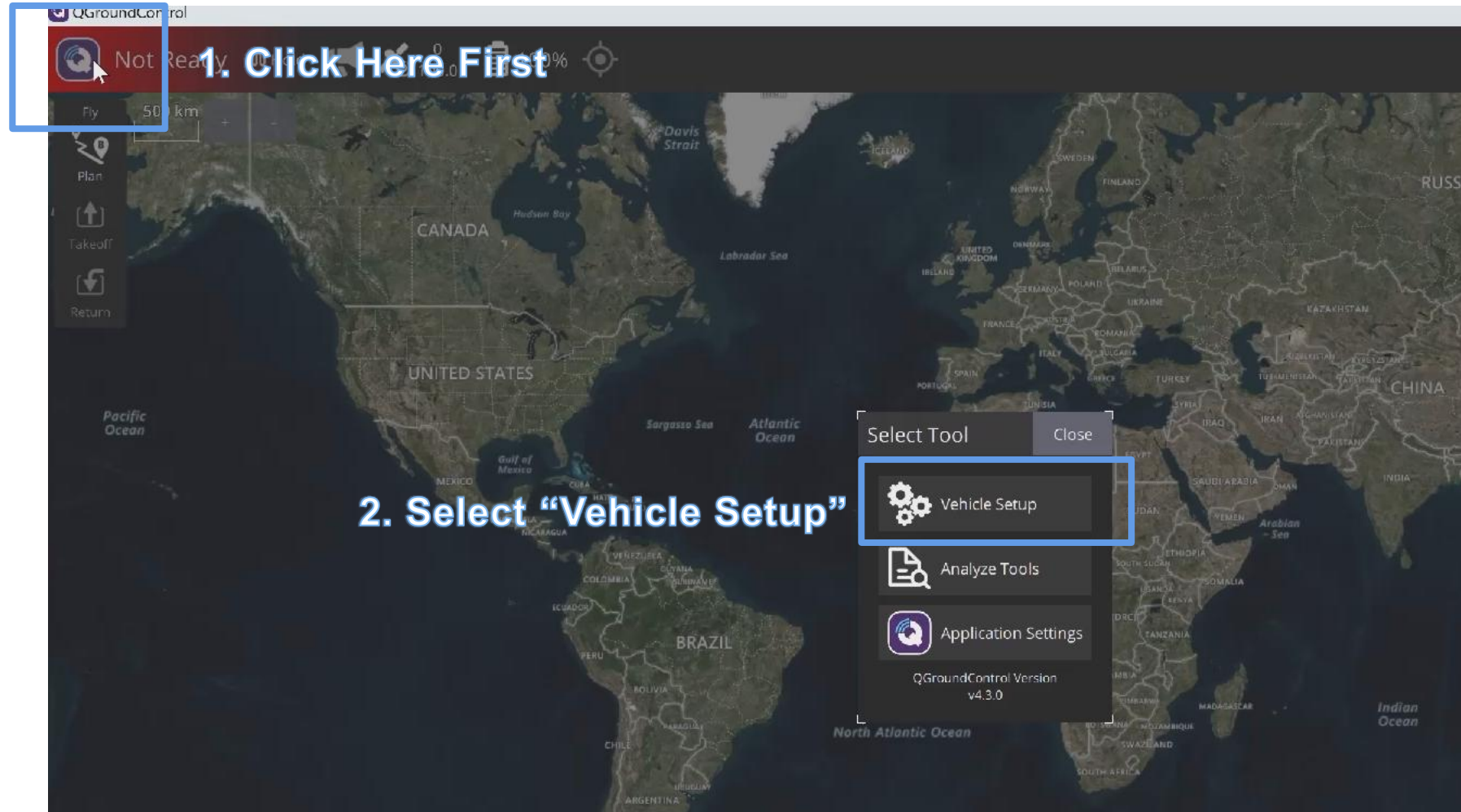
Wire like the figure.

Install QGC ([qgroundcontrol](http://qgroundcontrol.com)):

https://docs.qgroundcontrol.com/master/en/qgc-user-guide/getting_started/download_and_install.html

Set Up Flight Controller

Load Parameters: *next_5660.params*, which is provided by TAs.



Back <  Vehicle Setup

Summary	Search: Clear ☐ Show modified only
Firmware	Standard
Airframe	Battery Calibration
Sensors	Sensors
Radio	Camera Control
Flight Modes	Geometry
Power	Commander
Actuators	Multicopter Position Control
Safety	DShot
PID Tuning	EKF2
Flight Behavior	Events
Camera	Failure Detector
Parameters	Follow target
	FW TICS
	FW Rate Control
	FW Attitude Control
	Geofence
	Rover Position Control
	GPS

BAT1_A_PER_V	17.00000000	Battery 1 current per volt (A/V)
BAT1_CAPACITY	-1 mAh	Battery 1 capacity
BAT1_I_CHANNEL	-1	Battery 1 Current ADC Channel
BAT1_N_CELLS	6S Battery	Number of cells for battery 1
BAT1_R_INTERNAL	0.0050 Ohm	Explicitly defines the per cell internal resistance for battery 1
BAT1_SOURCE	Power Module	Battery 1 monitoring source
BAT1_V_CHANNEL	-1	Battery 1 Voltage ADC Channel
BAT1_V_CHARGED	4.20 V	Full cell voltage (5C load)
BAT1_V_DIV	10.11646938	Battery 1 voltage divider (V divider)
BAT1_V_EMPTY	3.20 V	Empty cell voltage (5C load)
BAT1_V_LOAD_DROP	0.10 V	Voltage drop per cell on full throttle
BAT_AVRG_CURRENT	15 A	Expected battery current in flight
BAT_CRIT_THR	7.00 %	Critical threshold
BAT_EMERGEN_THR	5.00 %	Emergency threshold
BAT_LOW_THR	15.00 %	Low threshold
BAT_V_OFFS_CURR	0.00000000	Offset in volt as seen by the ADC input of the current sensor

4.Click "Tools"

Refresh
Reset all to firmware's defaults
Reset to vehicle's configuration defaults
Load from file...
Save to file...
Clear all RC to Param
Reboot Vehicle

5.Select "Load From file"

Choose the nxt1.14.3_5660_2026.params

3. Click Parameters

QGroundControl

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Actuators

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Flight Behavior

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Parameters

Search: Clear Show modified only

Standard

Battery Calibration

Sensors

Camera Control

Geometry

Multicopter Position Control

DShot

EKF2

Events

Failure Detector

Follow target

FW TECS

FW Rate Control

FW Attitude Control

Geofence

Rover Position Control

GPS

Hover Thrust Estimator

BAT1_A_PER_V 17.00000000

BAT1_CAPACITY -1 mAh

BAT1_I_CHANNEL -1

BAT1_N_CELLS 6S Battery

BAT1_R_INTERNAL 0.0050 Ohm

BAT1_SOURCE Power Module

BAT1_V_CHARGED 4.20 V

BAT1_V_DIV 10.11646938

BAT1_V_EMPTY 3.20 V

BAT1_V_LOAD_DROP 0.10 V

BAT_AVRG_CURRENT 15 A

BAT_CRIT_THR 7.00 %

BAT_EMERGEN_THR 5.00 %

BAT_LOW_THR 15.00 %

BAT_V_OFFSET_CURR 0.00000000

Load Parameters Cancel Ok

The following parameters from the loaded file differ from what is currently set on the Vehicle. Click 'Ok' to update them on the Vehicle.

7. Click OK

Name	File	Vehicle
COM_FLTMODE1	Altitude	Stabilized
COM_FLTMODE2	Altitude	Stabilized
COM_FLTMODE3	Position	Altitude
COM_FLTMODE4	Position	Altitude
COM_FLTMODE5	Offboard	Position
COM_FLTMODE6	Offboard	Position
EKF2_HGT_REF	Range sensor	GPS
EKF2_OF_CTRL	Enabled	Disabled
IMU_GYRO_RATEMAX	2000 Hz Hz	1000 Hz Hz
IMU_INTEG_RATE	400 Hz Hz	200 Hz Hz
MAV_0_CONFIG	GPS 2	TELEM 1
MAV_0_FLOW_CTRL	Force off	Auto-detected
MAV_1_FORWARD	Disabled	Enabled
MAV_2_CONFIG	TELEM 1	Disabled
MAV_2_FLOW_CTRL	2	N/A
MAV_2_FORWARD	0	N/A
MAV_2_MODE	0	N/A
MAV_2_RADIO_CTL	1	N/A
MAV_2_RATE	0	N/A
MC_PITCHRATE_D	0.0015	0.0030
MC_PITCHRATE_I	0.101	0.200
MC_PITCHRATE_K	1.0000	0.4000
MC_PITCHRATE_P	0.080	0.150
MC_ROLLRATE_D	0.0006	0.0030
MC_ROLLRATE_I	0.039	0.200
MC_ROLLRATE_K	1.0000	0.3500
MC_ROLLRATE_P	0.030	0.150
MC_YAWRATE_D	0.00	0.00
MC_YAWRATE_I	0.12	0.10
MC_YAWRATE_K	1.0000	1.2000

QGroundControl

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Vehicle Setup

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Parameters

Search:

Clear

☐ Show modified only

Battery Calibration

Sensors

Camera Control

Geometry

Commander

Multicopter Position Control

DShot

EKF2

Events

Failure Detector

Follow target

FW TECS

FW Rate Control

FW Attitude Control

Geofence

Rover Position Control

GPS

BAT1_A_PER_V	17.00000000	Battery 1 current per volt (A/V)
BAT1_CAPACITY	-1 mAh	Battery 1 capacity
BAT1_I_CHANNEL	-1	Battery 1 Current ADC Channel
BAT1_N_CELLS	6S Battery	Number of cells for battery 1
BAT1_R_INTERNAL	0.0050 Ohm	Explicitly defines the internal resistance of battery
BAT1_SOURCE	Power Module	Battery 1 monitoring source
BAT1_V_CHANNEL	-1	Battery 1 Voltage ADC Channel
BAT1_V_CHARGED	4.20 V	Full cell voltage (5C load)
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BAT_AVRG_CURRENT	15 A	Expected battery current in flight
BAT_CRIT_THR	7.00 %	Critical threshold
BAT_EMERGEN_THR	5.00 %	Emergency threshold
BAT_LOW_THR	15.00 %	Low threshold
BAT_V_OFFS_CURR	0.00000000	Offset in volt as seen by the ADC input of the current sensor

Tools

8. Click "Tools"

9. Select "Reboot Vehicle"

Refresh

Reset all to firmware's defaults

Reset to vehicle's configuration defaults

Load from file...

Save to file...

Clear all RC to Param

Reboot Vehicle

Set up Serial

QGroundControl

Back < ⚙ Vehicle Setup

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Camera

Parameters

Search: Clear ☐ Show modified only

Mission	RC_PORT_CONFIG	Radio Controller	Serial Configuration for RC Input Driver
Mount	SER_GPS1_BAUD	Auto	Baudrate for the GPS 1 Serial Port
PWM Outputs	SER_GPS2_BAUD	921600 8N1	Baudrate for the GPS 2 Serial Port
Precision Land	SER_TEL1_BAUD	57600 8N1	Baudrate for the TELEM 1 Serial Port
Actuator Outputs	SER_TEL2_BAUD	115200 8N1	Baudrate for the TELEM 2 Serial Port
Radio Calibration			

Radio Switches

Serial

Return Mode

Return To Land

SD Logging

System

Thermal Compensation

Telemetry

Camera trigger

UXRCE-DDS Client

VTOL Takeoff

Keep the same:

SER_GPS2_BAUD: 921600 8N1
SER_TEL1_BAUD: 57600 8N1
SER_TEL2_BAUD: 115200 8N1

Disable Magnetometer

The screenshot shows the ArduPilot parameter interface. The left sidebar contains various categories: Summary, Firmware, Airframe, Sensors, Radio, Flight Modes, Power, Actuators, Safety, PID Tuning, Flight Behavior, Camera, and Parameters (highlighted in yellow). The main area displays a list of parameters. The 'System' tab is selected, and the 'SYS_HAS_MAG' parameter is highlighted with a blue box. The value for 'SYS_HAS_MAG' is 0, which is highlighted in red. A red text annotation 'Keep the same:' points to the 'SYS_HAS_MAG' row. Another red text annotation 'SYS_HAS_MAG: 0' is placed to the right of the parameter value. A third red text annotation 'Reboot Flight Controller' is located at the bottom right of the interface.

Parameter	Value	Description
Mission	SYS_AUTOCONFIG	Keep parameters
Mount	SYS_AUTOSTART	4001
PWM Outputs	SYS_CAL_ACCEL	0
Precision Land	SYS_CAL_BARO	0
Actuator Outputs	SYS_CAL_GYRO	0
Radio Calibration	SYS_CAL_TDEL	24 celcius
Radio Switches	SYS_CAL_TMAX	10 celcius
Serial	SYS_CAL_TMIN	5 celcius
Return Mode	SYS_DM_BACKEND	default (SD card)
Return To Land	SYS_FAC_CAL_MODE	Disabled
SD Logging	SYS_FAILURE_EN	Disabled
Sensor Calibration	SYS_HAS_BARO	Enabled
System	SYS_HAS_MAG	0
Thermal Compensation	SYS_HAS_NOM_DIST	0
Telemetry	SYS_HITL	HITL and SIH disabled
Camera trigger	SYS_MC_EST_GROUP	ekf2 (recommended)
UXRCE-DDS Client	SYS_RGB_MAXBRT	1.000 %
VTOL Takeoff	SYS_STCK_EN	Enabled

Keep the same:

SYS_HAS_MAG: 0

Reboot Flight Controller

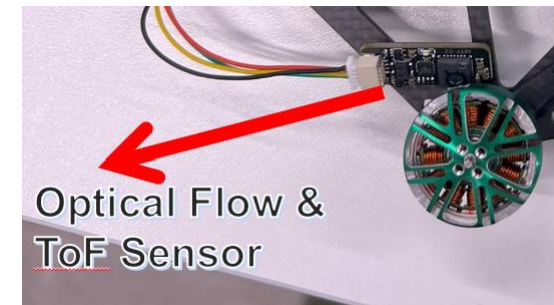
Set Optical Flow Direction

Firmware	Battery Calibration	SENS_EN_PAA3905	Disabled	PAA3905 Optical Flow
Airframe	Sensors	SENS_EN_PAW3902	Disabled	PAW3902/PAW3903 Optical Flow
Sensors	Camera Control	SENS_EN_PGA460	Disabled	PGA460 Ultrasonic driver (PGA460)
Radio	Geometry	SENS_EN_PMW3901	Disabled	PMW3901 Optical Flow
Flight Modes	Commander	SENS_EN_PX4FLOW	Disabled	PX4 Flow Optical Flow
Power	Multicopter Position Control	SENS_EN_SDP3X	Disabled	Sensirion SDP3X differential pressure sensor (external I2C)
Actuators	DShot	SENS_EN_SF1XX	Disabled	Lightware SF1xx/SF20/LW20 laser rangefinder (i2c)
Safety	EKF2	SENS_EN_SPL06	Disabled	Goertek SPL06 Barometer (external I2C)
PID Tuning	Events	SENS_EN_TF02PRO	Disabled	TF02 Pro Distance Sensor (i2c)
Flight Behavior	Failure Detector	SENS_EN_TRANGER	Disabled	TeraRanger Rangefinder (i2c)
Camera	Follow target	SENS_EN_VL53L0X	Disabled	VL53L0X Distance Sensor
Parameters	FW TECS	SENS_EN_VL53L1X	Disabled	VL53L1X Distance Sensor
	FW Rate Control	SENS_FLOW_RATE	70.000 Hz	Optical flow max rate
	FW Attitude Control	SENS_FLOW_ROT	Yaw 270°	Optical flow rotation
	Geofence	SENS_GPS_MASK	7	Multi GPS Blending Control Mask
	Rover Position Control	SENS_GPS_PRIME	0	Multi GPS primary instance
	GPS	SENS_GPS_TAU	10.0 s	Multi GPS Blending Time Constant
	Hover Thrust Estimator	SENS_LEDDAR1_CFG	Disabled	Serial Configuration for LeddarOne Rangefinder
		SENS_MAG_AUTOROT	Enabled	Automatically set external rotations
		SENS_MAG_RATE	15.000 Hz	Magnetometer max rate
		SENS_MAG_SIDES	Six side calibration	Bitfield selecting mag sides for calibration

Keep the same:

SENS_FLOW_ROT: Yaw 270

Tips: If you can not find this parameters, please check whether you have **plug in the optical flow sensor** and **set the correct serial**



Calibrate Gyroscope

The image shows the QGroundControl interface for vehicle setup. The 'Vehicle Setup' screen is open, with the 'Sensors Setup' tab selected. The 'Sensors Setup' screen displays a list of sensors on the left: Accelerometer, Level Horizon, Cancel, and Orientations. The 'Gyroscope' button is highlighted with a blue box. Below the list, a 'Calibrate Gyro' dialog box is shown, containing the text: 'For Gyroscope calibration you will need to place your vehicle on a surface and leave it still. Click Ok to start calibration.' The 'Ok' button is highlighted with a yellow box. To the right, a smaller inset shows the 'Sensors Setup' screen with the 'Gyroscope' button highlighted, and a green box around the 'Calibration complete' message and a 3D model of a plane on a surface, with the word 'Completed' below it.

QGroundControl

Back < Vehicle Setup

Summary

Sensors Setup

Sensors Setup is used to calibrate the sensors within your vehicle.

Start the individual calibration steps by clicking one of the buttons to the left.

Gyroscope

Accelerometer

Level Horizon

Cancel

Orientations

Calibrate Gyro

Cancel Ok

For Gyroscope calibration you will need to place your vehicle on a surface and leave it still.

Click Ok to start calibration.

Back < Vehicle Setup

Summary

Sensors Setup

Sensors Setup is used to calibrate the sensors within your vehicle.

Gyroscope

Accelerometer

Level Horizon

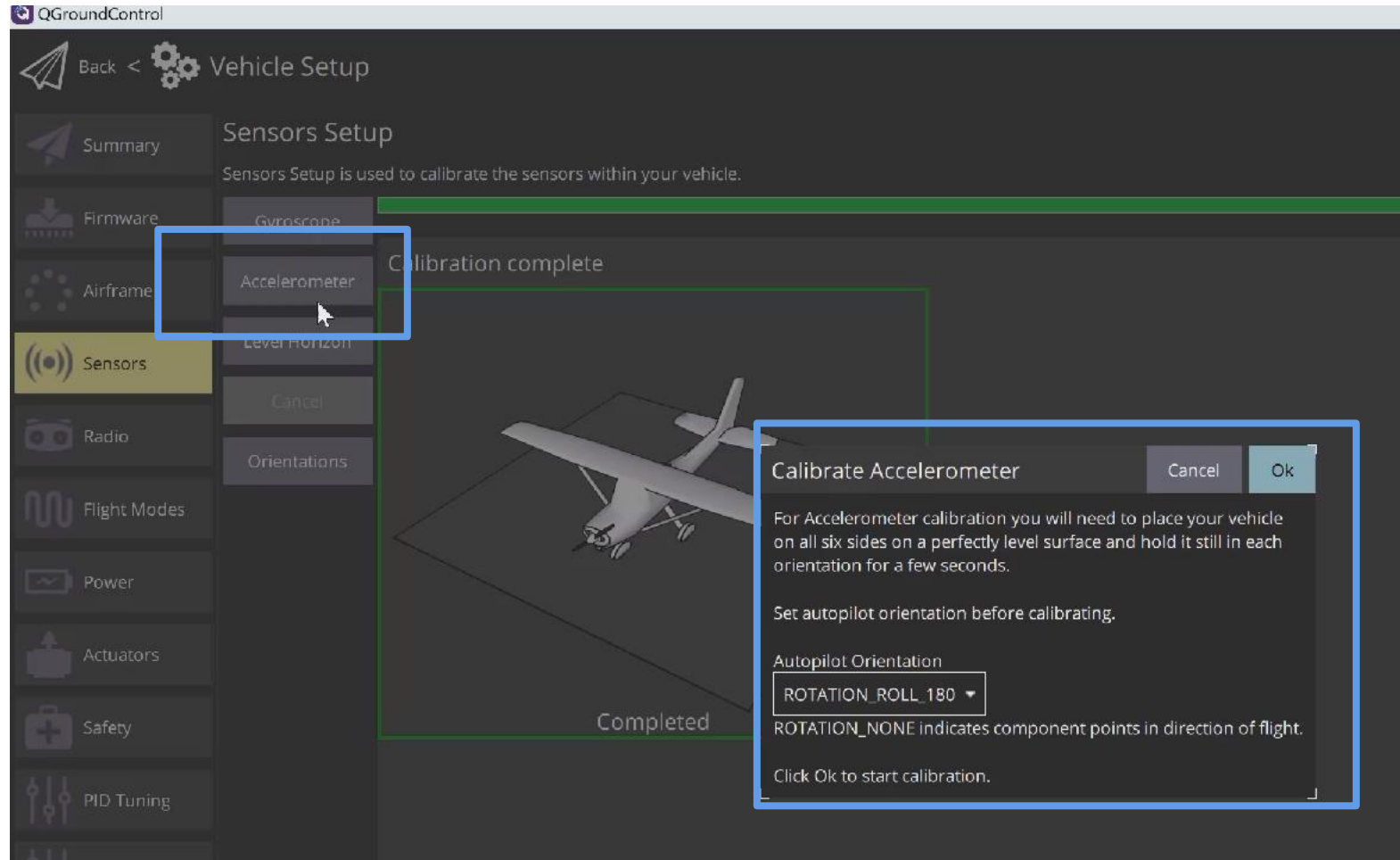
Cancel

Orientations

Calibration complete

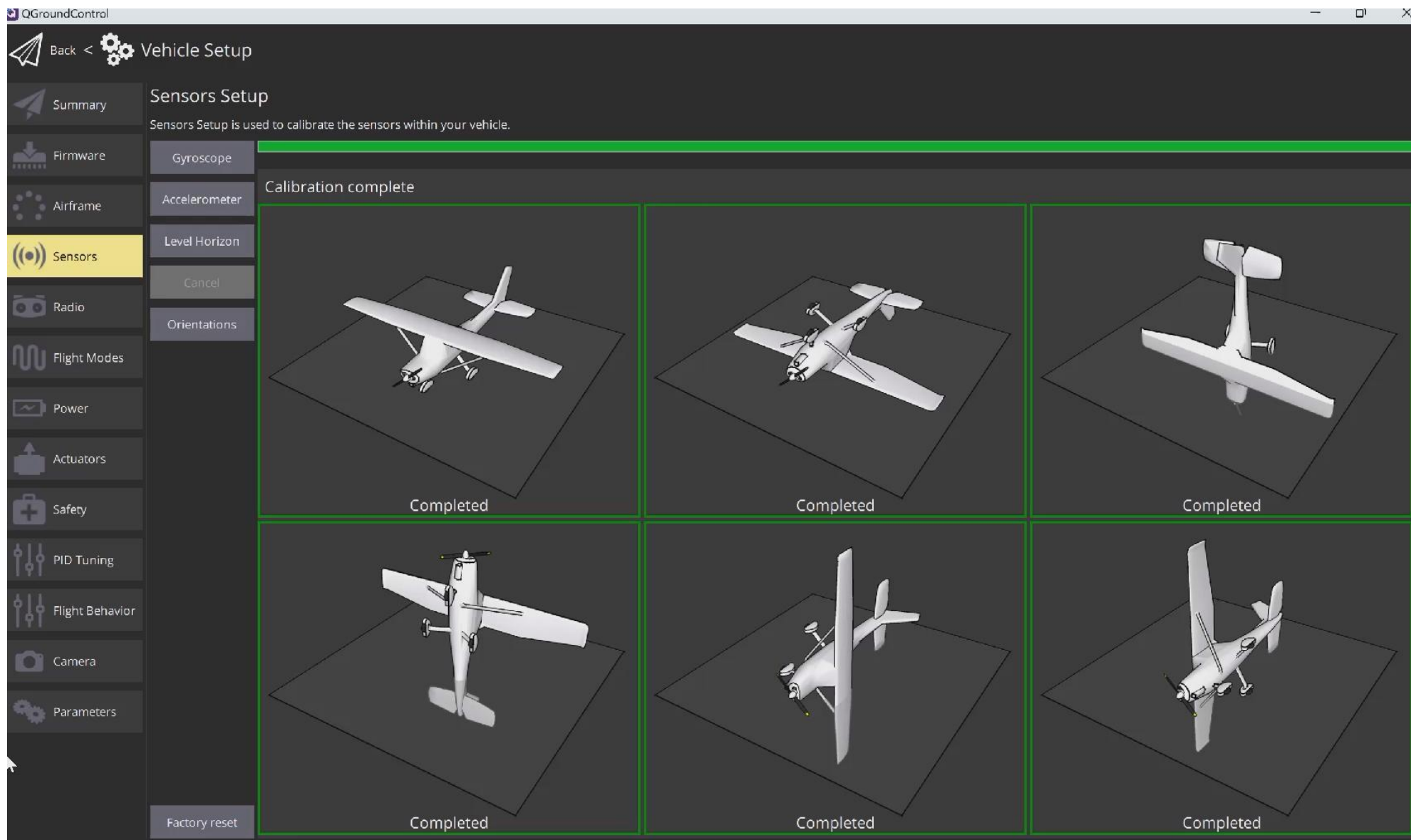
Completed

Calibrate Accelerometer

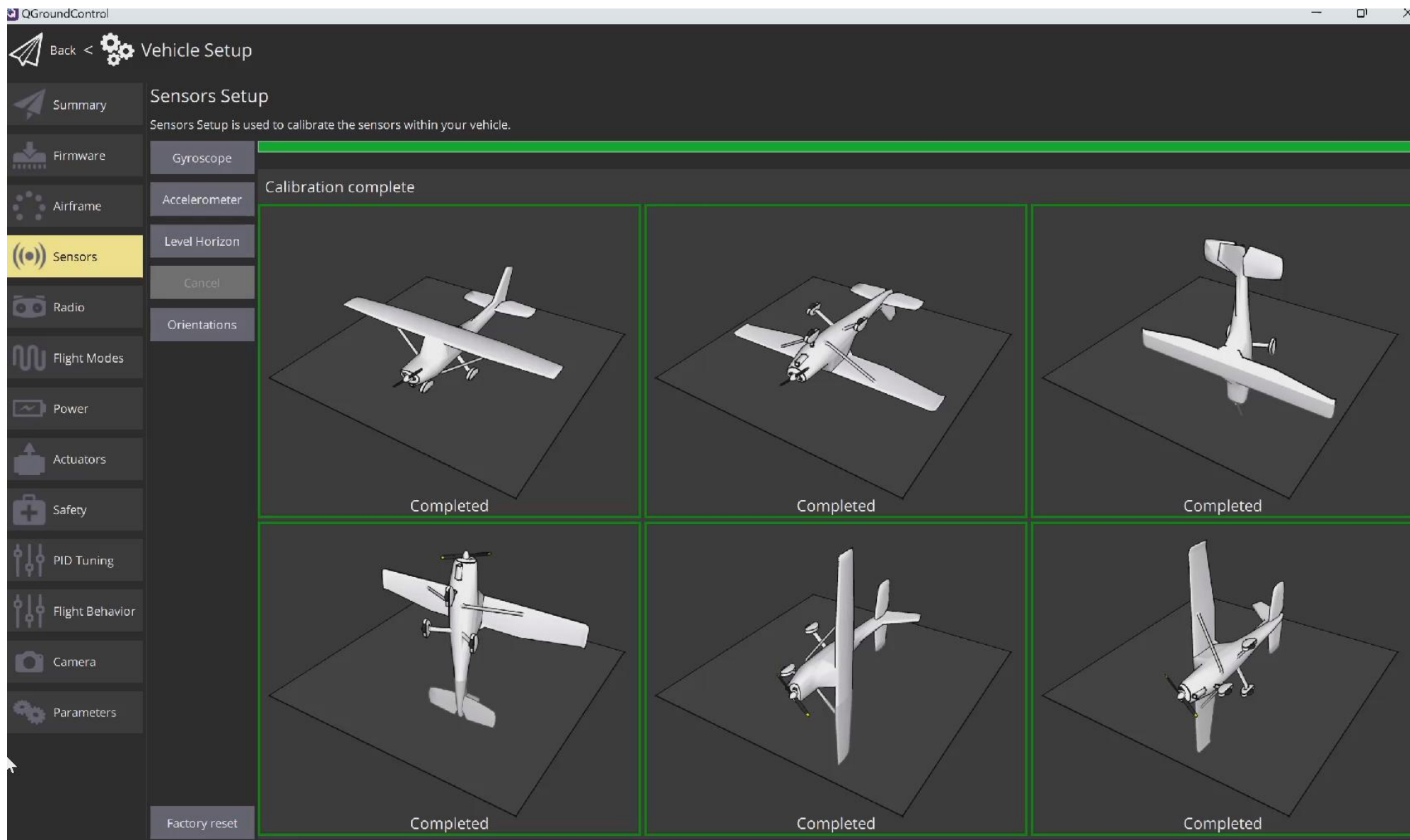


ROTATION depends on install direction

Calibrate Accelerometer



Calibrate Accelerometer



Calibrate Radio

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Radio
Flight Modes
Power
Actuators
Safety
PID Tuning

Radio Setup

Radio Setup is used to calibrate your transmitter. It also assign channels for Roll, Pitch, Yaw and Throttle vehicle control as well as determining whether they are reversed.

Attitude Controls

Roll

Pitch

Yaw

Throttle

Additional Radio setup:

AUX1 Passthrough RC channel AUX2 Passthrough RC channel

PARAM1 tuning channel PARAM2 tuning channel

PARAM3 tuning channel

Mode 1 ☒ Mode 2 ☐

Channel Monitor

1	<input type="range"/>	2	<input type="range"/>
3	<input type="range"/>	4	<input type="range"/>
5	<input type="range"/>	6	<input type="range"/>
7	<input type="range"/>	8	<input type="range"/>
9	<input type="range"/>	10	<input type="range"/>
11	<input type="range"/>	12	<input type="range"/>
13	<input type="range"/>	14	<input type="range"/>
15	<input type="range"/>	16	<input type="range"/>
17	<input type="range"/>	18	<input type="range"/>

Calibrate Radio

QGroundControl

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Flight Behavior

Radio Setup

Radio Setup is used to calibrate your transmitter. It also assign channels for Roll, Pitch, Yaw and Throttle vehicle control as well as determining whether they are reversed.

Attitude Controls

Roll

Pitch

Yaw

Throttle

Move the Throttle stick all the way down and leave it there...

Additional Radio Setup:

AUX1 Passthrough RC channel

AUX2 Passthrough RC channel

PARAM1 tuning channel

PARAM2 tuning channel

PARAM3 tuning channel

Mode 1 ☒ Mode 2 ☐

Channel Monitor

1		2	
3		4	
5		6	
7		8	
9		10	
11		12	
13		14	
15		16	
17		18	

Following the tips until finished

Calibrate Radio

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Safety
PID Tuning

Flight Modes Setup

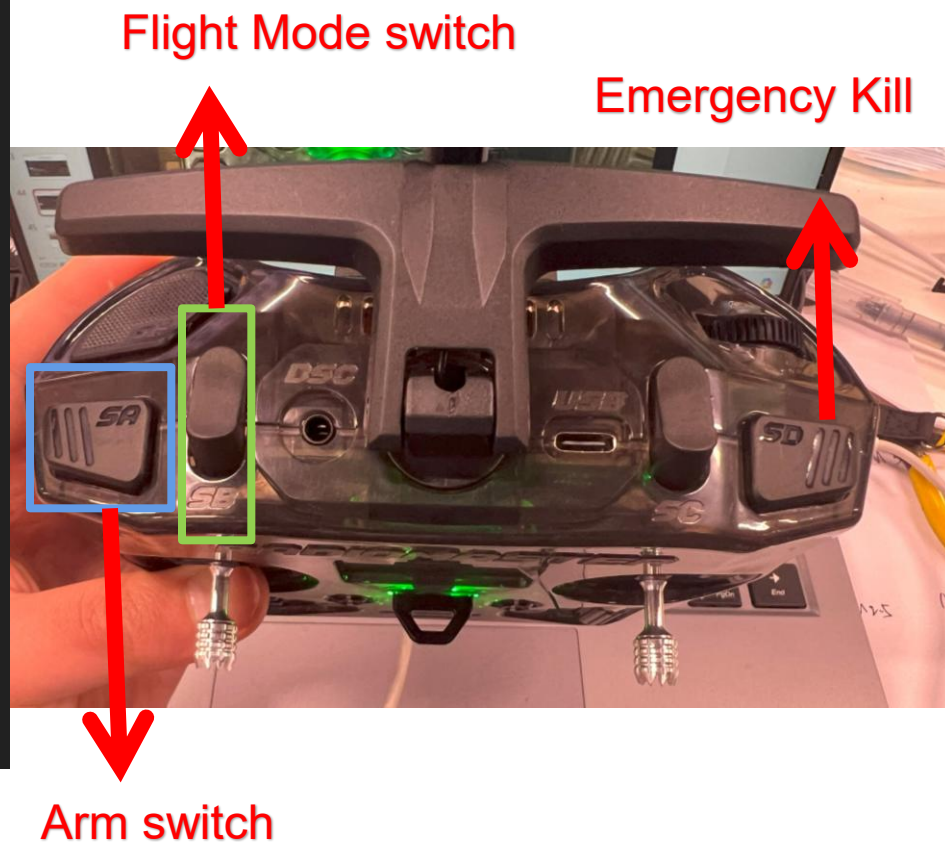
Flight Modes Setup is used to configure the transmitter switches associated with Flight Modes.

Flight Mode Settings		Switch Settings	
Mode Channel	Channel 6	Arm switch channel	Channel 5
Flight Mode 1	Altitude	Emergency Kill switch channel	Channel 8
Flight Mode 2	Altitude	Offboard switch channel	Unassigned
Flight Mode 3	Position	Landing gear switch channel	Unassigned
Flight Mode 4	Position	Loiter switch channel	Unassigned
Flight Mode 5	Position	Return switch channel	Unassigned
Flight Mode 6	Position		

Channel Monitor

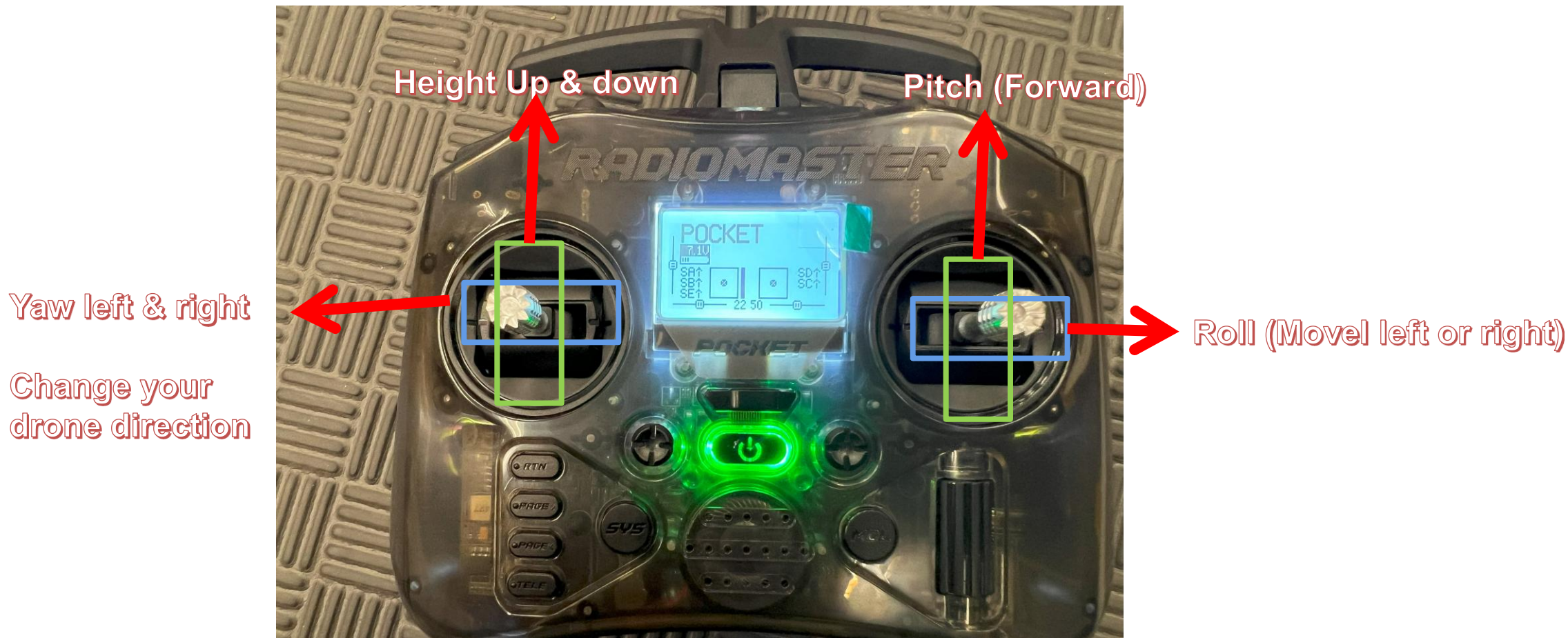
1		2	
3		4	
5		6	
7		8	
9		10	
11		12	
13		14	
15		16	
17		18	

Switch on the button, you will see the description becomes yellow, that means you switch on the function



We recommend you use the *Position mode* to fly. It is much friendly to beginner

Calibrate Radio



We recommend you do not change the yaw. It is much friendly to beginner

Check Motors & Direction

**Please Remove All
Propellers!!!!!!!**

**See the details of motors
in the next pages**

The screenshot displays the Betaflight configuration tool's 'Vehicle Setup' screen. The left sidebar contains navigation options: Summary, Firmware, Airframe, Sensors, Radio, Flight Modes, Power, **Actuators** (highlighted with a blue box), Safety, PID Tuning, Flight Behavior, Camera, and Parameters. The main area is divided into 'Geometry: Multirotor' and 'Actuator Outputs'.

Geometry: Multirotor

Motors: 4

	Position X	Position Y	Direction CCW
Motor 1:	0.15	0.15	☒
Motor 2:	-0.15	-0.15	☒
Motor 3:	0.15	-0.15	☐
Motor 4:	-0.15	0.15	☐

A diagram below the table shows a quadcopter layout with motors 1 (blue), 2 (blue), 3 (green), and 4 (green) at the corners.

Actuator Testing

☒ Careful: Actuator sliders are enabled

Below this, a row of sliders for 'All Motors', 'Motor 1', 'Motor 2', 'Motor 3', and 'Motor 4' is shown, with 'Motor 3' selected. At the bottom are buttons for 'Set Spin Direction 1' and 'Set Spin Direction 2'.

Actuator Outputs

PWM MAIN

Identify & Assign Motors

MAIN 1-4: DShot600

	Function	Rev Range (for Servos)
MAIN 1:	Motor 3	☐
MAIN 2:	Motor 2	☐
MAIN 3:	Motor 1	☐
MAIN 4:	Motor 4	☐

MAIN 5-6: PWM 400 Hz

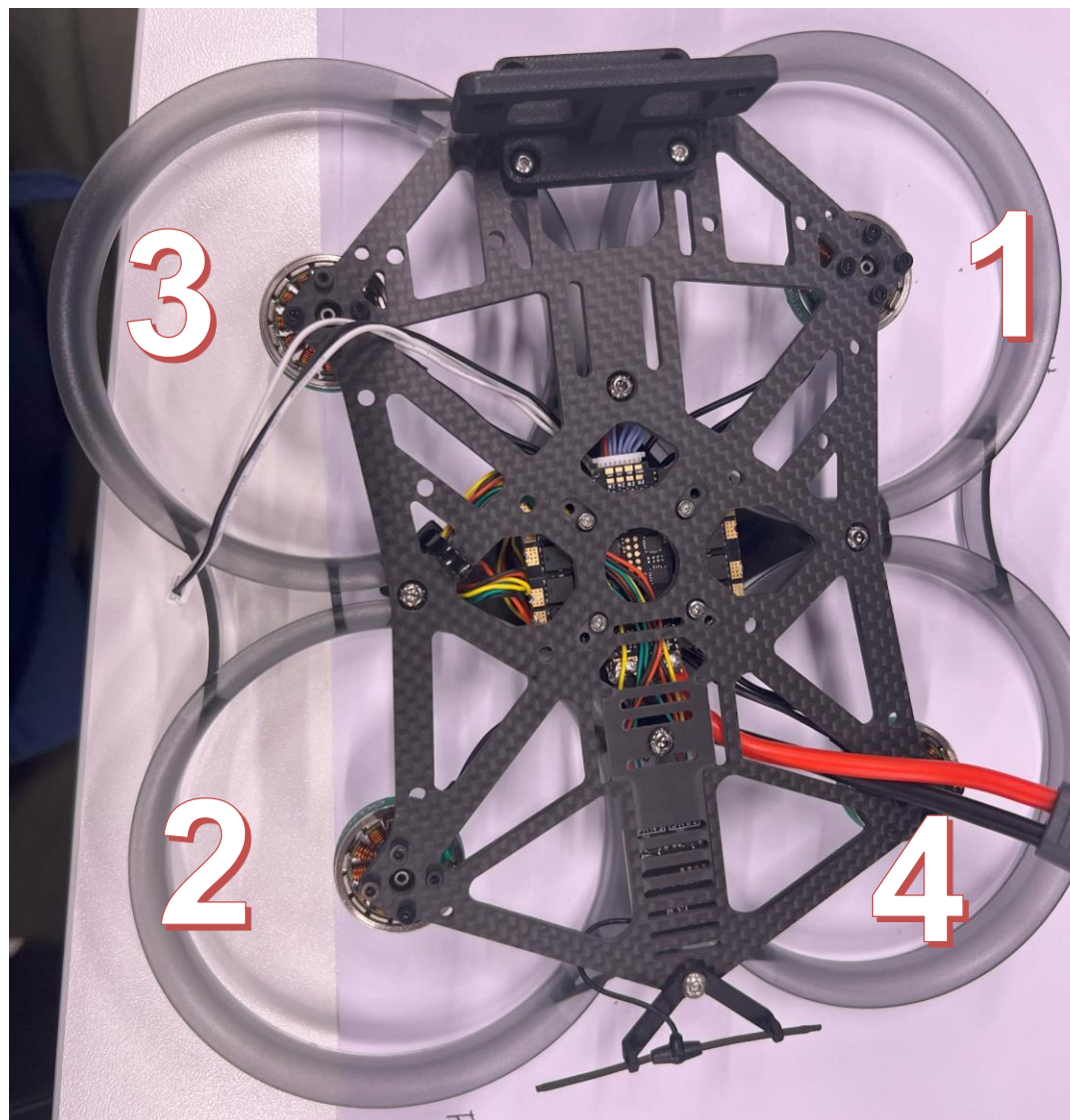
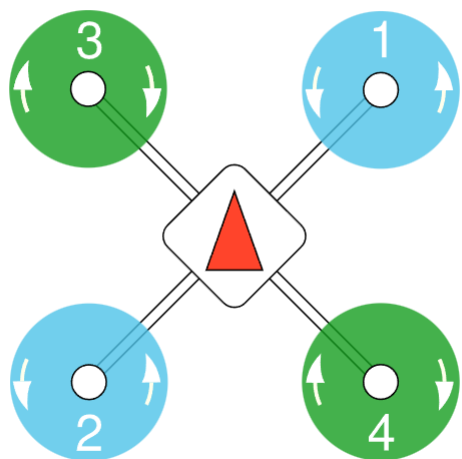
	Function	Disarmed	Minimum	Maximum	Rev Range (for Servos)
MAIN 5:	Disabled	1000	1000	2000	☐
MAIN 6:	Disabled	1000	1000	2000	☐

MAIN 7-8: PWM 400 Hz

	Function	Disarmed	Minimum	Maximum	Rev Range (for Servos)
MAIN 7:	Disabled	1000	1000	2000	☐
MAIN 8:	Disabled	1000	1000	2000	☐

Please Remove All Propellers!!!!!!!

Quadrotor x



Keep the number and the motor direction is the same as the left figure.

This step is very important, please check carefully

Remote Connections

Our quadrotor has remote telemetry.

After setup as before, you can connect the QGC with your radio controller.

Connect to computer with the type-c port in green.

“Disconnected” instruction on the next page.

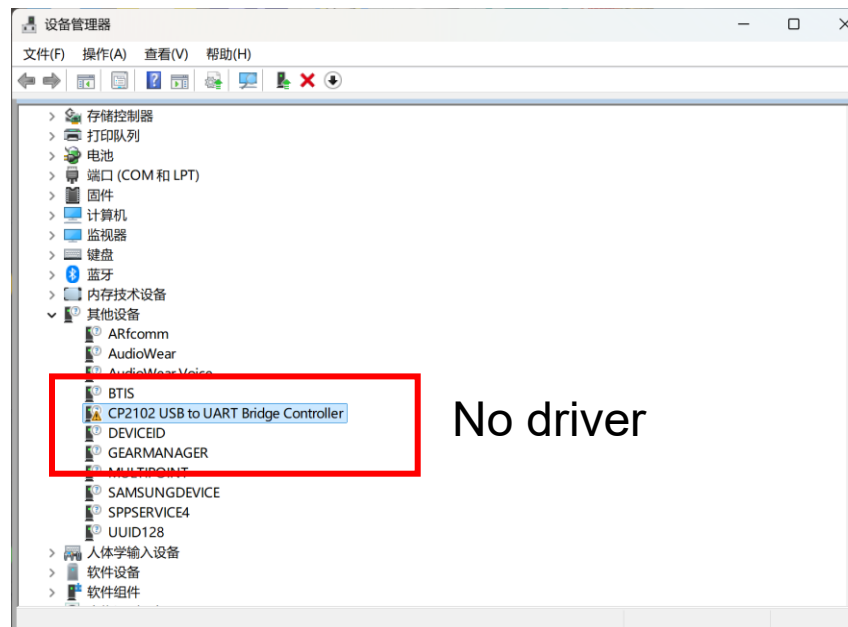
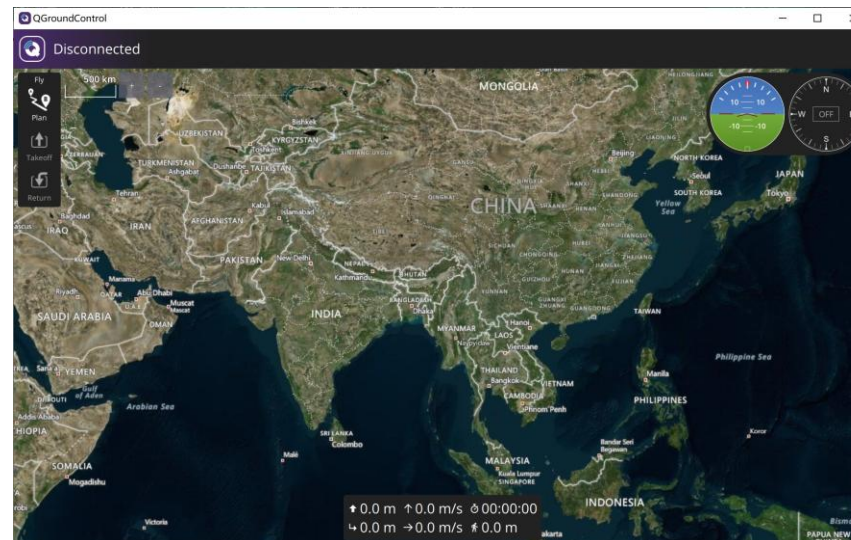


Remote Connections

If the QGC shows "Disconnected", you need to check whether your computer has the cp2012 driver.

If in the "Device Management" (Win), "CP2012 USB to UART " means there is no driver.

You can search "cp2012 driver" in your web browser.



No driver

Change the Stick Self-Centering



Introduction

Thank you for purchasing the RadioMaster POCKET radio system. Advanced and highly-versatile, POCKET is adaptable for pilots of any skill level. To ensure safe and reliable operations, please read this manual carefully prior to use.

As the software and hardware are constantly improved, the information contained in this manual is subject to change without prior notice. Please visit our website for up-to-date information and radio firmware.

POCKET is an open-sourced, ergonomic radio with powerful multi-protocol capability. Able to bind to most contemporary protocols on the market, this is one radio to rule them all! POCKET runs EdgeTX, an open-sourced operating system; for more information, please visit the EdgeTX link below.
-The RadioMaster team



Safety Information

Radio-controlled models commonly contain sharp components like propellers, rotating at high RPMs; when operating or maintaining the model, always use common sense and proceed with caution. Always disconnect power to the model AND remove propellers (if applicable) during the assembly or maintenance process.

Never operate the POCKET radio system in the following conditions:

- In severe weather or strong windy conditions, such as rain, hail, snow, storms or electromagnetic environments.
- In limited visibility.
- In the proximity of other people, property, high-voltage power lines, public roads, vehicles or animals.
- When feeling unwell, or under the influence of drugs or alcohol.
- If the POCKET radio system/model is damaged or showing signs of interference.
- In areas with high 2.4GHz interference or where 2.4GHz radios are prohibited.
- When the TX/RX battery is low.
- In areas where local regulations prohibit the use of R/C aircraft.



Manual and firmware download

POCKET is pre-installed with factory approved EdgeTX firmware. To download the latest manual or stock factory firmware, please visit the RadioMaster website: <https://www.radiomasterrc.com>
To download the latest EdgeTX firmware for your POCKET radio system, please visit the EdgeTX website: <https://www.edgetx.org>

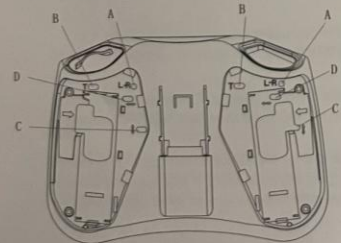


Important!

POCKET is pre-installed with the most stable firmware at the factory at time of release. please only attempt to update the firmware if you are confident in the process. Incorrect firmware updates may cause the remote control to become inoperable. Damage as a result of improper firmware update may not be covered under warranty.



Gimbal adjustment



- A) L-R to switch M1/M2. Turn clockwise to disable self-centering (Throttle). Turn counterclockwise to enable self-centering (Elevator).
- B) T to adjust the throttle stick tension. Turn clockwise to increase stick tension. Turn counterclockwise to decrease stick tension.
- C) (Up-Down) Turn clockwise to increase vertical stick tension. Turn counterclockwise to decrease stick tension.
- D) (Left-Right) Turn clockwise to increase horizontal stick tension. Turn counterclockwise to decrease stick tension.



简介

感谢您购买RadioMaster POCKET遥控系统。该系统用途广泛，可供初学者和专业人士使用。为了确保正确、安全地使用本产品，请在使用前仔细阅读本使用说明书。由于版本升级，已经进行了更改。本手册中包含的信息如有更改，恕不另行通知。
POCKET遥控器适用于所有类型的固定翼、滑翔机、直升机和多旋翼飞机。可以根据使用的航空器选择型号类型，并可以使用各种混合功能。

-RadioMaster 团队敬上。



安全须知

许多遥控模型都配备了强大的电机和锋利的螺旋桨。使用模型时，请谨慎行事。进行组装或维护时，请确保已断开模型的电源并卸下螺旋桨。

在以下情况下，请勿操作POCKET遥控系统：

- 在恶劣天气或强风条件下，例如雨，冰雹，下雪，暴风雨或电磁环境中。
- 在能见度有限的任何情况下。
- 在可能存在人员、财产、电力高压线、公共道路、有车辆或动物的区域。
- 如果您感到疲倦或不适，或在药物或酒精的影响下。
- 如果遥控器或模型似乎已损坏或无法正常工作。
- 在2.4GHz干扰较大的区域或禁止使用2.4GHz无线电的地方。
- 当电池电压太低而无法使用时。
- 在当地法规禁止使用航空模型的区域。



说明书和固件下载

POCKET支持强大的开源固件EdgeTX。要下载最新的软件手册，请访问RadioMaster官方网站：
<https://www.radiomasterrc.com>
要为您的POCKET遥控器下载最新的EdgeTX固件，请访问EdgeTX网站：<https://www.edgetx.org>

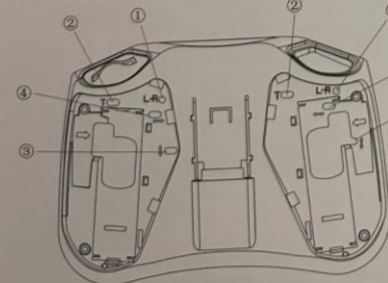


警告！

POCKET出厂时预装最稳定的固件。如果您有经验并且有信心更新系统固件，请仅更新固件。不正确的更新可能会导致遥控器无法操作。



摇杆调节



- ①，L-R孔调节左右手互换，顺时针打到底，摇杆上下旋转无弹力（油门）。
- ②，T孔调节油门手感轻重，顺时针油门手感变重，逆时针油门手感变轻。
- ③，孔调节摇杆上下旋转弹力，顺时针弹力变大，逆时针弹力变小。
- ④，孔调节摇杆左右旋转弹力，顺时针弹力变大，逆时针弹力变小。

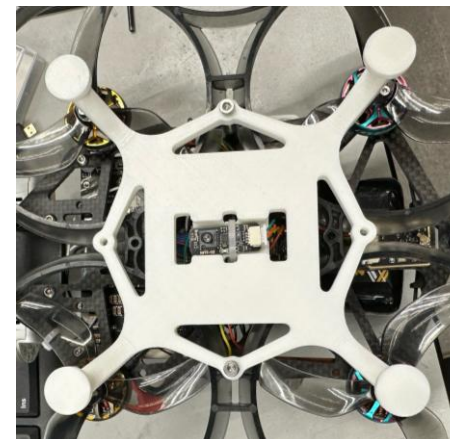
Part 4

Quadrotor Assembly Part 4

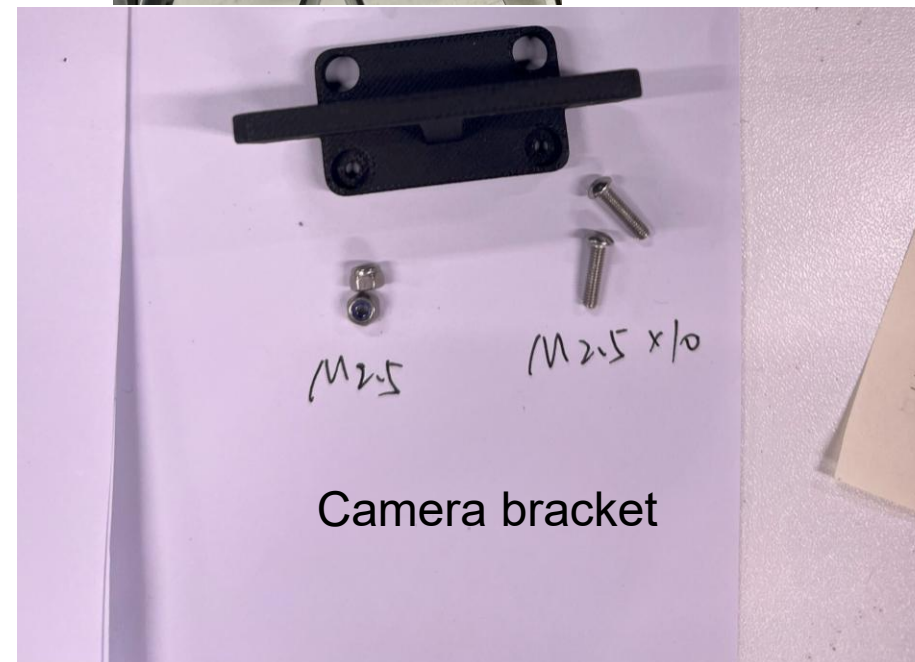
- Equipment List:

Item	Quant.
Camera bracket	1
Circle bracket	1
Optical Flow shield	1
Antenna bracket	1
M3 * 14 Screw	7
M3* 6 Screw	6
M3* 10 Screw	1
M3 * 15 studs	7
M2.5 nuts	2
M2.5*10 Screw	2
M2*8 Screw (For propellers)	8

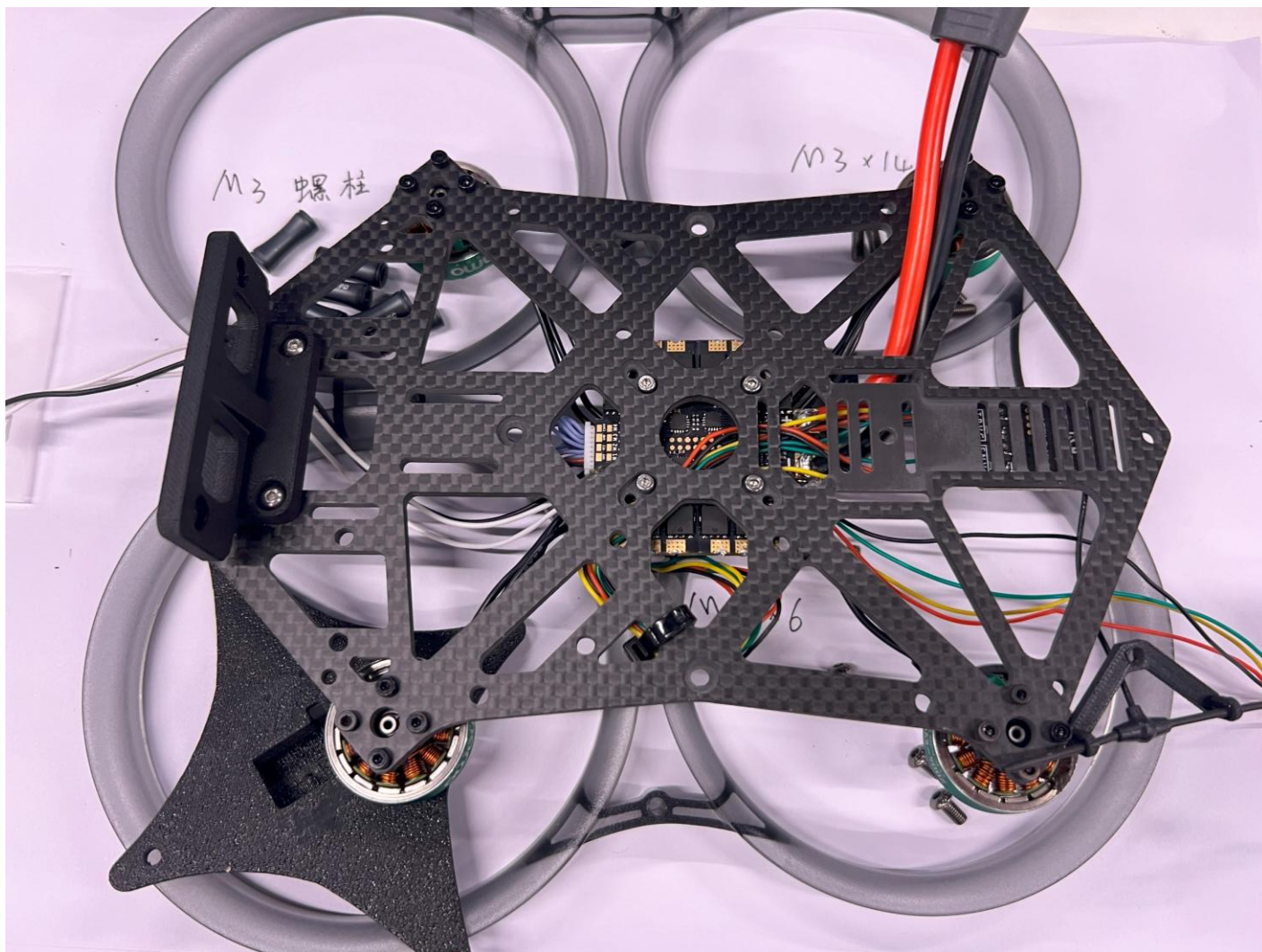
Assemble the base frame



Optical Flow & ToF
Shield

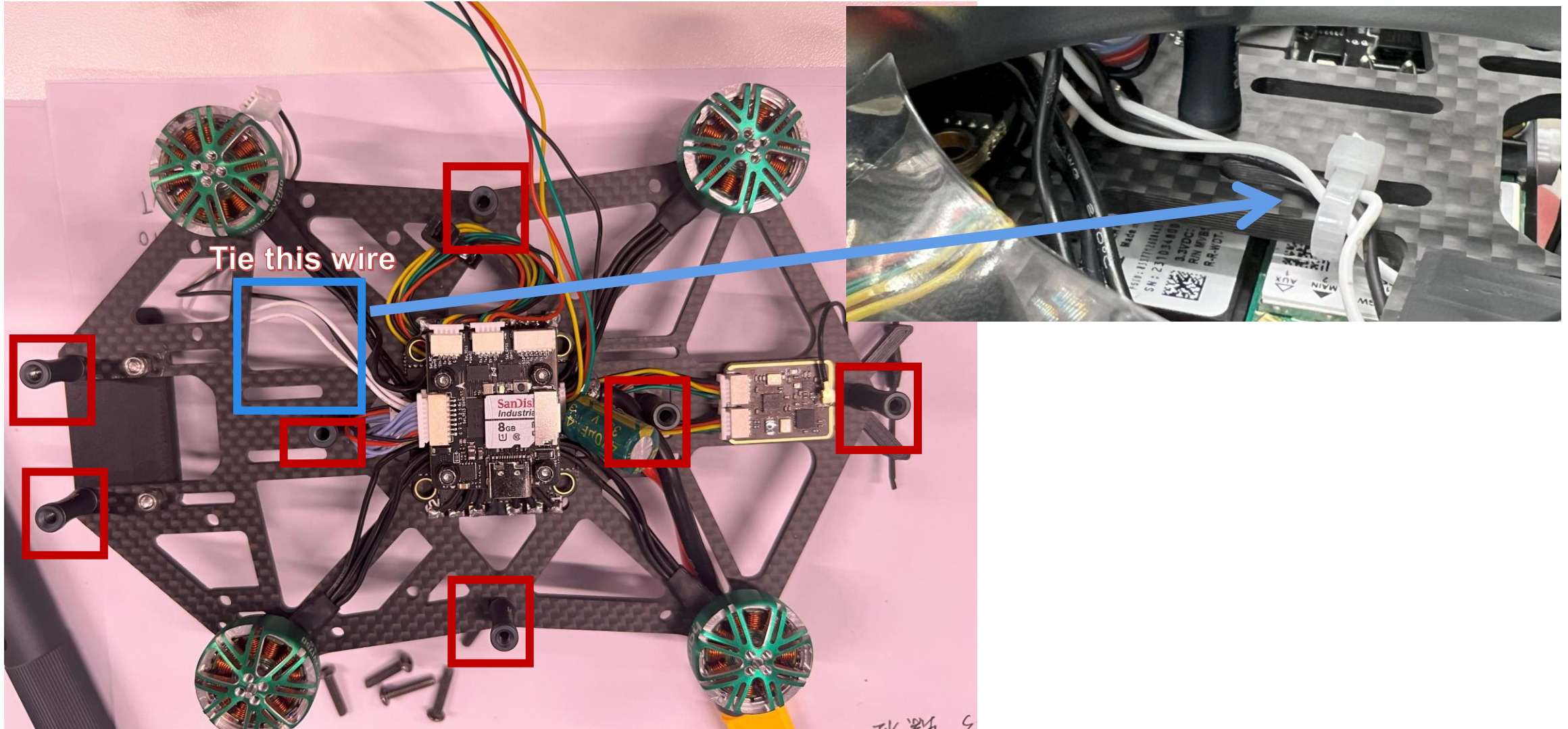


Install Camera Bracket



Do not install camera in this step, here we just for demonstration

Install studs

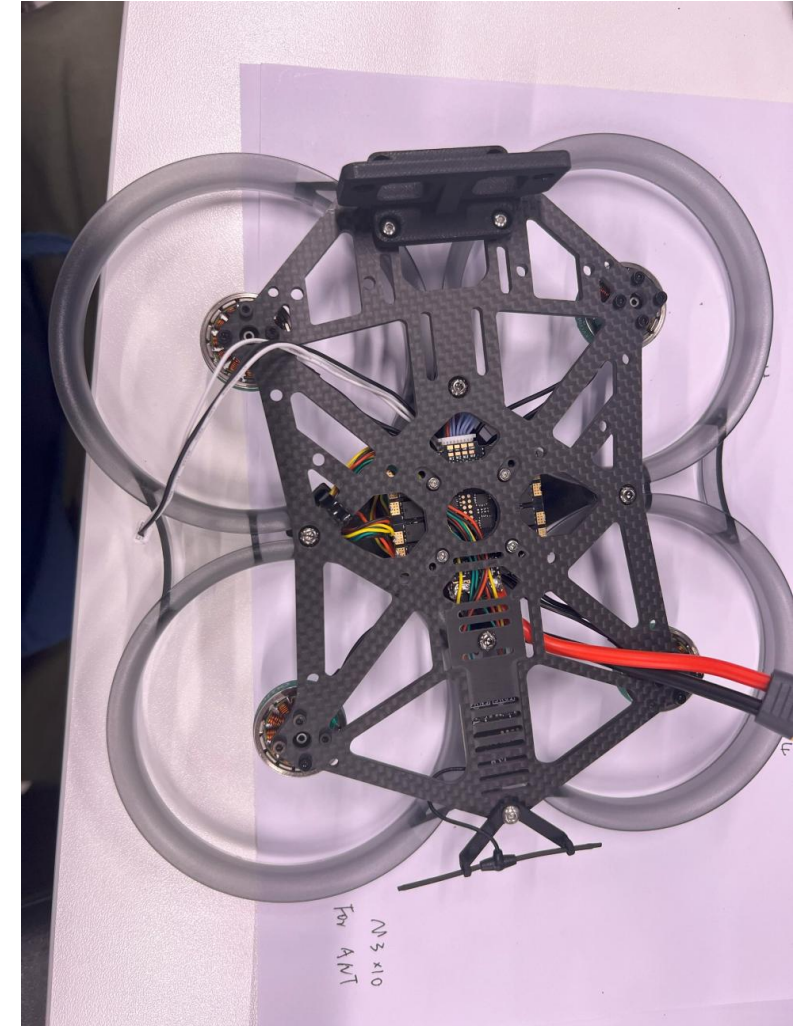
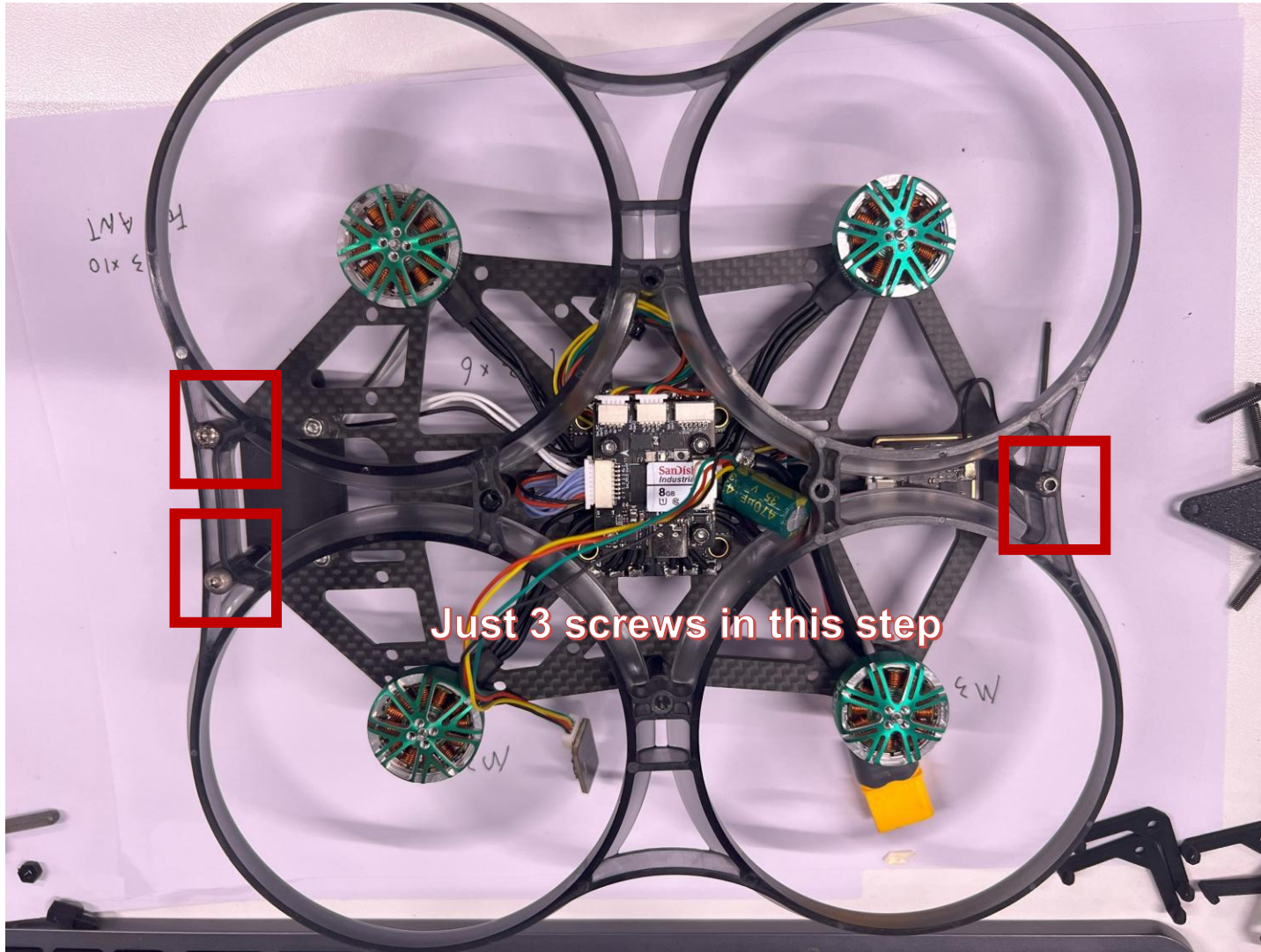


Install Type C Extension Cable

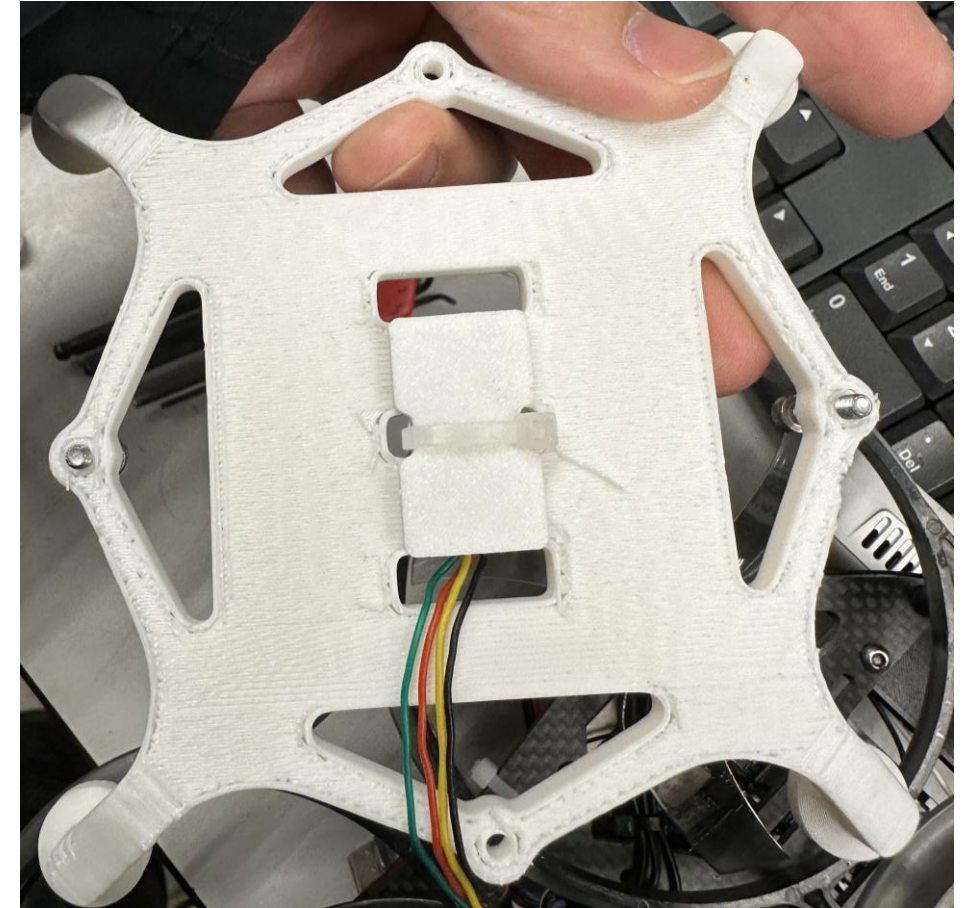
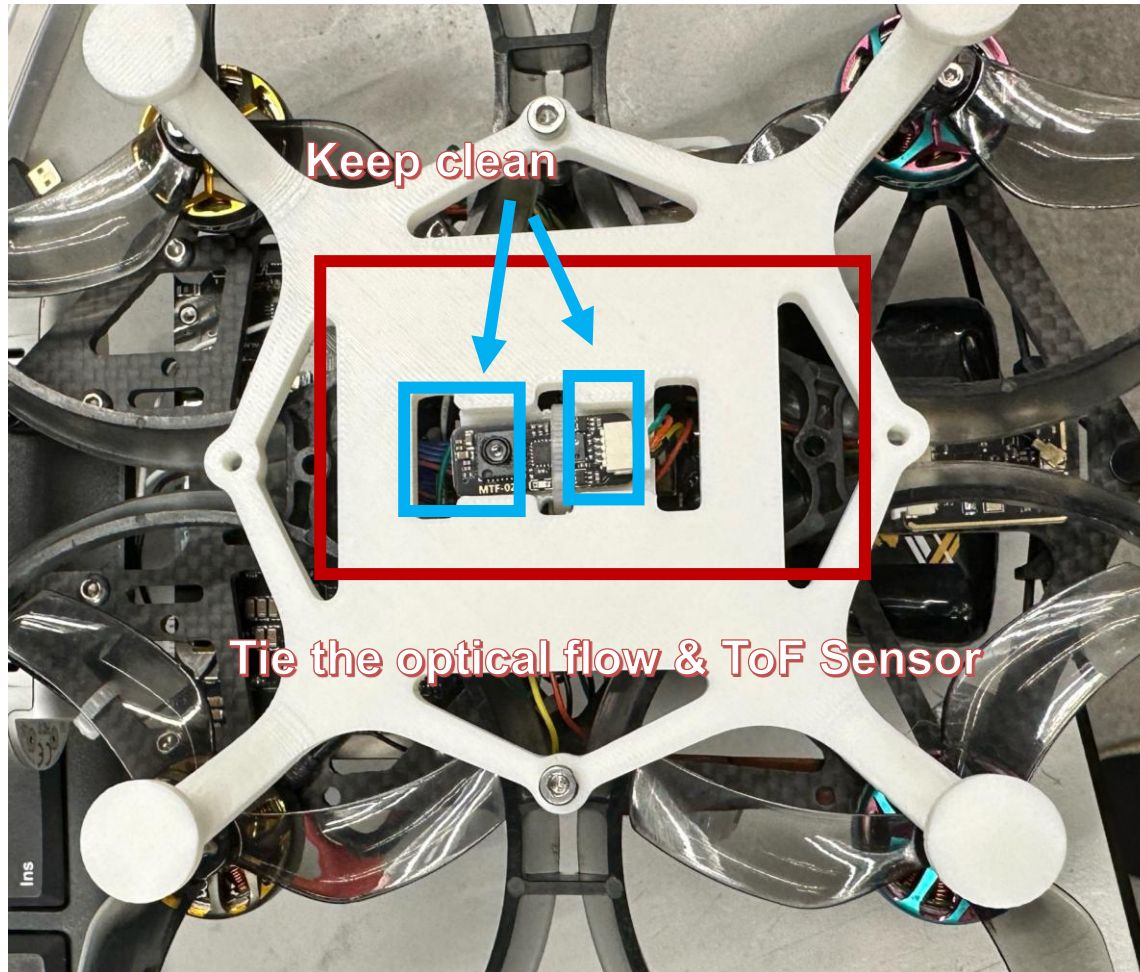


The Type-C extension cable makes possible subsequent parameter adjustments simpler.

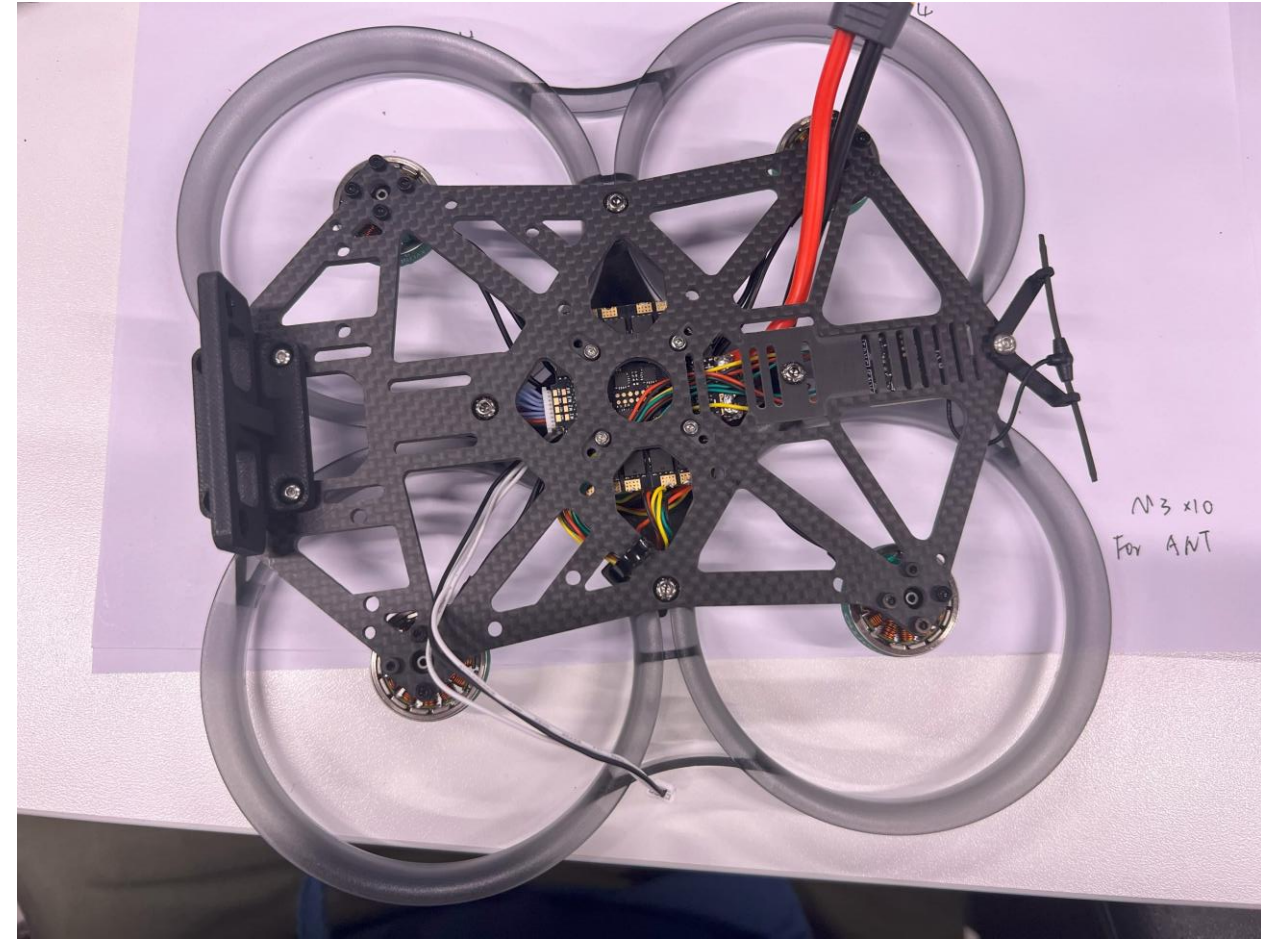
Install Circles



Install Optical Flow & ToF Sensor



Install Circles



Before Flight

Make sure ALL your work is correct

Any mistake will be very likely to cause
drone crash

!!! Let TA check your work before flight

Acquire propellers from TA

Install propellers.



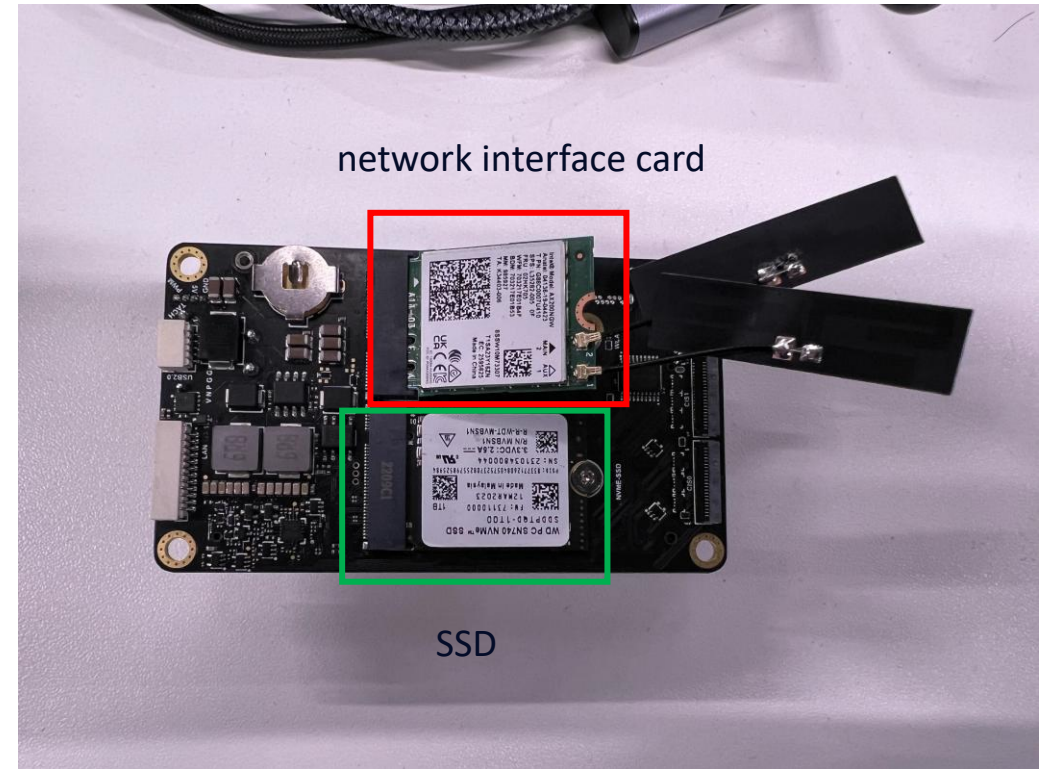
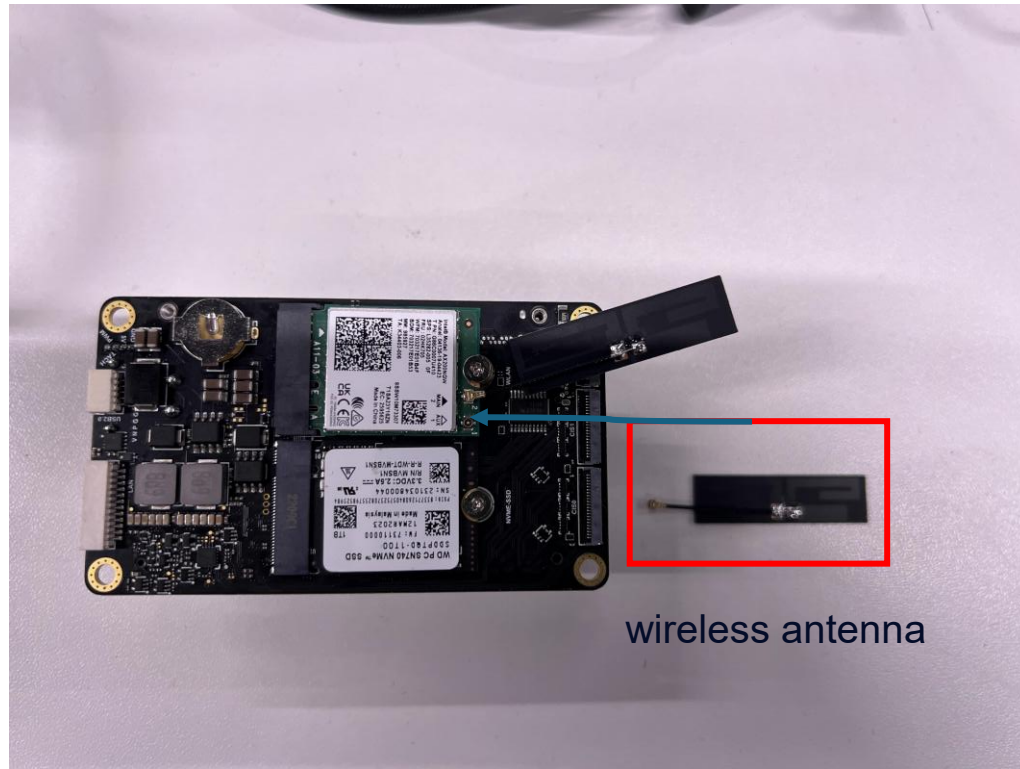


Part 5: Jetson (Xavier NX / Orin Nano) Setup Guidance

Quadrotor Assembly Part 5

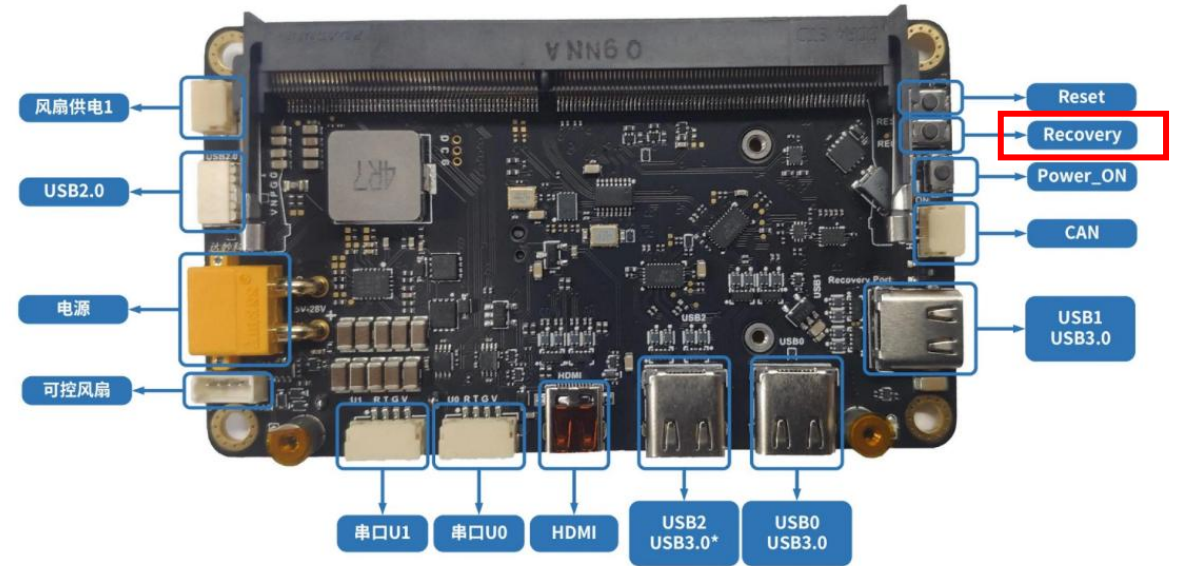
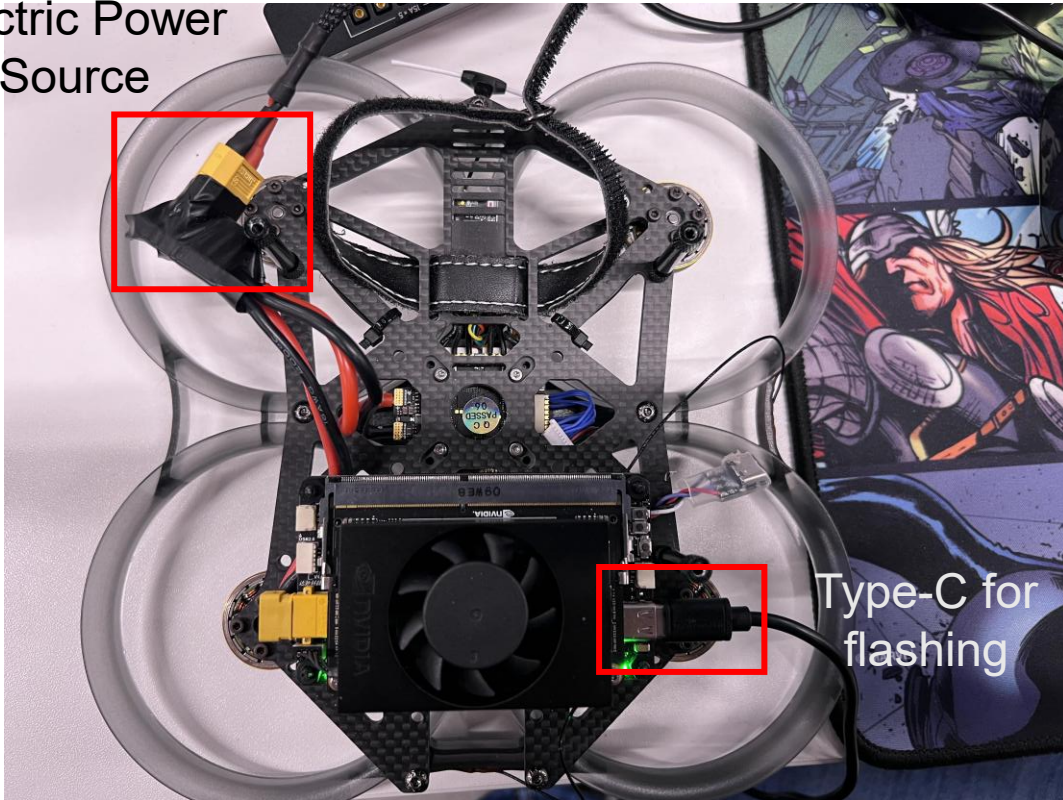
- Equipment List:

Item	Quant.
Nvidia Jetson (Xavier NX / Orin Nano)	1
Jetson carrier board (DM Tech.)	1
SSD	1
Wireless network interface card	1
Wireless antenna	2



1. Press the wireless antenna into the holes on network interface card like right figure.
2. Setup SSD and network interface card.

Electric Power
Source



1. Press and hold the **recovery** button until power is applied to the Jetson.
2. Connect Type-C to your own laptop (**Ubuntu** operating system).

er

Need to Set Up Your Environment

Provides an end-to-end development environment for DRIVE, Jetson, Holoscan, Rivermax, DOCA for both host and target devices.

Download NVIDIA SDK Manager 2.0.0



.deb
Ubuntu



.rpm CentOS/RHEL



Docker Image
Ubuntu 18.04



Docker Image
Ubuntu 20.04



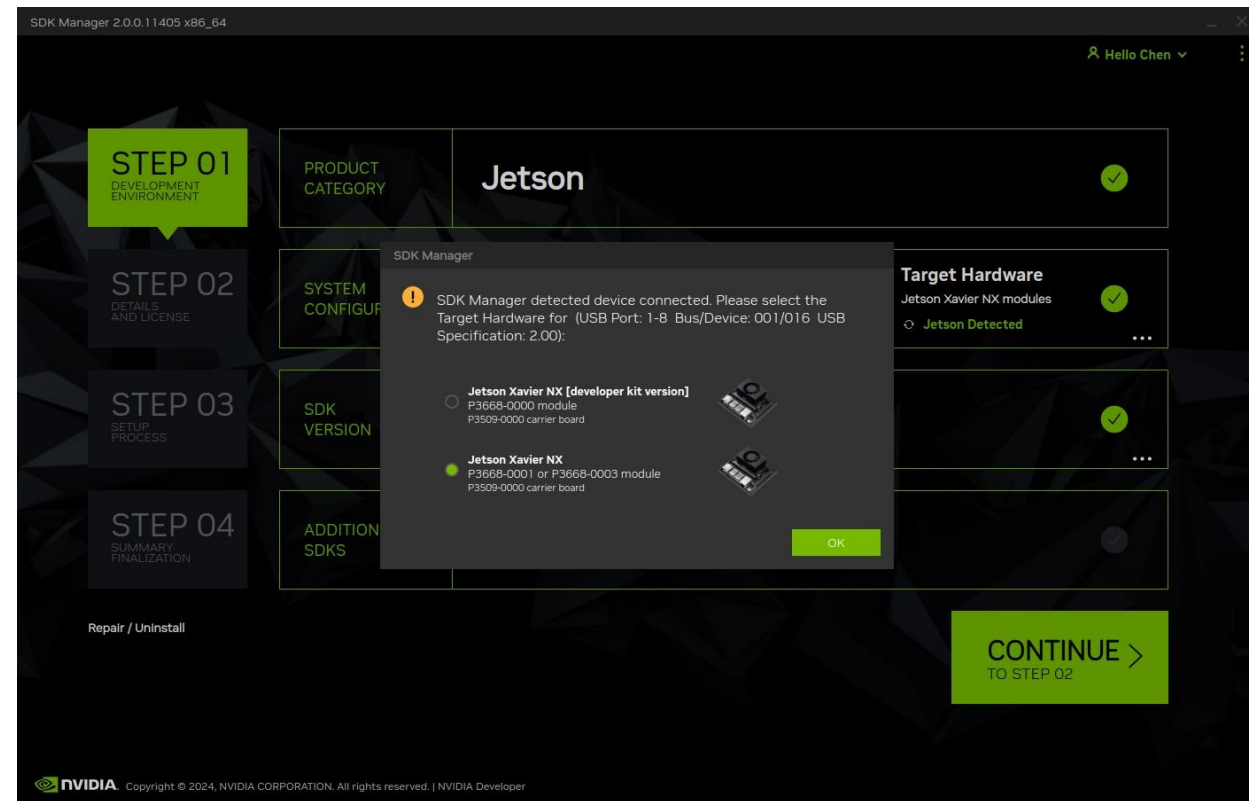
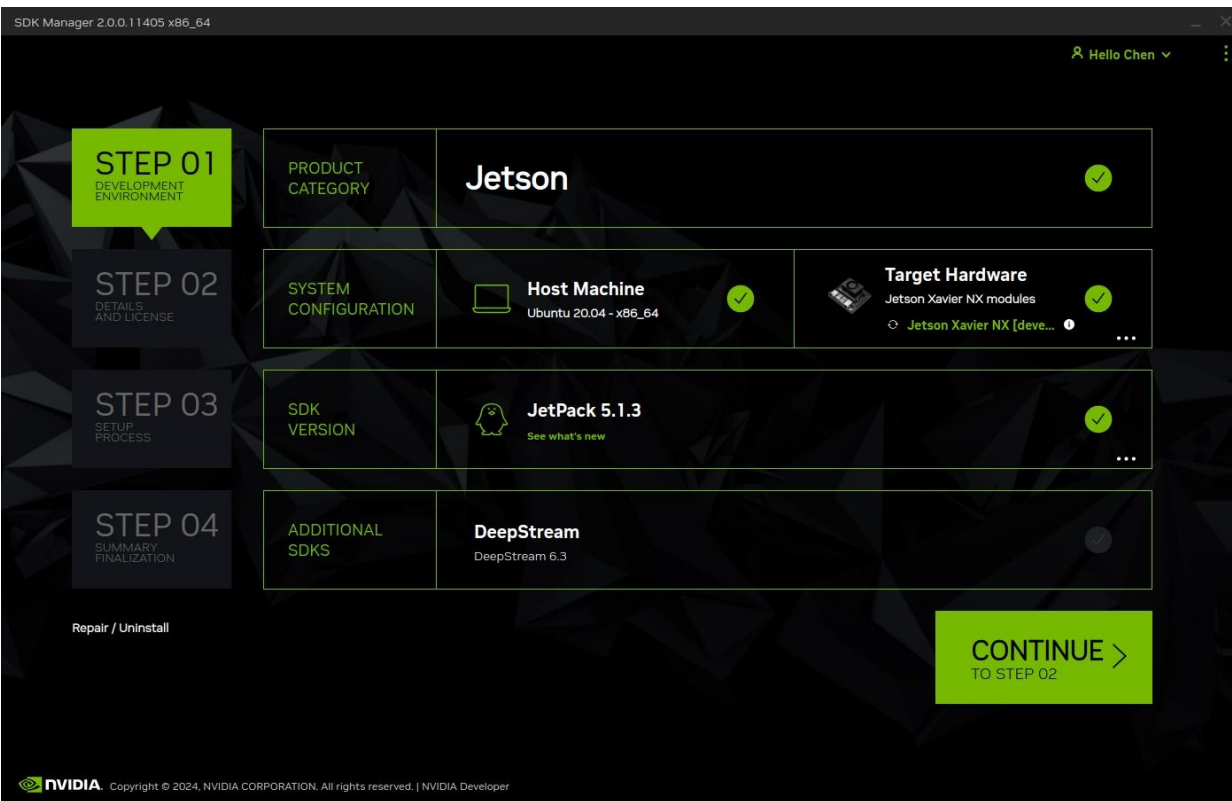
Docker Image
Ubuntu 22.04

[SDK Manager User Guide](#)

SDK Packages

DEVELOPMENT ENVIRONMENT

1. Download nvidia sdk manager on your laptop in <https://developer.nvidia.com/sdk-manager>.
2. Install this sdk using “sudo dpkg -i sdkmanager....deb” in the terminal.



1. After connecting type-C to your laptop, select Jetson Xavier NX.
2. In the SDK version, select 5.1.6

SDK Manager 2.0.0.11405 x86_64

Hello Chen

STEP 01
DEVELOPMENT
ENVIRONMENT

STEP 02
DETAILS
AND LICENSE

STEP 03
SETUP
PROCESS

STEP 04
SUMMARY
FINALIZATION

JETPACK 5.1.3 LINUX FOR JETSON XAVIER NX MODULES

[Expand all](#)

HOST COMPONENTS	DOWNLOAD SIZE	STATUS
Computer Vision	97.5 MB	Downloaded
Developer Tools	1,261 MB	

TARGET COMPONENTS	DOWNLOAD SIZE	STATUS
☒ Jetson Linux		
> Jetson Linux image	2,177 MB	
> Flash Jetson Linux	0 MB	
☐ Jetson Runtime Components		
> CUDA Runtime	1,435 MB	
> CUDA X-AI Runtime	1,416 MB	
> Computer Vision Runtime	61.3 MB	
> NVIDIA Container Runtime	2.6 MB	
> Multimedia	71.5 MB	

System requires up to 36GB (host) and 0GB (target) of available disk space during setup.

Download folder: /home/eason/Downloads/nvidia/sdkm_downloads [change](#) (7GB required)

Target HW image folder: /home/eason/nvidia/nvidia_sdk [change](#) (29GB required)

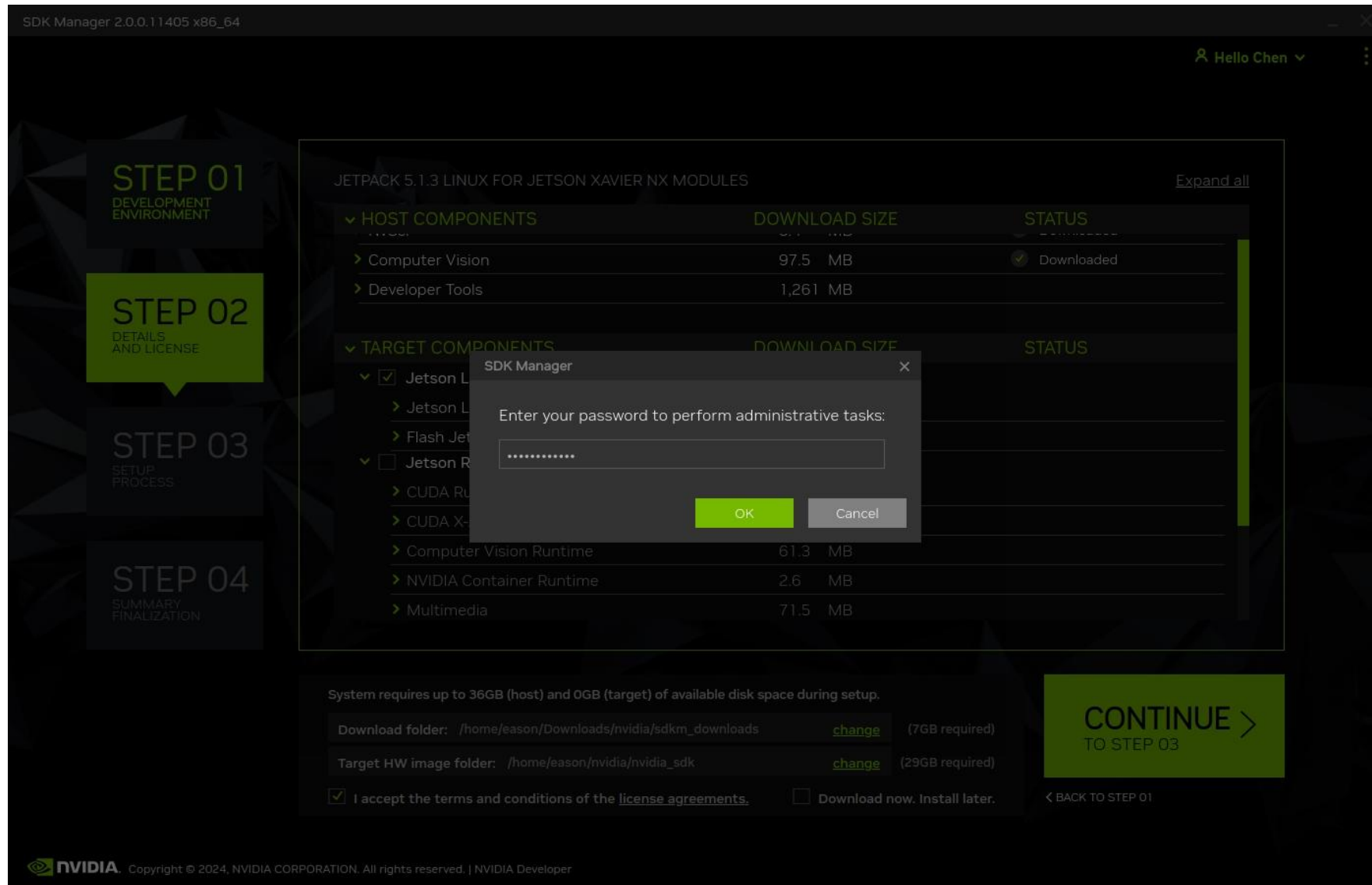
☒ I accept the terms and conditions of the [license agreements](#). ☐ Download now. Install later.

CONTINUE >
TO STEP 03

[BACK TO STEP 01](#)

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1. Only select Jetson Linux.
2. Continue.



1. Input you laptop password.

SDK Manager 2.0.0.11405 x86_64

Hello Chen

STEP 01
DEVELOPMENT
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AND LICENSE

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DETAILS

TERMINAL

JETPACK 5.1.3 LINUX FOR JETSON XAVIER NX MODULES [Expand all](#)

HOST COMPONENTS	DOWNLOAD SIZE	STATUS
> CUDA	3,269 MB	Downloading - 0%
> NvSci	0.4 MB	Install Pending
> Computer Vision	97.5 MB	Install Pending
> Developer Tools	1,261 MB	Downloading - 29%

TARGET COMPONENTS	DOWNLOAD SIZE	STATUS
> Jetson Linux		
> Jetson Linux image	2,177 MB	Downloading - 26%
> Flash Jetson Linux	0 MB	Flash Pending

Downloading: 22.46% (4.95MB/s)

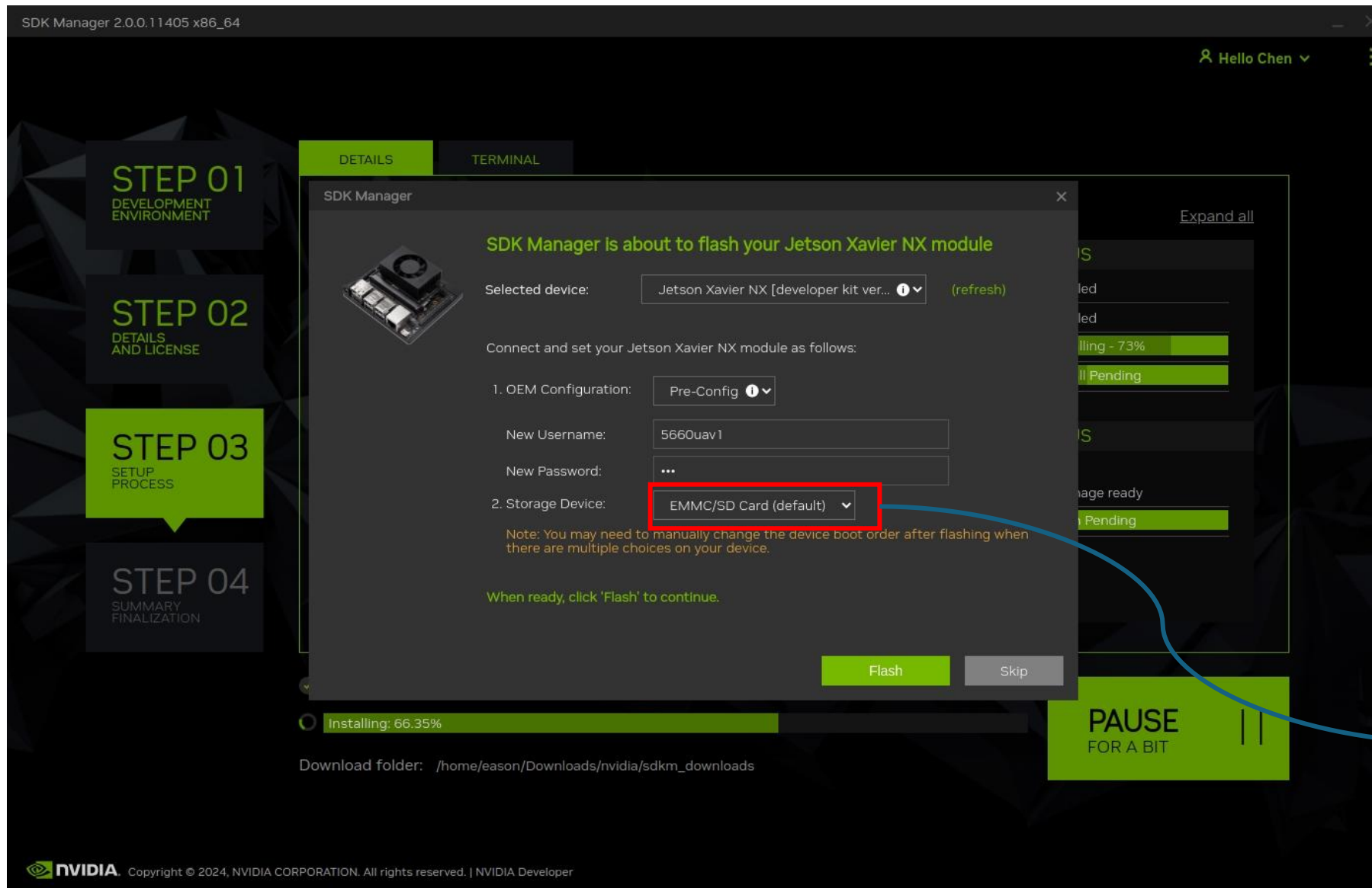
Installing: 4.17%

Download folder: /home/eason/Downloads/nvidia/sdkm_downloads

PAUSE
FOR A BIT

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1. Wait the download and installation process (around 30 mins).



Select
"NVMe"

1. Set username and password, select "NVMe" in storage device.

STEP 01

DEVELOPMENT
ENVIRONMENT

STEP 02

DETAILS
AND LICENSE

STEP 03

SETUP
PROCESS

STEP 04

SUMMARY
FINALIZATION

DETAILS

TERMINAL

Component Name: All

Message Type: All

Filter text here...

Find

```
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/glm/doc/api/a00048.html
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/glm/doc/api/open.png
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/glm/doc/api/folderopen.png
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/FreeImage-3.15.4/macosx/libfreeimage.a
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/FreeImage-3.15.4/linux64/
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/FreeImage-3.15.4/linux64/libfreeimage.a
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/FreeImage-3.15.4/FreeImage.h
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/FreeImage-3.15.4/win64/FreeImage.dll
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/FreeImage-3.15.4/linuxaarch64/libfreeimage.a
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/FreeImage-3.15.4/win32/
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/FreeImage-3.15.4/win32/FreeImage.dll
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/FreeImage-3.15.4/win32/FreeImage.lib
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/gli/core/make_texture.inl
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/GL/glxext.h
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/glfw-3.3.8/docs/extra.css.map
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/glfw-3.3.8/src/monitor.c
15:28:49 INFO: Nsight Perf SDK - host: ./Samples/OpenGL/gtruc/ogl-samples/external/glfw-3.3.8/deps/glad/vulkan.h
15:28:49 INFO: Nsight Perf SDK - host: ./redist/NvPerfUtility/include/NvPerfRangeProfilerD3D12.h
15:28:49 INFO: Nsight Perf SDK - host: ./redist/NvPerfUtility/build/ReportGenerator/pub/turing/report_tu11x.py
15:28:49 INFO: Nsight Perf SDK - host: command finished successfully
15:28:49 SUMMARY: Nsight Perf SDK - host: Install completed successfully.
15:28:49 INFO: All done!
```

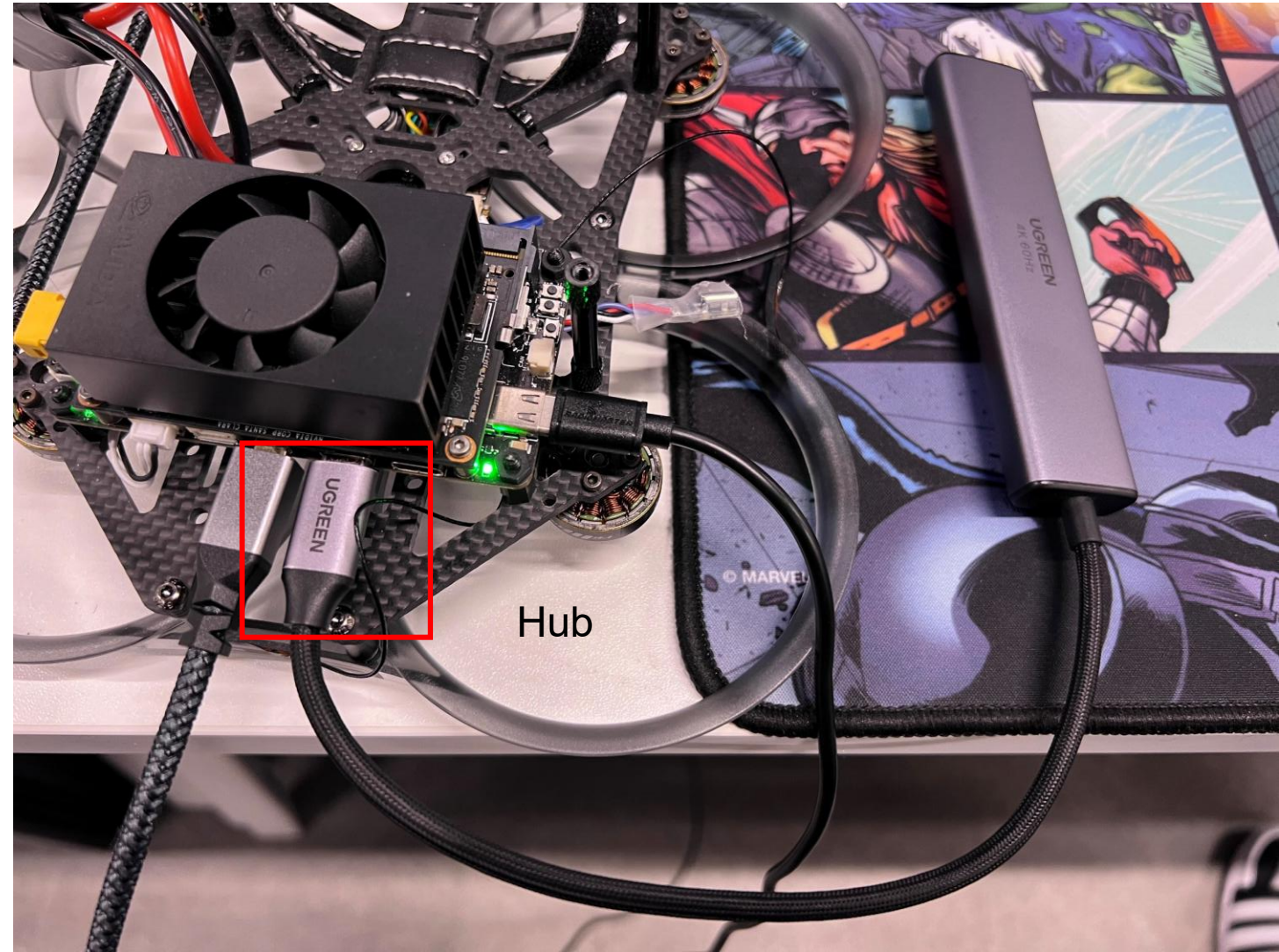
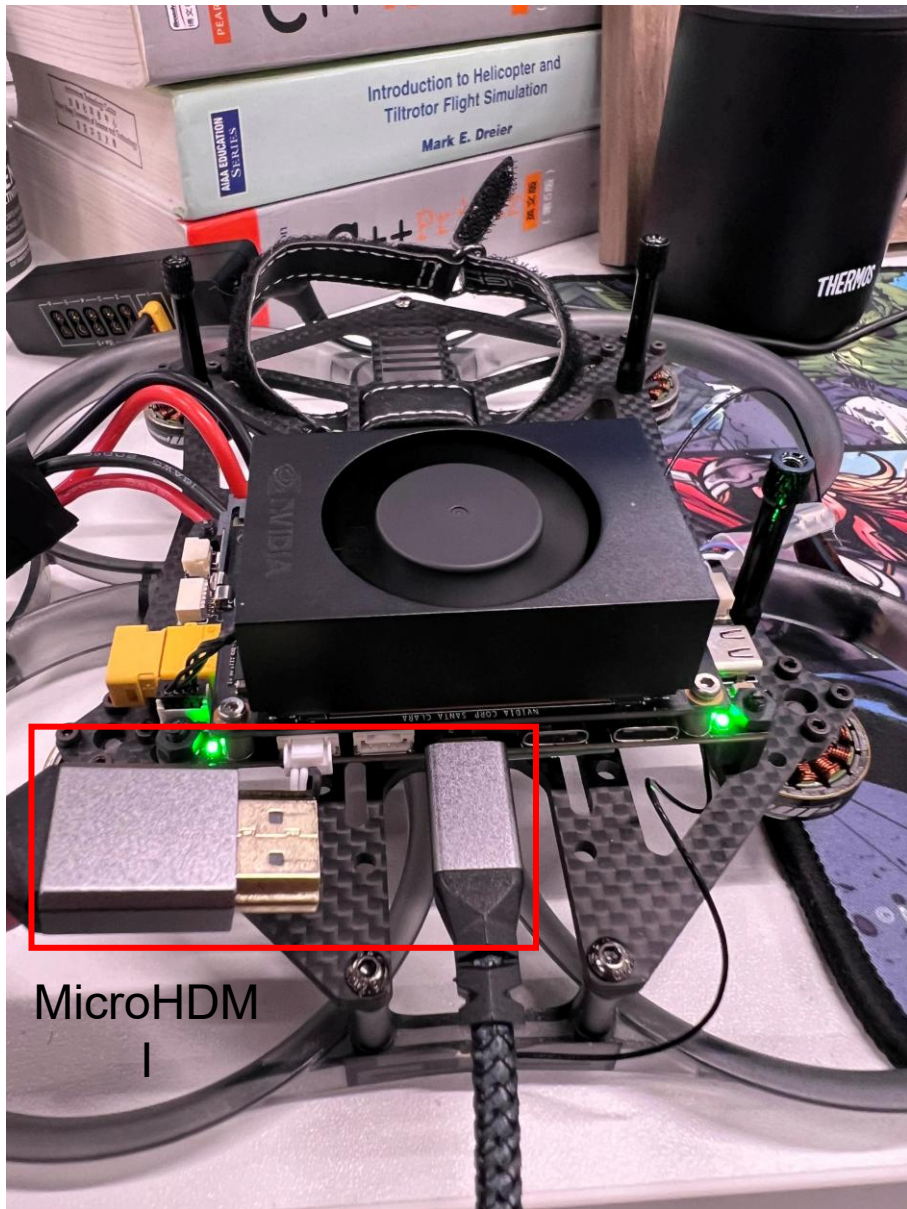
**INSTALLATION COMPLETED SUCCESSFULLY.**

EXPORT LOGS

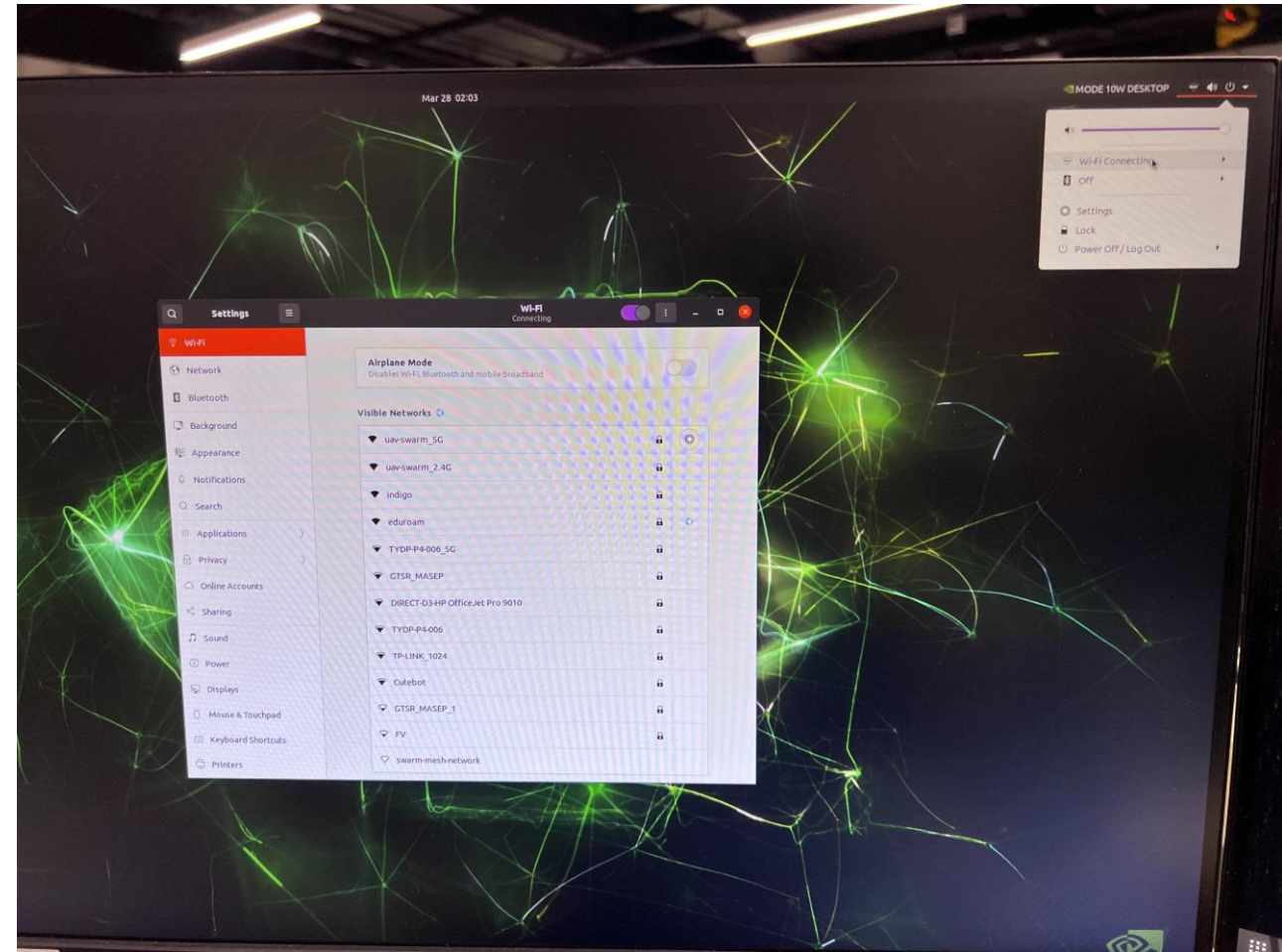
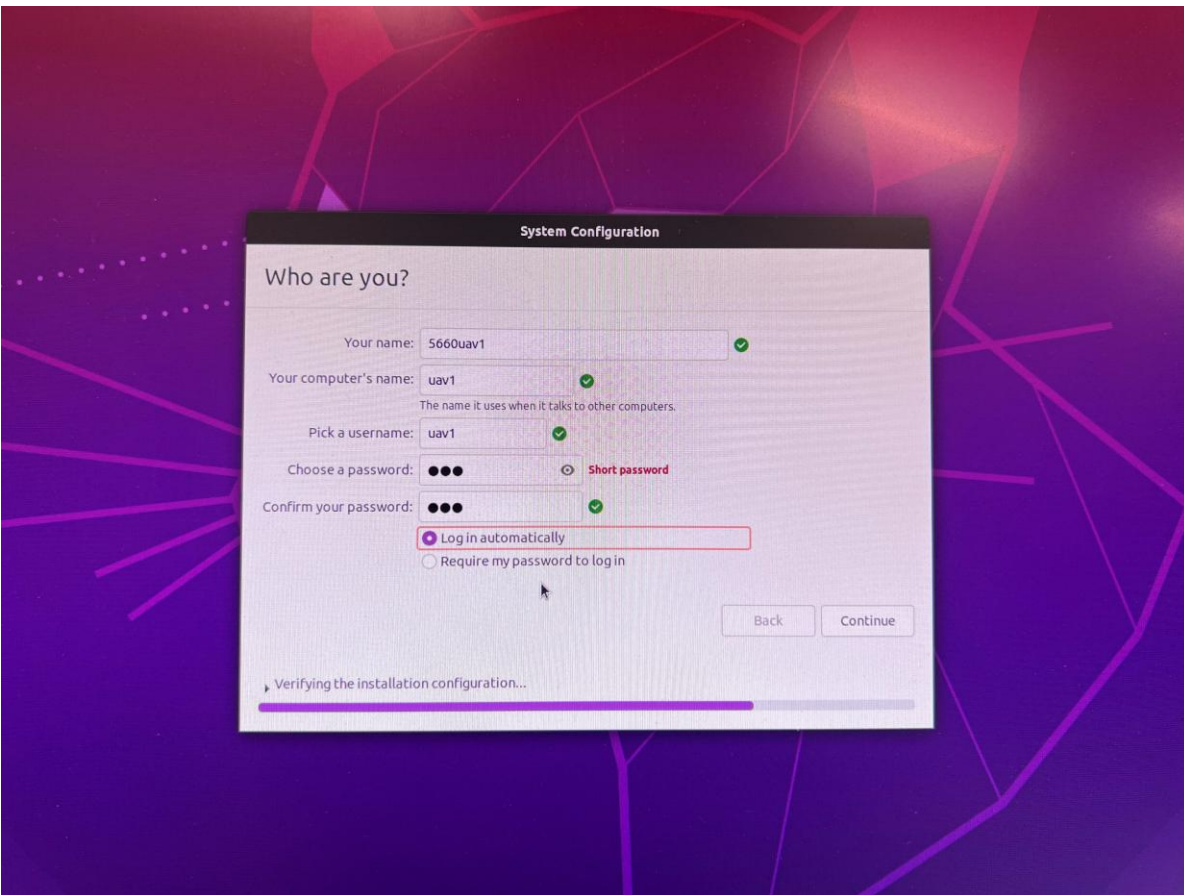
FINISH
AND EXIT

< BACK TO STEP 01

1. Finish the flashing of Xavier NX.



1. Connect to screen, mouse, and keyboard. (MicroHDMI and Hub will be provided.)



1. Setup system configuration.
2. Connect to internet using WiFi. (Network: uav-swarm-5G, Password: hkustdji)

NoMachine for ARM - aarch64



Version:	8.11.3_3
Package size:	48.31 MB
Package type:	TAR.GZ
MD5 signature:	0b7011f14a09f773ed54d0a016ce5022
For:	Ubuntu 14.04/16.04/18.04/20.04, Debian 8/9/10, Fedora 26/27/28/29/30/31/32/33/34, RHEL 7/8, OpenSUSE Tumbleweed



Although your ARMv8 device may not be listed here, we encourage you to try the packages. Please consult the installation and configuration [notes](#) about Linux for ARM packages for more details about devices and specific distributions we have tested.

Download



Downloading and Installing

TAR.GZ version

1. Click on Download and save the TAR.GZ file in the /usr directory.
2. Extract the TAR.GZ archive by running from command line:

```
$ sudo tar zxvf nomachine_8.11.3_3_aarch64.tar.gz
```

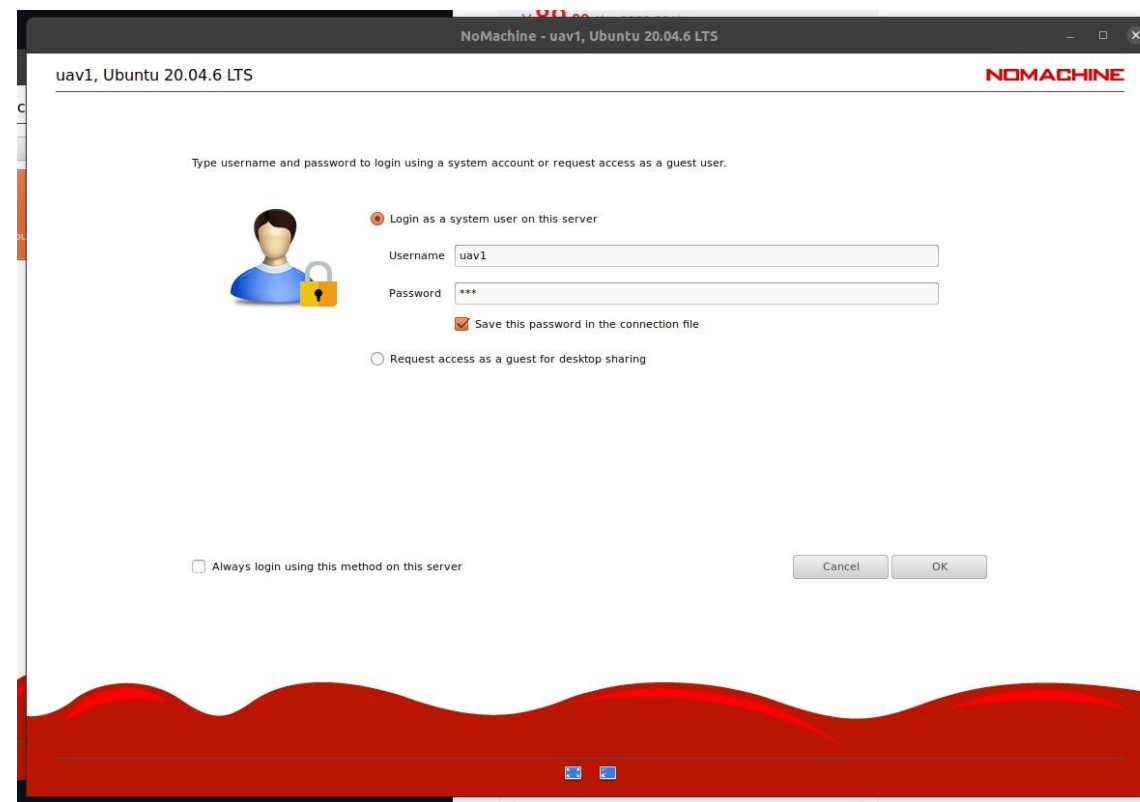
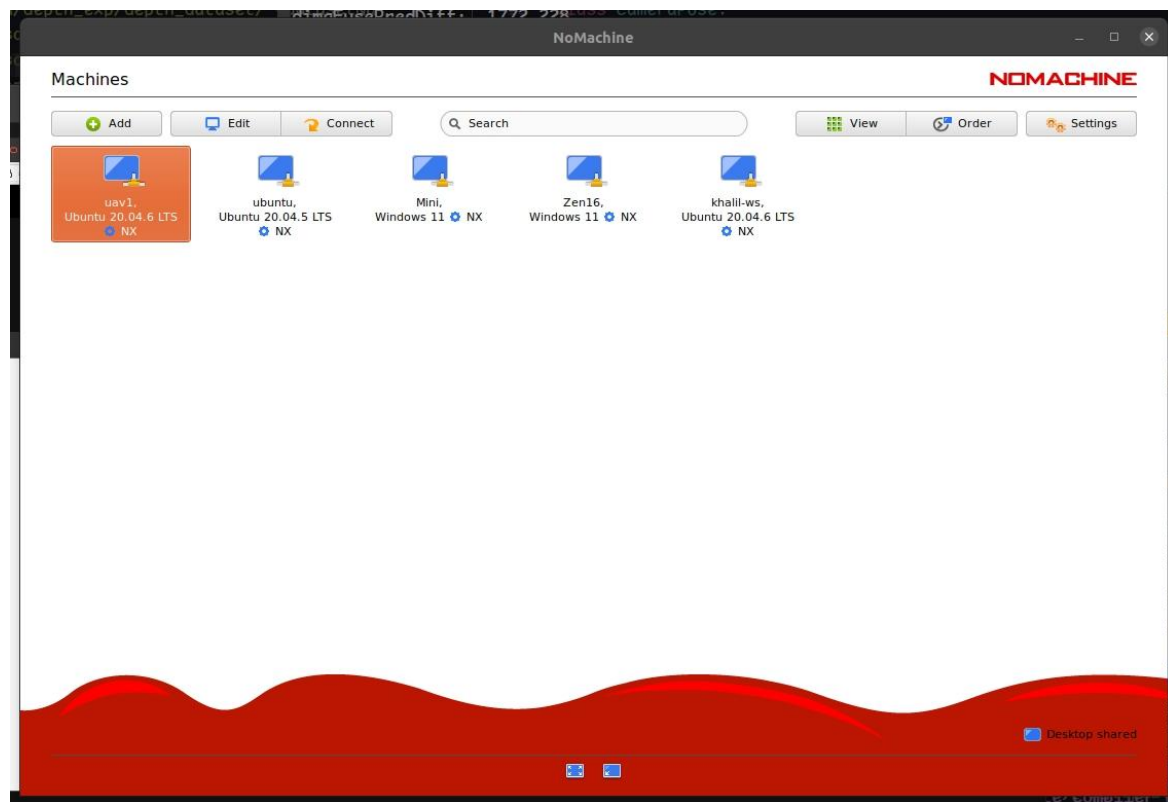
3. Install NoMachine by executing:

```
$ sudo /usr/NX/nxserver --install
```

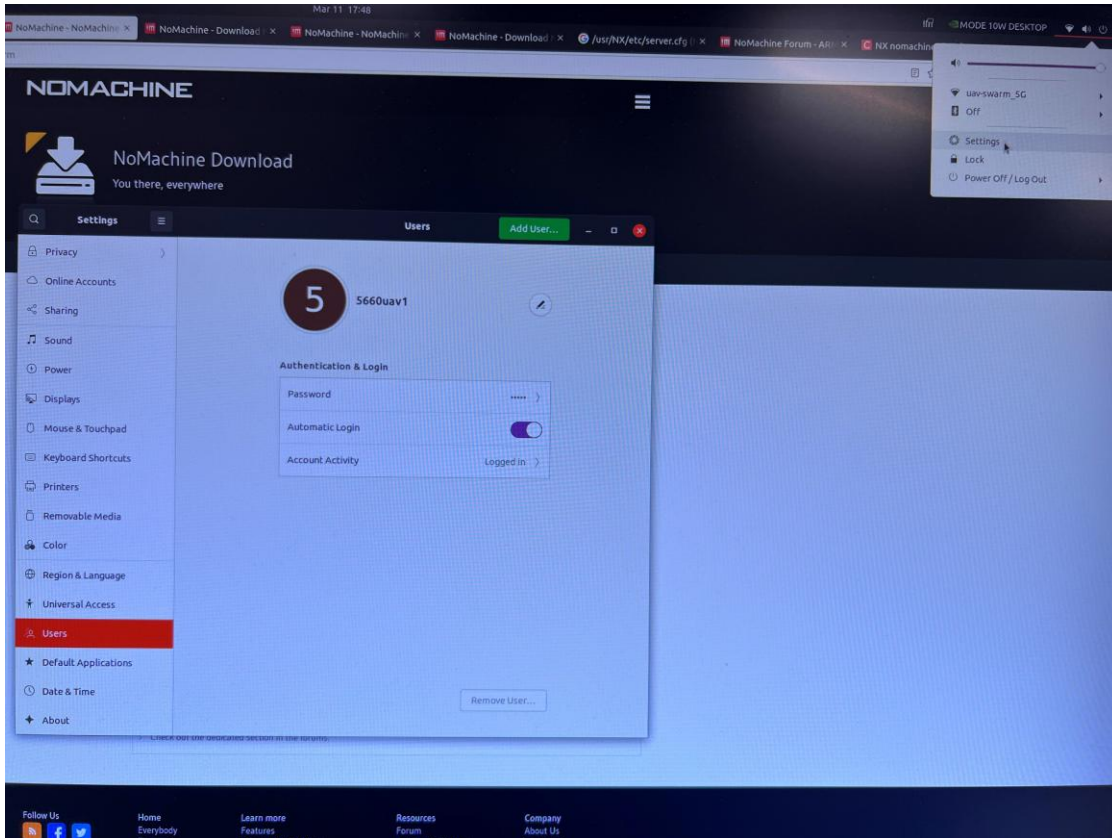
If you don't have the sudo utility installed, log on as superuser ("root") and run the command without sudo.

NOTE: click [here](#) for detailed instructions on how to install the NoMachine packages.

1. Download nomachine in the Xavier NX from <https://downloads.nomachine.com/download/?id=115&distro=ARM>.
2. Follow the given instruction to install this software.



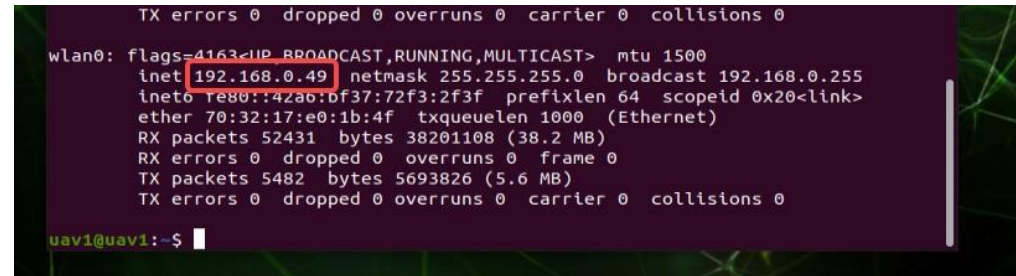
1. Make sure nomachine is also installed in your laptop similar to the last step. Notably, choose right version compatible with your operating system (e.g. Linux).
2. Make sure your laptop and Xavier NX connect the same internet, e.g., uav-swarm-5G.



In your NX, Open terminal on "~/"
Input "touch remote.sh"
Edit this shell by

```
sudo init 3  
sudo /usr/NX/bin/nxserver --restart
```

Afterwards, input "ifconfig" in the terminal to obtain the IP of NX



In the terminal of your laptop, connect NX using
"ssh YOUR_NX_NAME@YOUR_NX_IP"
Then, run the shell by "./remote.sh".

1. Open settings and turn automatic login on.
2. After the operation of right part, you have no need to connect screen, mouse, and keyboard in the next time.
3. When you turn on Xavier NX, wait your NX in nomachine on your laptop and connect it in your own laptop!

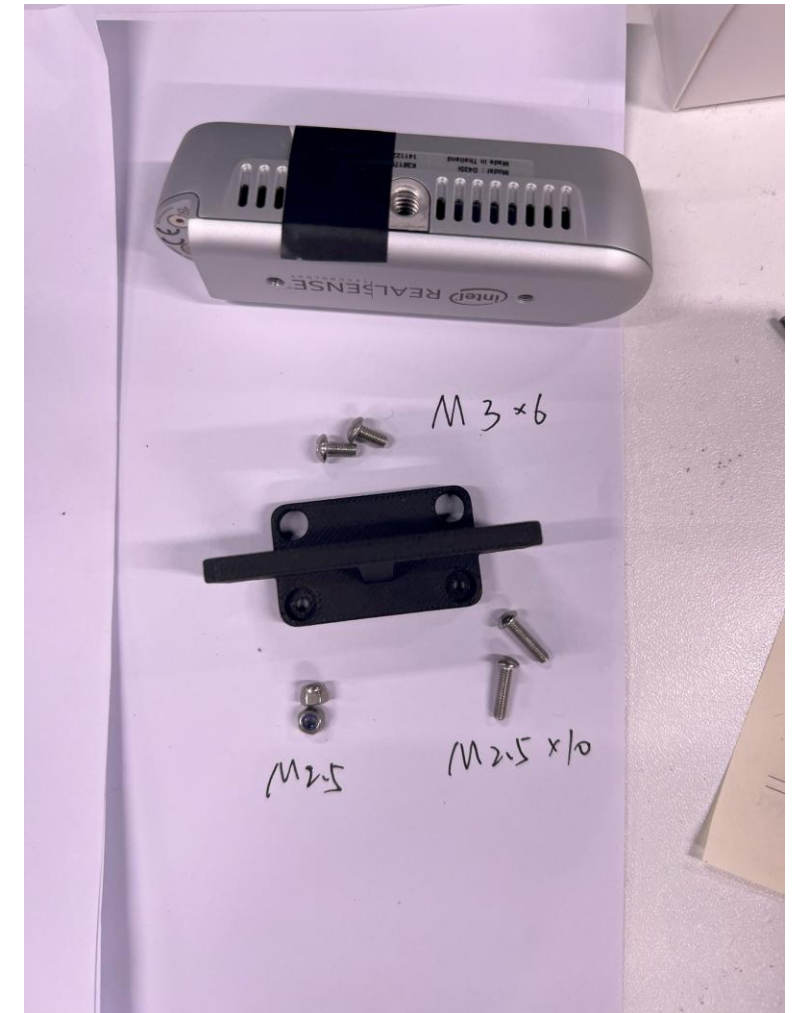
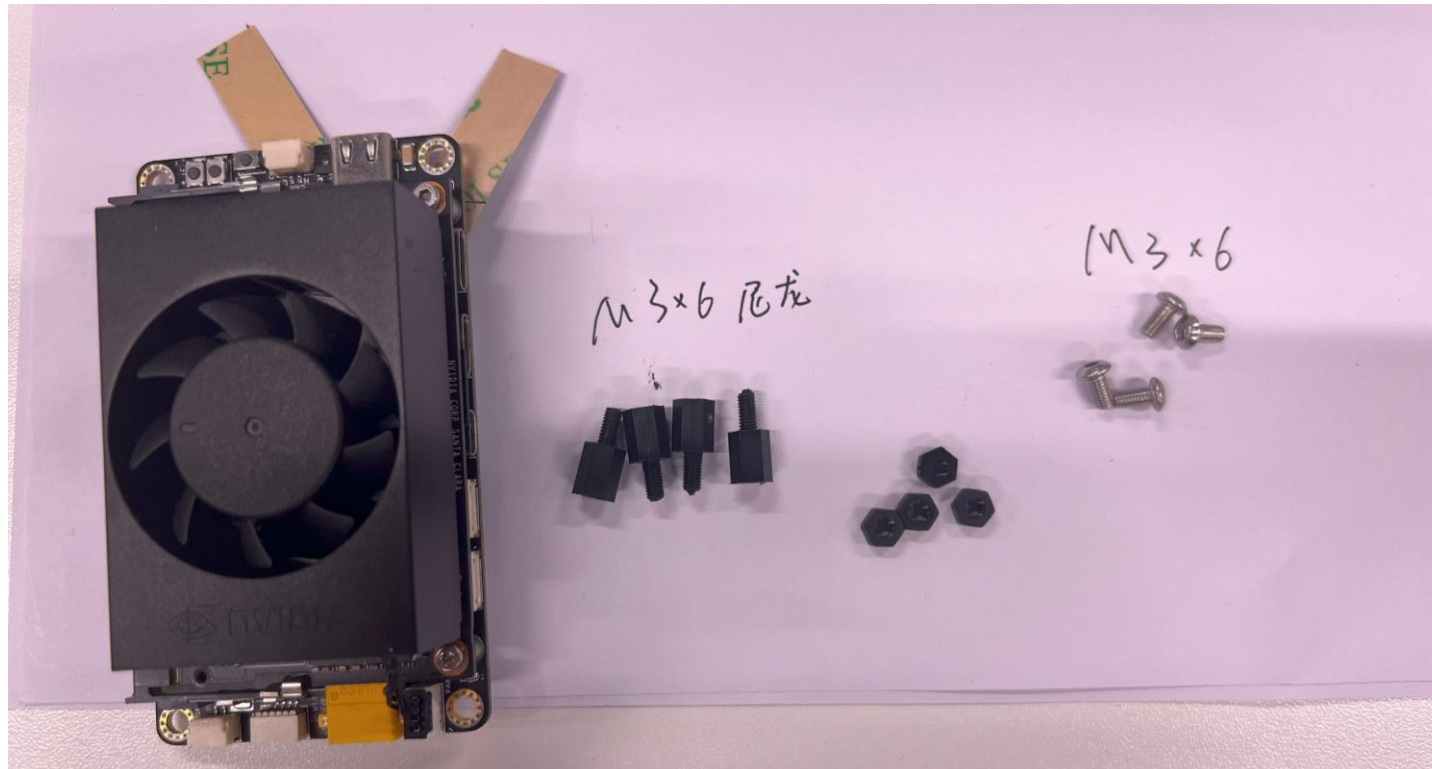
Part 6: Install Camera & NX

Quadrotor Assembly Part 6

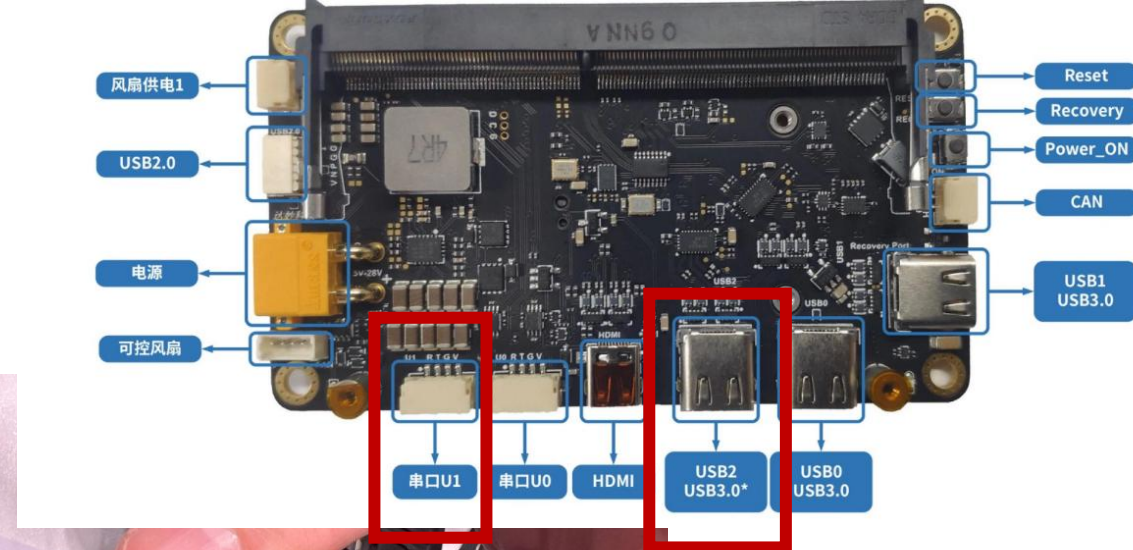
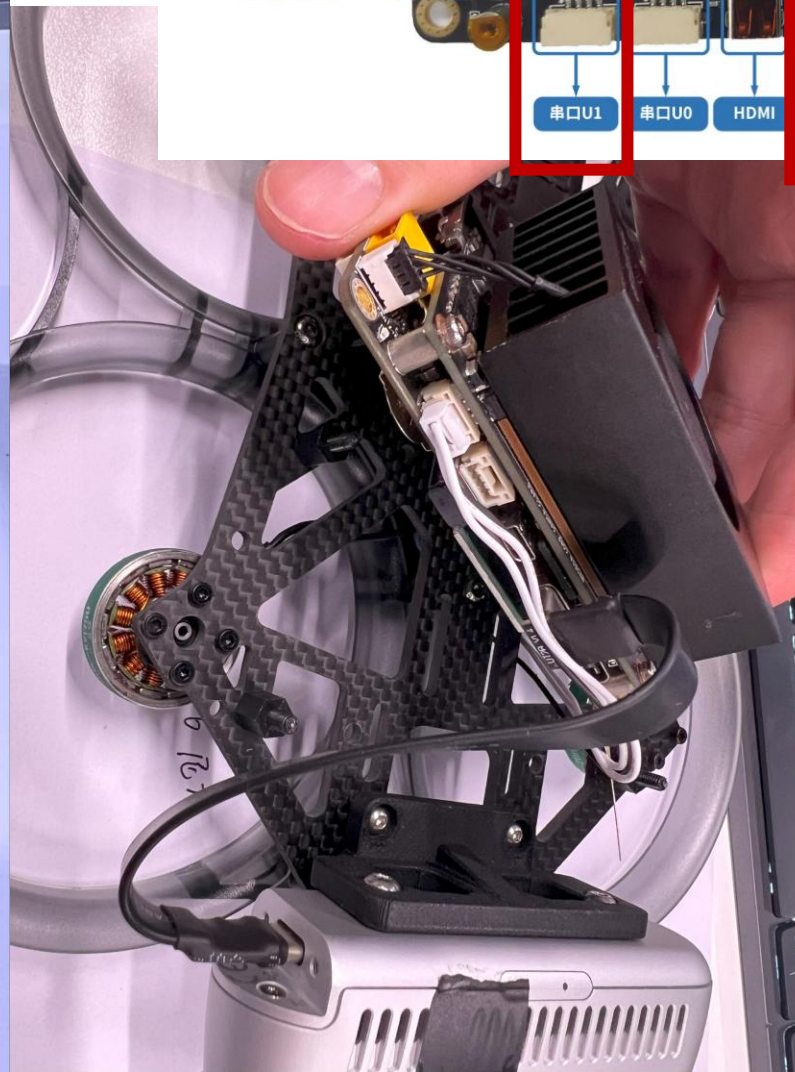
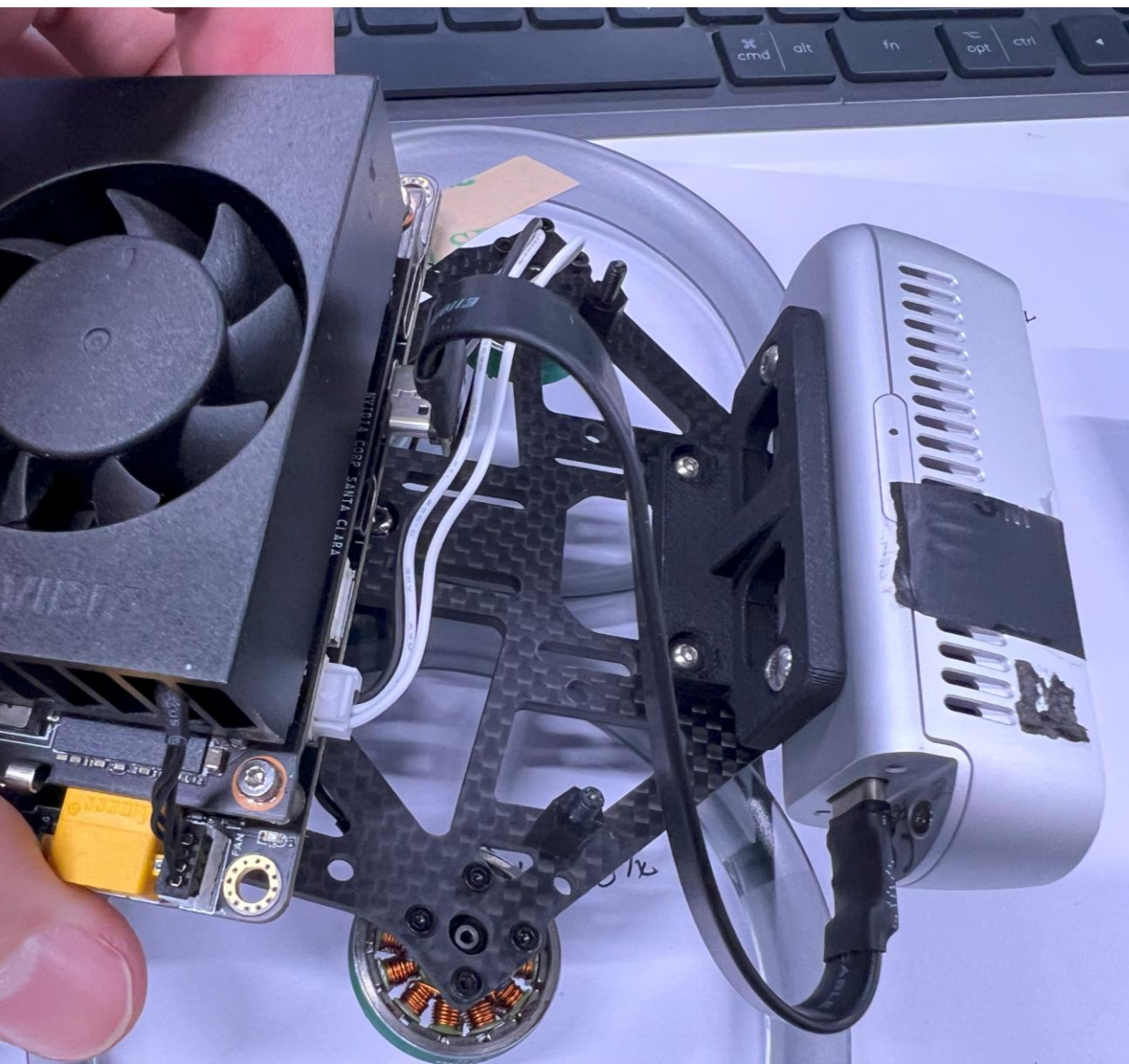
- Equipment List:

Item	Quant.
NX With carrier	1
M3 * 6 Stubs	4
M3 * 6 Screw	6
M3 Nuts	4
Battery Tie	1
Power port board	1
RealSense Camera	1
USB 3.0 Wire	1

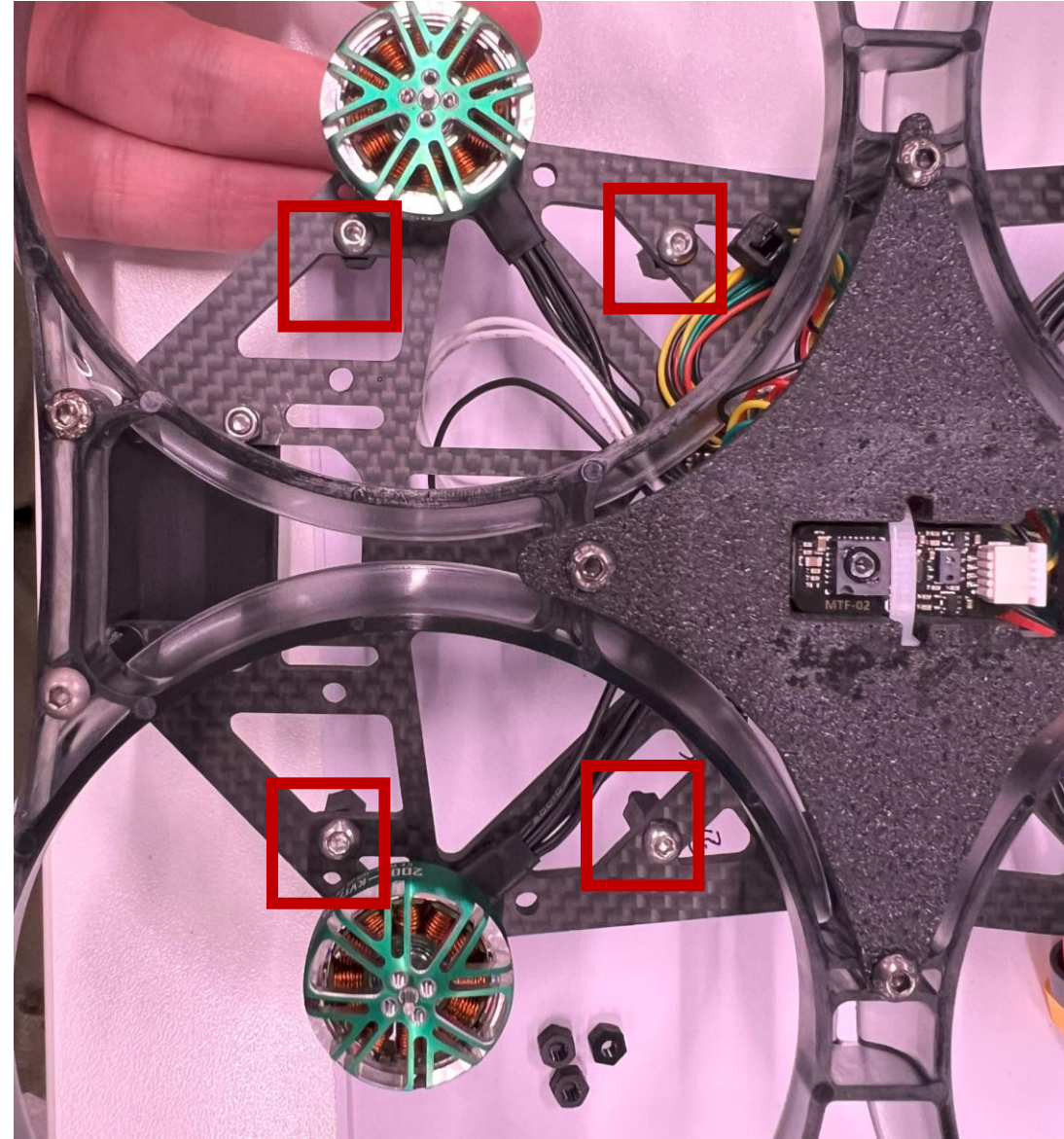
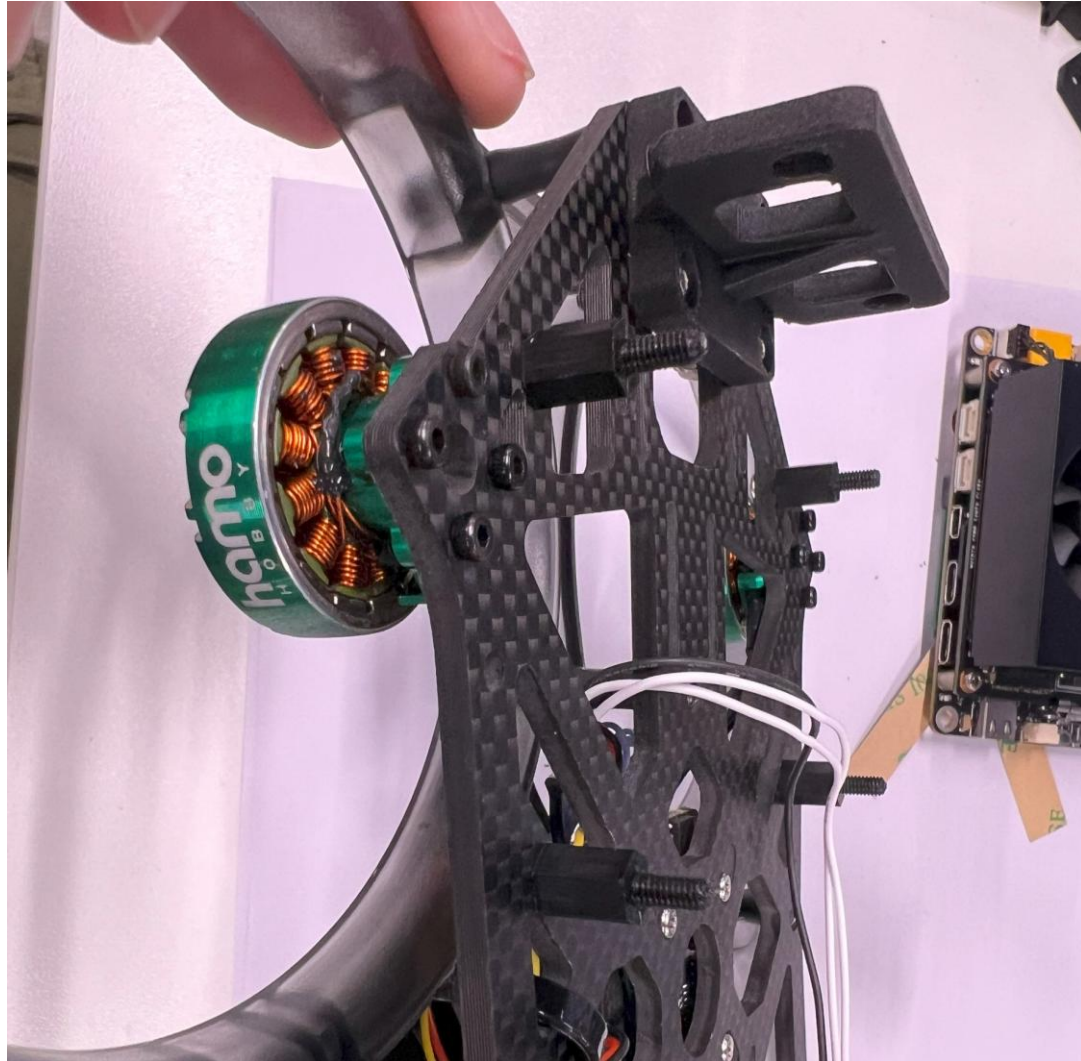
Quadrotor Assembly Part 4



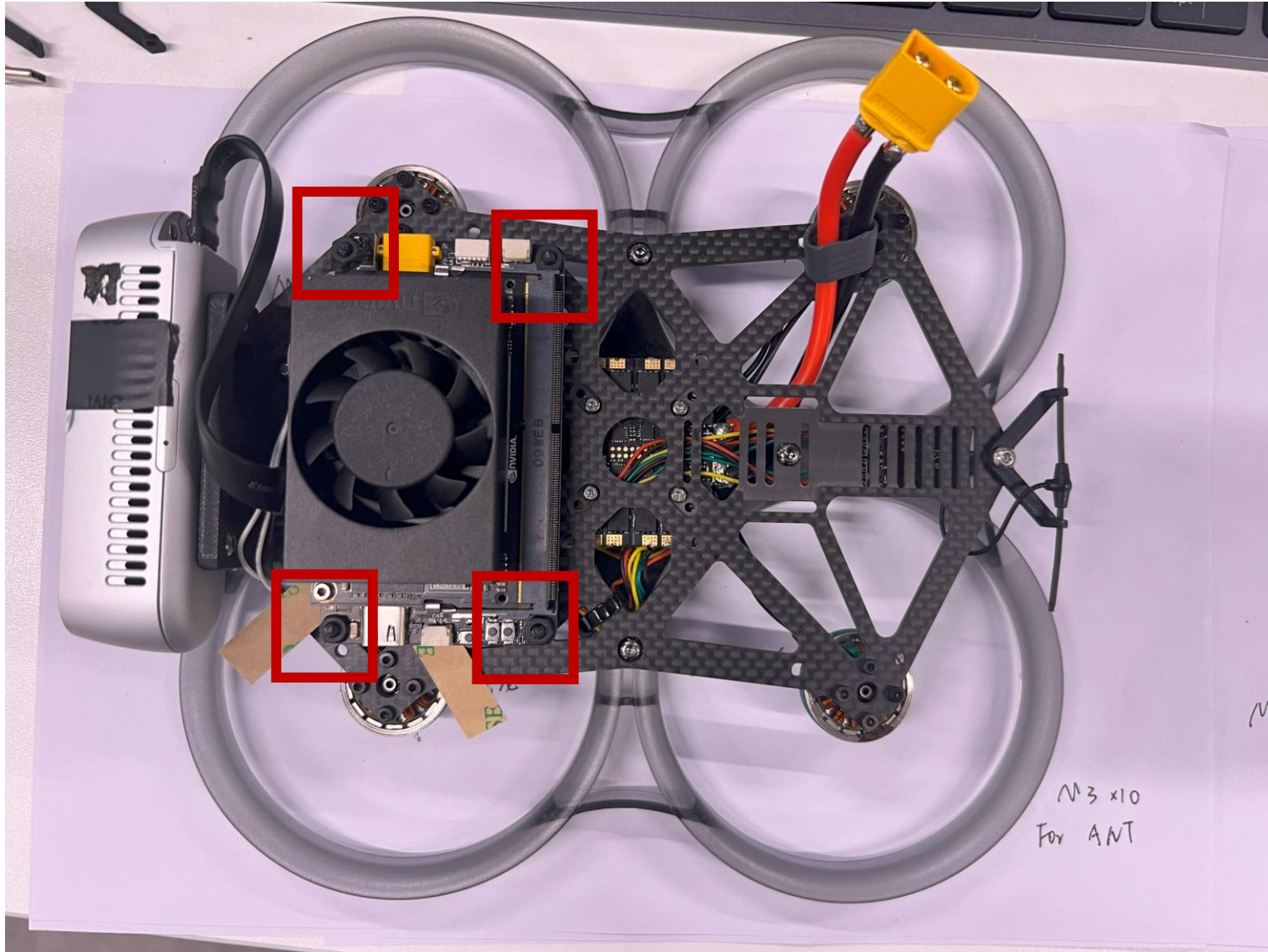
Connect Wire



Install Screw

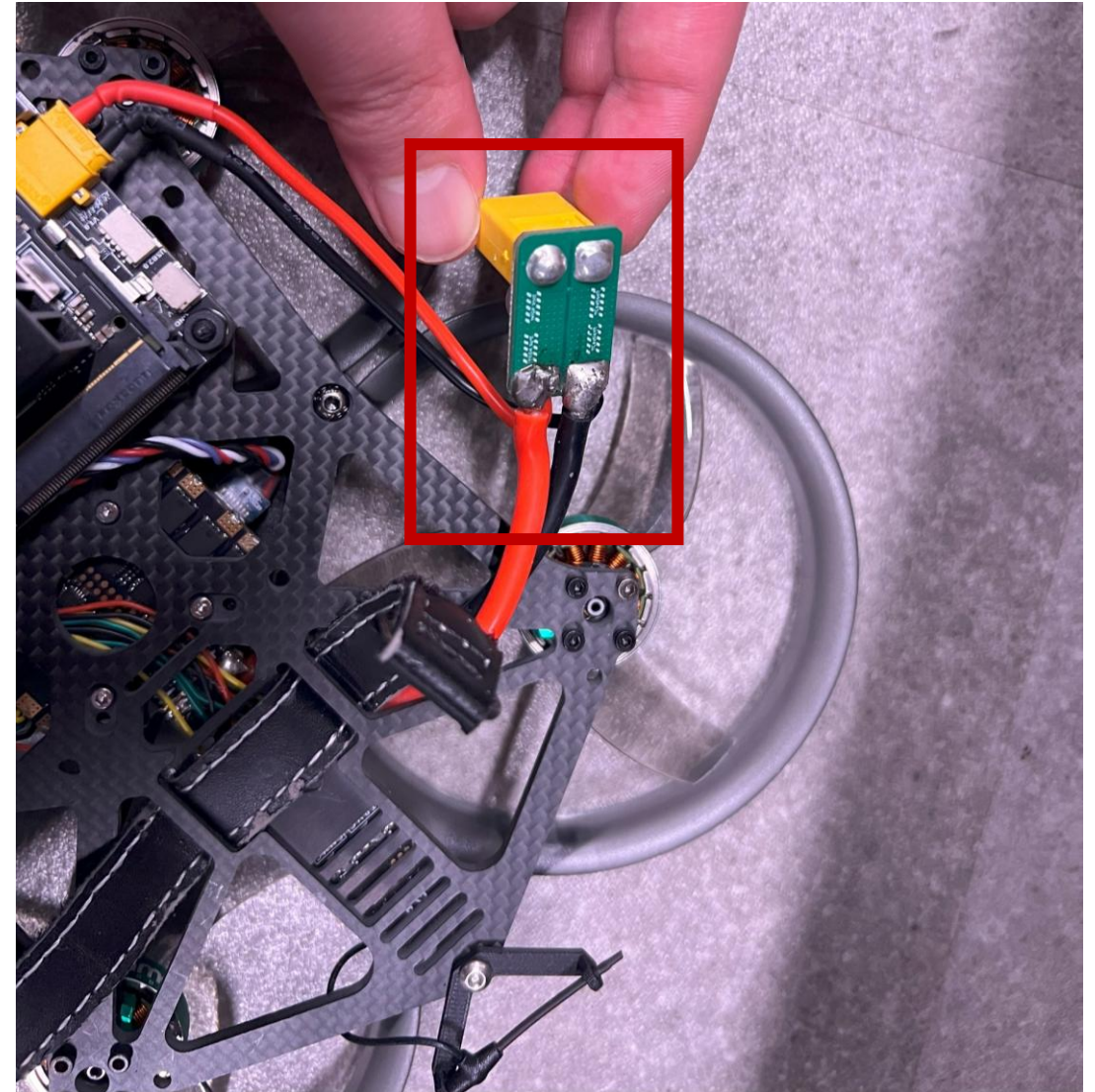
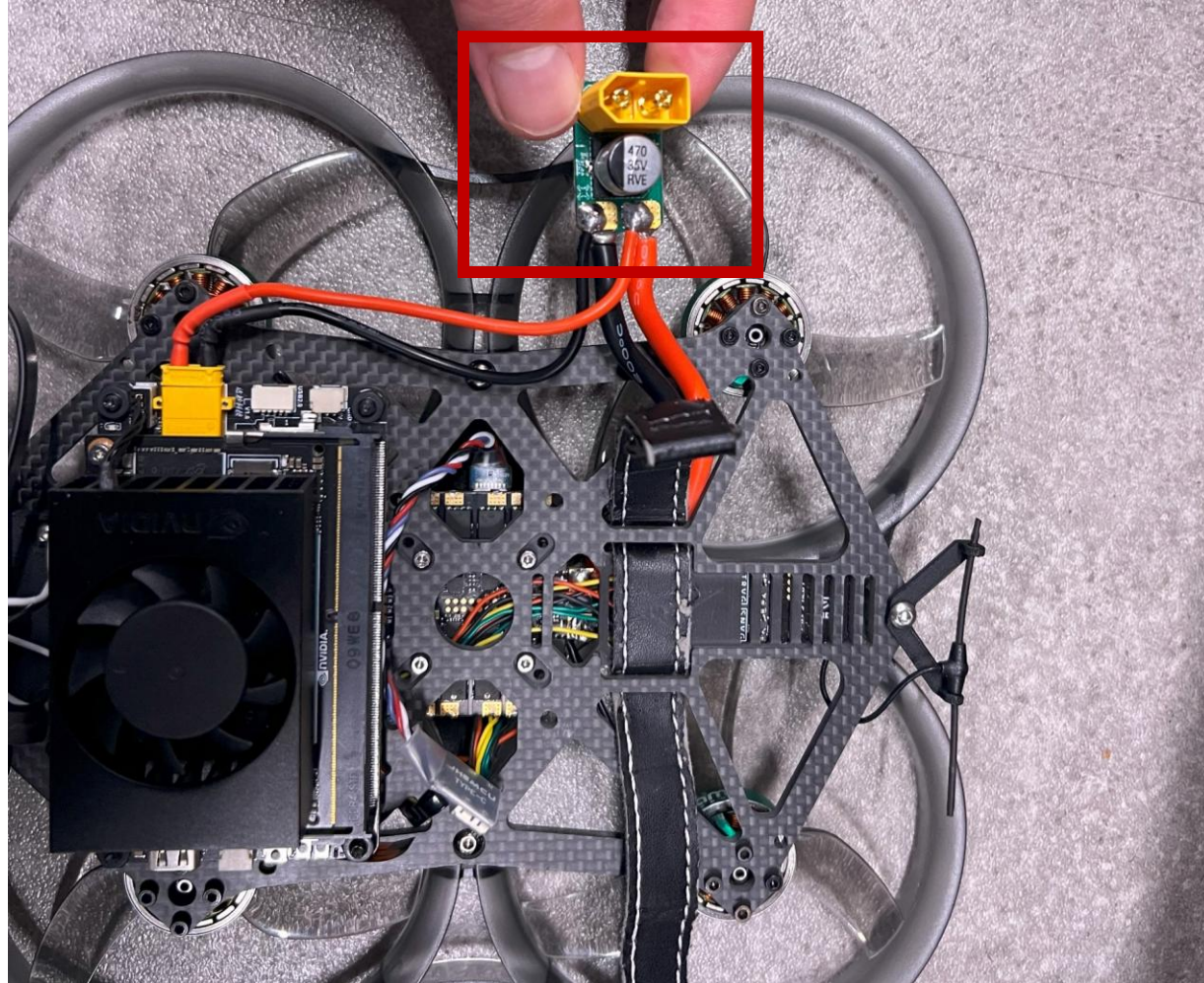


Install Screw

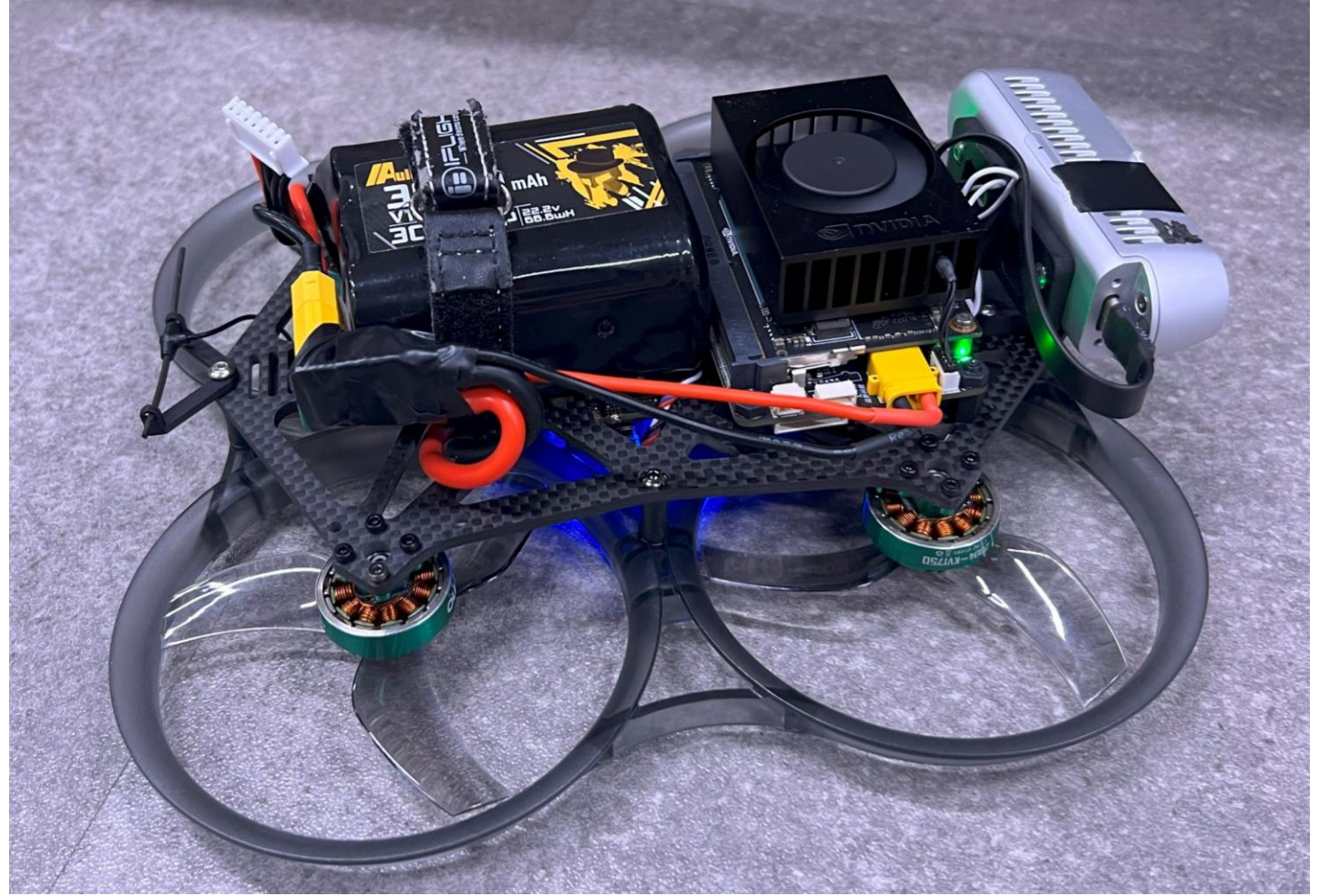


Install Power Line

Don't forget to wrap the power line with tape to insulate it



Finished





READY FOR TAKEOFF